

GATE

GENERAL APTITUDE

Previous Questions with solutions
Subject wise & Chapter wise
(2010 – 2016)

VANI
PUBLICATIONS

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Preface

Dear Reader,

First and foremost let me discuss about GATE exam. GATE is basically an objective type examination, conducts IITs and IISc in the month of February every year. Now days GATE examination gained lot of importance because, not only for M.Tech admission but also for Jobs in PSUs. These PSUs are providing fascinating career to young engineering graduates with excellent packages.

So now the question is how to crack this exam? For this exam one need to prepare according to syllabus provided in notification. In GATE examination basically examiners test your basics and concepts in each and every subject according to their weightages. So, one need to know clearly what to prepare for securing good rank, for this Vani Institute is providing simple solution with this book.

In this book we are providing stream wise, all previous years GATE General Aptitude Questions with solutions. One can use this book for quick revision and to understand clearly what to focus in each subject.

We developed to the best of our knowledge, in case any mistake and suggestions please feel free to inform us.

Director,

VANI INSTITUTE

GATE - GENERAL APTITUDE SYLLABUS

Syllabus for General Aptitude (GA) (COMMON TO ALL PAPERS)

Verbal Ability: English grammar, sentence completion, verbal analogies, word groups, instructions, critical reasoning and verbal deduction.

Numerical Ability: Numerical computation, numerical estimation, numerical reasoning and data interpretation.

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GENERAL APTITUDE

EEE PAPER – I

- 1) The man who is now Municipal Commissioner worked as _____
a) the security guard at a university b) a security guard at the university
c) a security guard at university d) the security guard at the university

Ans: (b)

Sol: Option (a) and (d) cannot be the answers because of the word 'the security'

- 2) Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia.
Choose the option which is closest in meaning to the underlined phrase in the above sentence.

- a) put up with b) put in with c) put down to d) put up against

Ans: (a)

Sol: 'cope with' means put up with

- 3) Find the odd one in the following group of words.
mock, deride, praise, jeer

- a) mock b) deride c) praise d) jeer

Ans: (c)

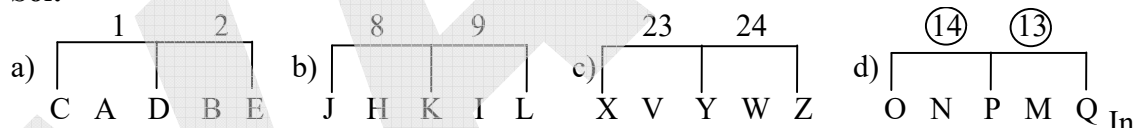
Sol: mock, deride and jeer are synonyms which means mockery. Therefore, the odd one is 'praise'

- 4) Pick the odd one from the following options.

- a) CADBE b) JHKIL c) XVEWZ d) ONPMQ

Ans: (d)

Sol:



options a, b and c, the letters skipped between consecutive pair of letters in the English alphabet is in increasing order (i.e) A and B, H and I and V and W but in option 'd' N and M are present instead of M and N so, option 'd' is odd one from the group.

- 5) In a quadratic function, the value of the product of the roots (α , β) is 4. Find the value of $\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$

- a) n^4 b) 4^n c) 2^{2n-1} d) 4^{n-1}

Ans: (b)

Sol:
$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}} = \frac{\alpha^n + \beta^n}{\frac{1}{\alpha^n} + \frac{1}{\beta^n}} = \frac{\alpha^n + \beta^n}{\frac{\beta^n + \alpha^n}{\alpha^n \times \beta^n}} = \alpha^n \times \beta^n$$

$$\alpha^n \times \beta^n = (\alpha \times \beta)^n = (4)^n$$

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Q.No 6-Q.No10 Carry two marks each

- 6) Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____
- a) 35 b) 45 c) 65 d) 90

Ans: (a)

Sol: Total faculty members = 150

The faculty members having facebook account = FB = 55

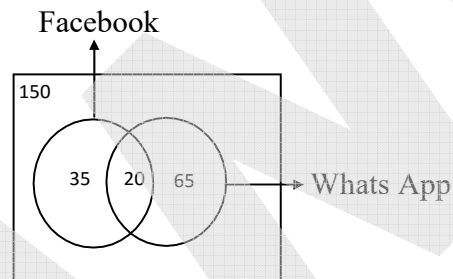
The faculty members having whatsapp = W = 85

The faculty members do not have face book (or) Whats App accounts = 30

The faculty members having any account = $150 - 30 = 120$

The faculty members having both the accounts = $(FB + W) - 120 = (55 + 85) - 120 = 20$

∴ The number of faculty members connected only through Facebook accounts
= $55 - 20 = 35$



- 7) Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required.
- Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?
- (i) The author believes that computers are not good for us.
(ii) Mobile computers and the internet are both intended inventions.
- a) (i) only b) (ii) only c) both (i) and (ii) d) neither (i) nor (ii)

Ans: (d)

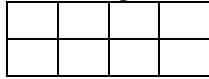
- 8) All hill-stations have a lake. Ooty has two lakes.
- Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?
- (i) Ooty is not a hill-station.
(ii) No hill-station can have more than one lake.
- a) (i) only b) (ii) only c) both (i) and (ii) d) neither (i) nor (ii)

Ans: (d)

Sol: Statement (i) is not true because Ooty is a hill station due Ooty has two lakes
statement(ii) is also not true, because in given statements, for hill station one lake is compulsory but not mentioned about number of lakes.

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- 9) In a 2×4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?



- a) 21 b) 27 c) 30 d) 36

Ans: (c)

Sol: In given 2×4 rectangle grid, the following type of rectangles are present.

One figured rectangles = 8

Two figured rectangles = 10

Three figured rectangles = 4

Four figured rectangles = 5

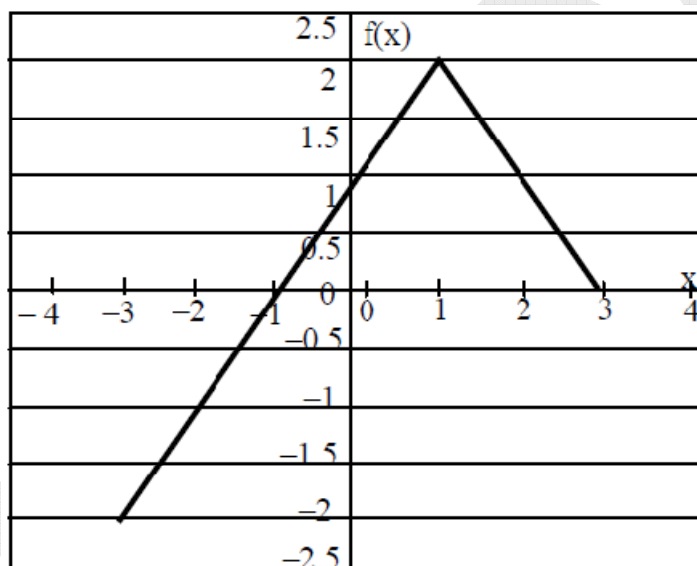
Six figured rectangles = 2

Eight figured rectangles = 1

Total No. of rectangles = 30

$\therefore$ The No. of rectangles observed in the given grid = 30.

10)



Choose the correct expression for $f(x)$ given in the graph.

- a) $f(x) = 1 - |x - 1|$ b) $f(x) = 1 + |x - 1|$
c) $f(x) = 2 - |x - 1|$ d) $f(x) = 2 + |x - 1|$

Ans: (c)

Sol: @ $x = 3$ from given graph $f(x)$ must be equals to zero

Option A

$$f(x) = 1 - |x - 1|$$

$$@ x = 3$$

$$f(x) = 1 - |3 - 1|$$

$$= 1 - 2 = -1$$

So, it is not

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Option B

$$f(x) = 1 + |x - 1|$$

$$@ x = 3$$

$$\begin{aligned} f(x) &= 1 + |3 - 1| \\ &= 1 + 2 = 3 \end{aligned}$$

So, it is not

Option C

$$f(x) = 2 - |x - 1|$$

$$@ x = 3$$

$$\begin{aligned} f(x) &= 2 - |3 - 1| \\ &= 2 - 2 = 0 \end{aligned}$$

So, it is true

Option D

$$f(x) = 2 + |x - 1|$$

$$@ x = 3$$

$$\begin{aligned} f(x) &= 2 + |3 - 1| \\ &= 2 + 2 = 4 \end{aligned}$$

So, it is not

GENERAL APTITUDE

EEE PAPER – II

- 1) The chairman requested the aggrieved shareholders to _____ him.
a) bare with b) bore with c) bear with d) bare
Ans: (c)
Sol: 'bear with' means to be patient with some one
- 2) Identify the correct spelling out of the given options:
a) Managable b) Manageable c) Mangaable d) Managible
Ans: (b)
Sol: 'manageable' is the correct spelling
- 3) Pick the odd one out in the following:
13, 23, 33, 43, 53
a) 23 b) 33 c) 43 d) 53
Ans: (b)
Sol: In the group of given numbers, all are prime numbers but 33 is not
- 4) R2D2 is a robot. R2D2 can repair aeroplanes. No other robot can repair aeroplanes. Which of the following can be logically inferred from the above statements?
a) R2D2 is a robot which can only repair aeroplanes.
b) R2D2 is the only robot which can repair aeroplanes.
c) R2D2 is a robot which can repair only aeroplanes.
d) Only R2D2 is a robot.
Ans: (b)
Sol: The statement 'No other robot can repair aeroplanes' means R2D2 is the only robot which can repair aeroplanes so option (b) is the best inference
- 5) If $|9y - 6| = 3$, then $y^2 - 4y/3$ is _____.
a) 0 b) $+1/3$ c) $-1/3$ d) undefined

Ans: (c)

Sol: $|9y - 6| = 3$

$9y - 6 = \pm 3$

Case I: $9y - 6 = 3$

$$\Rightarrow y = 1$$

Case II: $9y - 6 = -3$

$$\Rightarrow y = \frac{1}{3}$$

Substitute $y = 1$

$$y^2 - \frac{4y}{3} = 1^2 - \frac{4(1)}{3} = 1 - \frac{4}{3} = -\frac{1}{3}$$

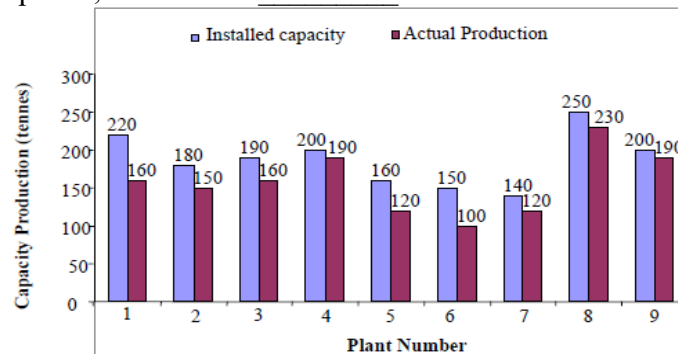
Substitute $y = \frac{1}{3}$

$$y^2 - \frac{4y}{3} = \left(\frac{1}{3}\right)^2 - \frac{4(1/3)}{3} = \frac{1-4}{9} = -\frac{1}{3}$$

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Q. 6 to Q. 10 carry two marks each

- 6) The following graph represents the installed capacity for cement production (in tonnes) and the actual production (in tonnes) of nine cement plants of a cement company. Capacity utilization of a plant is defined as ratio of actual production of cement to installed capacity. A plant with installed capacity of at least 200 tonnes is called a large plant and a plant with lesser capacity is called a small plant. The difference between total production of large plants and small plants, in tonnes is _____



Ans: 120

Sol: Installed capacity ≥ 200 tonnes $\Rightarrow$ large plant

Installed capacity < 200 tonnes $\Rightarrow$ small plant

From given multiple pie chart, the large plants are 1, 4, 8 & 9

Total production of large plants = $160 + 190 + 230 + 190 = 770$ tonnes

Total production of small plants = $150 + 160 + 120 + 100 + 120 = 650$ tonnes

$\therefore$ The difference between total production of large plants and small plants in tonnes
 $= 770 - 650 = 120$

- 7) A poll of students appearing for masters in engineering indicating that 60% of the students believed that mechanical engineering is a profession unsuitable for women. A research study on women with master or higher degrees in mechanical engineering found that 90% of such women were successful in their professions.

Which of the following can be logically inferred from the above paragraph?

- a) Many students have misconceptions regarding various engineering disciplines.
- b) Men with advanced degrees in mechanical engineering believe women are well suited to be mechanical engineers.
- c) Mechanical engineering is a profession well suited for women with masters or higher degrees in mechanical engineering.
- d) The number of women pursuing higher degrees in mechanical engineering is small.

Ans: (c)

Sol: A poll says that women with masters or higher degrees in mechanical engineers are successful in their professions. This statement leads to the option 'c' which is the best inference

- 8) Sourya committee had proposed the establishment of sourya institutes of Technology (SITs) in line with Indian Institutes of Technology (IITs) to cater to the technological and industrial needs of a developing country

Which of the following can be logically inferred from the above sentence?

Based on the proposal,

- (i) In the initial years, SIT students will get degrees from IIT.
- (ii) SITs will have a distinct national objective.

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(iii) SITs like institutions can only be established in consultation with IIT.

(iv) SITs will serve technological needs of a developing country.

a) (iii) and (iv) only b) (i) and (iv) only c) (ii) and (iv) only d) (ii) and (iii) only

Ans: (c)

Sol: Option (i) and (iii) state phrases like 'in the initial years' and 'SIT like institutions can only be established in consultation with IIT' cannot be logically inferred so (ii) and (iv) are the best inferences i.e. option (c)

- 9) Shaquille O' Neal is a 60% career free throw shooter, meaning that he successfully makes 60 free throws out of 100 attempts on average. What is the probability that he will successfully make exactly 6 free throws in 10 attempts?

a) 0.2508 b) 0.2816 c) 0.2934 d) 0.6000

Ans: (a)

Sol: n = no. of attempts = 10

x = free throws successfully = 6

∴ The probability that he will successfully make exactly 6 free throws in 10 attempts

$${}^n C_x P^x \cdot q^{n-x} \quad [\because q = 1 - p]$$

$$P = 60\% = \frac{60}{100} = 0.6$$

$$= {}^{10} C_6 \times (0.6)^6 \times (0.4)^4$$

$$= 210 \times 0.046656 \times 0.0256$$

$$= 0.2508$$

- 10) The numeral in the units position of $211^{870} + 146^{127} \times 3^{424}$ is _____

Ans: 7

Sol: $211^{870} + 146^{127} \times 3^{424} = (1)^2 + (6)^3 \times 1$

$$= 1 + 216 \times 1$$

$$= 217$$

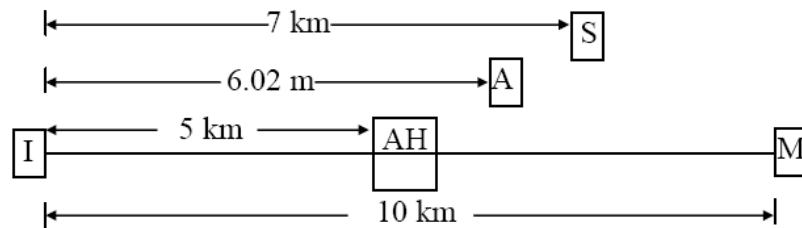
∴ Units digit = '7'

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ECE PAPER – I

- 1) Which of the following is CORRECT with respect to grammar and usage?
Mount Everest is _____
(A) the highest peak in the world (B) highest peak in the world
(C) one of highest peak in the world (D) one of the highest peak in the world
Ans: (A)
Sol: Before superlative article 'the' has to be used. 'one of' the expression should take plural noun and so option 'C' and 'D' can't be the answer.
- 2) The policeman asked the victim of a theft, "What did you _____?"
(A) loose (B) lose (C) loss (D) louse
Ans: (B)
Sol: 'lose' is verb.
- 3) Despite the new medicine's _____ in treating diabetes, it is not _____ widely.
(A) effectiveness --- prescribed (B) availability ---- used
(B) prescription ---- available (D) acceptable--- proscribed
Ans: (A)
Sol: 'effectiveness' is noun and 'prescribed' is verb. These words are apt and befitting with the word 'medicine.'
- 4) In a huge pile of apples and oranges, both ripe and unripe mixed together, 15% are unripe fruits, of the unripe fruits, 45% are apples, of the ripe ones, 66% are oranges. If the pile contains a total of 5692000 fruits, how many of them are apples?
(A) 2029198 (B) 2467482 (C) 2789080 (D) 3577422
Ans: (A)
Sol: Total no. of fruits = 5692000
Unripe type of apples = 45% of 15% of 5692000
$$= \frac{45}{100} \times \frac{15}{100} \times 5692000$$
$$= 384210$$
$$\text{Ripe type of apples} = \frac{34}{100} \times \frac{85}{100} \times 5692000 = 1644988$$
$$\therefore \text{Total no. of apples} = 384210 + 1644988 = 2029198$$
- 5) Michael lives 10 km away from where I live. Ahmed lives 5km away and Susan lives 7 km away from where I live. Arun is farther away than Ahmed but closer than Susan from where I live. From the information provided here, what is one possible distance (in km) at which I live from Arun's place?
(A) 3.00 (B) 4.99 (C) 6.02 (D) 7.01
Ans: (c)
Sol: From given data, the following diagram is possible.

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I = I live

AH = Ahmed lives; M = Michael lives

S = Susan lives; A = Arun lives

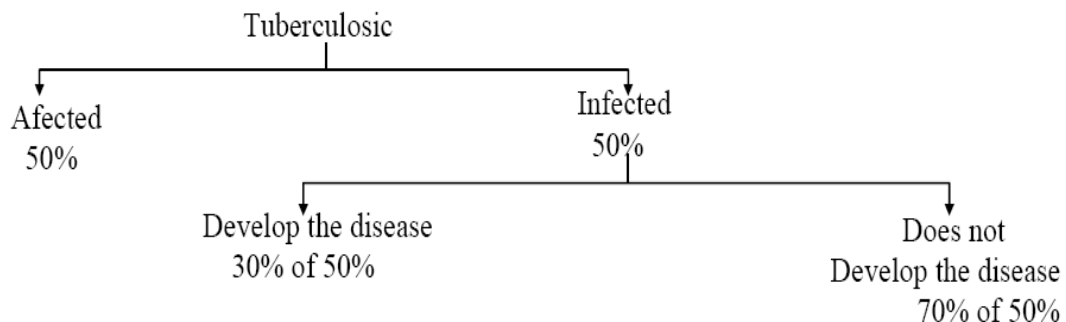
→ Arun lives farther away than Ahmed means more than 5 km but closer than Susan means less than 7 km, from given alternatives, option 'C' only possible.

Q.6 – Q.10 Carry two marks each.

- 6) A person moving through a tuberculosis prone zone has a 50% probability of becoming infected. However, only 30% of infected people develop the disease. What percentage of people moving through a tuberculosis prone zone remains infected but does not shows symptoms of disease?
- (A) 15 (B) 33 (C) 35 (D) 37

Ans: (C)

Sol:



- 7) In a world filled with uncertainty, he was glad to have many good friends. He had always assisted them in times of need and was confident that they would reciprocate. However, the events of the last week proved him wrong. Which of the following inference(s) is/are logically valid and can be inferred from the above passage?
- (i) His friends were always asking him to help them.
 (ii) He felt that when in need of help, his friends would let him down.
 (iii) He was sure that his friends would help him when in need.
 (iv) His friends did not help him last week.

(A) (i) and (ii) (B) (iii) and (iv) (C) (iii) only (D) (iv) only

Ans: (B)

Sol: The words 'was confident that they would reciprocate' and 'last week proved him wrong' lead to statements iii and iv as logically valid inferences.

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- 8) Leela is older than her cousin Pavithra, Pavithra's brother Shiva is older than Leela. When Pavithra and Shiva are visiting Leela, all three like to play chess. Pavithra wins more often than Leela does. Which one of the following statements must be TRUE based on the above?

(A) When Shiva plays chess with Leela and Pavithra, He often loses.
(B) Leela is the oldest of the three.
(C) Shiva is a better chess player than Pavithra.
(D) Pavithra is the youngest of the three.

Ans: (D)

Sol: From given data, the following arrangement is possible.

Shiva

Leela

Pavithra

Among four alternatives, option D is TRUE.

- 9) If $q^{-a} = \frac{1}{r}$ and $r^{-b} = \frac{1}{s}$ and $s^{-c} = \frac{1}{q}$, the value of abc is _____.

(A) $(rqs)^{-1}$

(B) 0

(C) 1

(D) $r+q+s$

Ans: (C)

Sol: $q^{-a} = \frac{1}{r} \Rightarrow \frac{1}{q^a} = \frac{1}{r} \Rightarrow q^a = r$

$$r^{-b} = \frac{1}{s} \Rightarrow \frac{1}{r^b} = \frac{1}{s} \Rightarrow s = r^b$$

$$s^{-c} = \frac{1}{q} \Rightarrow \frac{1}{s^c} = \frac{1}{q} \Rightarrow s^c = q$$

$$q^a = r \Rightarrow (s^c)^a = r \Rightarrow s^{ac} = r$$

$$(s^{ac})^b = s$$

$$s^{abc} = s$$

$$\therefore abc = 1$$

- 10) P, Q, R and S are working on a project. Q can finish the task in 25 days, working alone for 12 hours a day. R can finish the task in 50 days, working alone for 12 hours per day. Q worked 12 hours a day but took sick leave in the beginning for two days. R worked 18 hours a day on all days. What is the ratio of work done by Q and R after 7 days from the start of the project?

(A) 10:11

(B) 11:10

(C) 20:21

(D) 21:20

Ans: (C)

Sol: Q can finish the task = 25 days, 12 hrs/day

$$= 300 \text{ hrs, } 1 \text{ hr} = \frac{1}{300} \text{ th}$$



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R can finish the task = 50 days, 12 hrs/day

$$= 50 \times 12$$

$$= 600 \text{ hrs, } 1 \text{ hr} = \frac{1}{600} \text{ th}$$

Q working hours $\Rightarrow (7 - 2) \times 12 = 60 \text{ hrs}$

R working hours $\Rightarrow 7 \times 18 = 126 \text{ hrs}$

After 7 days, the ratio of work done by Q and R

Q : R

$$\frac{60}{300} : \frac{126}{600}$$

$$20 : 21$$

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ECE PAPER – II

- 1) Based on the given statements, select the appropriate option with respect to grammar and usage

Statements

- (i) The height of Mr. X is 6 feet.
- (ii) The height of Mr. Y is 5 feet
- (A) Mr. X is longer than Mr. Y
- (B) Mr. X is more elongated than Mr. Y
- (C) Mr. X is taller than Mr. Y
- (D) Mr. X is lengthier than Mr. Y

Ans:(C)

Sol: In degrees of comparison Mr. X is taller than Mr. Y is apt.

Positive degree	-	tall
Comparative degree	-	taller
Superlative degree	-	tallest

- 2) The students _____ the teacher on teachers day for twenty years of dedicated teaching.
(A) facilitated (B) felicitated (C) fantasized (D) facillitated

Ans: (B)

Sol: Felicitate means honour.

- 3) After India's cricket world cup victory in 1985, Shrotria who was playing both tennis and cricket till then, decided to concentrate only on cricket. And the restis history. What does the underlined phrase mean in this context?
(A) history will rest in peace (B) rest is recorded in history books
(C) rest is well known (D) rest in archaic

Ans: (C)

Sol: 'rest is history' is an idiomatic expression which means 'rest is well known'.

- 4) Given $(9 \text{ inches})^{1/2} = (0.25 \text{ yards})^{1/2}$, which one of the following statements is TRUE?
(A) 3 inches = 0.5 yards (B) 9 inches = 1.5 yards
(C) 9 inches = 0.25 yards (D) 81 inches = 0.0625 yards

Ans: (C)

Sol: Given $(9 \text{ inches})^{1/2} = (0.25 \text{ yards})^{1/2}$
 $9 \text{ inches} = 0.25 \text{ yards}$

- 5) S, M, E and F are working in shifts in a team to finish a project. M works with twice the efficiency of others but for half as many days as E worked. S and M have 6 hour shifts in a day, whereas E and F have 12 hours shifts. What is the ratio of contribution of M to contribution of E in the project?
(A) 1:1 (B) 1:2 (C) 1:4 (D) 2:1

Ans:(B)

Sol: M efficiency = 2 [efficiency of S, E and F]

Contribution of M in the project = x days $\times$ 6 hrs $\times$ 2

Contribution of E in the project = 2x days $\times$ 12 hrs $\times$ 1



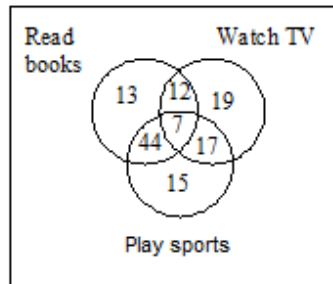
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Contribution of M : Contribution of E

$$\begin{aligned} x \times 6 \times 2 : 2x \times 12 \times 1 \\ 1 : 2 \end{aligned}$$

Q.6 – Q.10 Carry two marks each

- 6) The Venn diagram shows the preference of the student population for leisure activities.



From the data given, the number of students who like to read books or play sports is ____
(A) 44 (B) 51 (C) 79 (D) 108

Ans: (D)

Sol: Read books = $n(R)$
 $= 12 + 44 + 7 + 13$
 $= 76$

Play sports = $n(S)$
 $= 44 + 7 + 17 + 15$
 $= 83$

$n(R \cap S) = 44 + 7$
 $= 51$

$n(R \cup S) = n(R) + n(S) - n(R \cap S)$
 $= 76 + 83 - 51$
 $= 108$

- 7) Social science disciplines were in existence in an amorphous form until the colonial period when they were institutionalized. In varying degrees, they were intended to further the colonial interest. In the time of globalization and the economic rise of postcolonial countries like India, conventional ways of knowledge production have become obsolete.

Which of the following can be logically inferred from the above statements?

- (i) Social science disciplines have become obsolete.
- (ii) Social science disciplines had a pre-colonial origin
- (iii) Social science disciplines always promote colonialism
- (iv) Social science must maintain disciplinary boundaries

- (A) (ii) only (B) (i) and (iii) only
(C) (ii) and (iv) only (D) (iii) and (iv) only

Ans: (A)

Sol: Until the colonial period means pre-colonial origin. Other options can't be inferred.

GENERAL APTITUDE

- 8) Two and a quarter hours back, when seen in a mirror, the reflection of a wall clock without number markings seemed to show 1:30. What is the actual current time shown by the clock?

(A) 8:15 (B) 11:15 (C) 12:15 (D) 12:45

Ans: (D)

Sol: Time back = $2\frac{1}{4} = 2 \text{ hrs } 15 \text{ min}$

Clock time (C.T) + Mirror Time (M.T) = 12

↓
1.30

$$\therefore \text{C.T} = \begin{array}{r} 60 \\ 12.00 \\ 1.30 \\ \hline 10.30 \end{array}$$

$\therefore$ The actual time shown by the clock = $10.30 + 2.15 = 12.45$

- 9) M and N start from the same location. M travels 10 km East and then 10 km North-East. N travels 5 km South and then 4 km South-East. What is the shortest distance (in km) between M and N at the end of their travel?

(A) 18.60 (B) 22.50 (C) 20.61 (D) 25.00

Ans: (C)

Sol: From the given data, the following diagram is possible

$$\cos 45^\circ = \frac{DE}{4}$$

$$DE = \cos 45^\circ \times 4 = 2.828 \text{ km}$$

$$\sin 45^\circ = \frac{EN}{4}$$

$$EN = \sin 45^\circ \times 4 = 2.828 \text{ km}$$

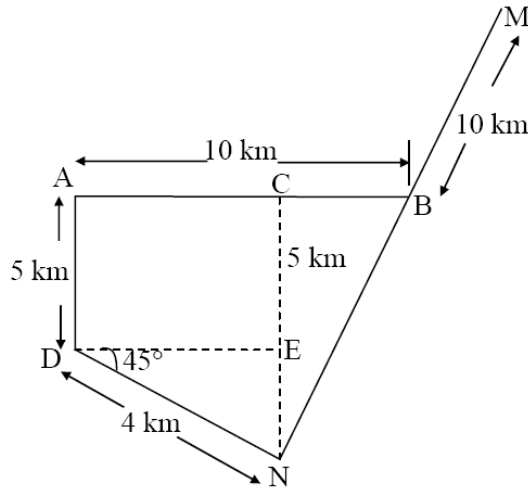
$$CN = NE + CE = 2.828 + 5 = 7.828 \text{ km}$$

$$CB = AB - AC = 10 - 2.828 = 7.171 \text{ km}$$

$$\begin{aligned} (NB)^2 &= (NC)^2 + (BC)^2 \\ &= (7.828)^2 + (7.171)^2 \end{aligned}$$

$$\therefore NB = \sqrt{(7.828)^2 + (7.171)^2} = 10.616 \text{ km}$$

$$\therefore NM = NB + BN = 10.616 + 10 = 20.61 \text{ km}$$



GENERAL APTITUDE

10. A wire of length 340 mm is to be cut into two parts. One of the parts is to be made into a square and the other into a rectangle where sides are in the ratio of 1:2. What is the length of the side of the square (in mm) such that the combined area of the square and the rectangle is a MINIMUM?

(A) 30

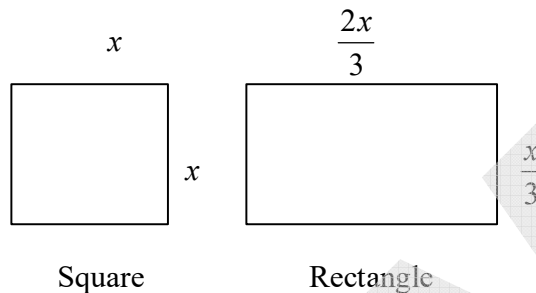
(B) 40

(C) 120

(D) 180

Ans: (B)

Sol: Length of the wire = 340 m



$$\text{Perimeter of rectangle} = 2 \left[\frac{x}{3} + \frac{2x}{3} \right]$$

$$= 2x$$

$$\text{Side of square} = 340 - 2x$$

$$\text{Side of square} = \frac{340 - 2x}{4}$$

Total area = Area of square + Area of rectangle

$$= \left[\frac{340 - 2x}{4} \right]^2 + \frac{x}{2} \times \frac{2x}{3} = \left[\frac{340 - 2x}{4} \right]^2 + \frac{2x^2}{9}$$

Combined area of square + rectangle = minimum

$$f'(x) = 0$$

$$f(x) = \left[\frac{340 - 2x}{4} \right]^2 + \frac{2}{9}x^2$$

$$f'(x) = \frac{4}{9}x - \frac{340 - 2x}{4} = 0$$

Another method:

Elimination procedure from alternatives option [C] and [D] are not possible because area may be maximum.

Option (A)

Side of the square = $x = 30$ mm

Perimeter of the square = $30 + 30 + 30 + 30 = 120$ mm

$\therefore$ Perimeter of the rectangle = $340 - 120 = 220$ mm

$$2x + 2 \times 2x = 220$$

GENERAL APTITUDE

$$x = 37$$

$$2x = 37 \times 2 = 74$$

$$\text{Area of square} = x^2 = (37)^2 = 900$$

$$\text{Area of rectangle} = x \times 2x = 37 \times 74 = 2738$$

$$\text{Total area} = 900 + 2738 = 3638 \text{ mm}^2$$

Option (B)

$$\text{Side of the square} = x = 40 \text{ mm}$$

$$\text{Perimeter of the square} = 340 - 160 = 180 \text{ mm}$$

$$2x + 2 \times 2x = 180 \text{ mm}$$

$$6x = 180 \text{ mm}$$

$$x = 30 \text{ mm}$$

$$\text{Area of the square} = 40 \times 40 = 1600 \text{ mm}^2$$

$$\text{Area of the rectangle} = 30 \times 2 \times 30 = 1800 \text{ mm}^2$$

$$\therefore \text{Total area} = 1600 + 1800 = 3400 \text{ mm}^2$$

$$\therefore 3400 \text{ mm}^2 < 3638 \text{ mm}^2$$

Option B is correct.

GENERAL APTITUDE

ECE PAPER - III

- 1) An apple costs Rs.10. An onion costs Rs.8.
Select the most suitable sentence with respect to grammar and usage.
(A) The price of an apple is greater than an onion
(B) The price of an apple is more than onion
(C) The price of an apple is greater than that of an onion
(D) Apples are more costlier than onions
Ans: (C)
Sol:Based on the given sentences option 'C' is the correct sentence which is in the comparative degree. Option 'A' and 'B' convey the wrong comparison and 'D' has double comparative and so they are wrong.
- 2) The Buddha said, "Holding on to anger is like grasping a hot coal with the intent of throwing it at someone else; you are the one who gets burnt."
(A) Burning (B) igniting (C) clutching (D) flinging
Ans: (C)
Sol:The underlined word 'grasping' means clutching or holding something tightly.
- 3) M has a son Q and a daughter R. He has no other children. E is the mother of P and daughter-in-law of M. How is P related to M?
(A) P is the son-in-law of M (B) P is the grandchild of M
(C) P is the daughter in law of M (D) P is the grandfather of M
Ans: (B)
Sol:Q and R are the son and daughter of M, E is the mother of P and daughter-in-law of M means Q and E are married couples in the family.
 $\therefore$ P is the grandchild of M.
- 4) The number that least fits this set: (324, 441, 97 and 64) is _____.
(A) 324 (B) 441 (C) 97 (D) 64
Ans: (C)
Sol:In the given set of numbers, all are perfect squares but 97 is not
 324 is square of $18 \Rightarrow (18)^2 = 324$
 441 is square of $21 \Rightarrow (21)^2 = 441$
 64 is square of $8 \Rightarrow (8)^2 = 64$
 97 is not the square of any number
 $\therefore$ The number that least fits in given set is 97.
- 5) It takes 10s and 15s, respectively, for two trains travelling at different constant speeds to completely pass a telegraph post. The length of the first train is 120 m and that of the second train is 150 m. The magnitude of the difference in the speeds of the two trains (in m/s) is _____.
(A) 2.0 (B) 10.0 (C) 12.0 (D) 22.0
Ans: (A)

GENERAL APTITUDE

Sol: Speed of the train (ST) = $\frac{\text{length of the train (LT)} + \text{Distance (D)}}{\text{Time (T)}}$

$$(ST) = \frac{LT + D}{T}$$

D = Distance (or) length of the platform = 0

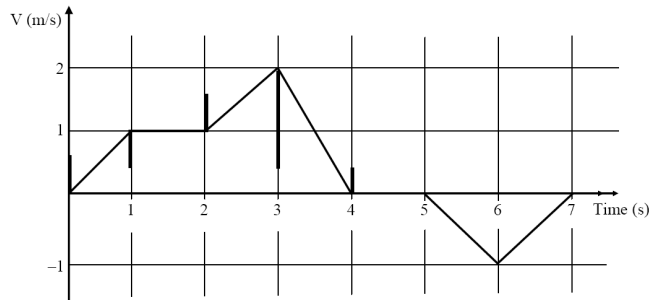
$$\therefore \text{Speed of the first train (ST}_1\text{)} = \frac{120}{10} = 12 \text{ m/s}$$

$$\text{Speed of the second train (ST}_2\text{)} = \frac{150}{15} = 10 \text{ m/s}$$

$\therefore$ The magnitude of the difference in the speeds of the two trains (m/s) = $12 - 10 = 2$

Q.6 – Q.10 carry Two marks each

- 6) The velocity V of a vehicle along a straight line is measured in m/s and plotted as shown with respect to time in seconds. At the end of the 7 seconds, how much will the odometer reading increase by (in m)?



- (A) 0 (B) 3 (C) 4 (D) 5

Ans: (D)

Sol: The odometer reading increases from starting point to end point

Area of the given diagram = Odometer reading

Area of the velocity and time graph per second

$$1^{\text{st}} \text{ sec} \Rightarrow \text{triangle} = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

$$2^{\text{nd}} \text{ sec} \Rightarrow \text{square} = 1 \times 1 = 1$$

$$3^{\text{rd}} \text{ sec} \Rightarrow \text{square} + \text{triangle} = 1 \times 1 + \frac{1}{2} \times 1 \times 1 = 1\frac{1}{2}$$

$$4^{\text{th}} \text{ sec} \Rightarrow \text{triangle} = \frac{1}{2} \times 1 \times 2 = 1$$

$$5^{\text{th}} \text{ sec} \Rightarrow \text{straight line} = 0$$

$$6^{\text{th}} \text{ sec} \Rightarrow \text{triangle} = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

$$7^{\text{th}} \text{ sec} \Rightarrow \text{triangle} = \frac{1}{2} \times 1 \times 1 = \frac{1}{2}$$

Total Odometer reading at 7 seconds

$$= \frac{1}{2} + 1 + 1\frac{1}{2} + 1 + 0 + \frac{1}{2} + \frac{1}{2} = 5$$

GENERAL APTITUDE

- 7) The overwhelming number of people infected with rabies in India has been flagged by the World Health Organization as a source of concern. It is estimated that inoculating 70% of pets and stray dogs against rabies can lead to a significant reduction in the number of people infected with rabies. Which of the following can be logically inferred from the above sentences?
- (A) The number of people in India infected with rabies is high.
(B) The number of people in other parts of the world who are infected with rabies is low.
(C) Rabies can be eradicated in India by vaccinating 70% of stray dogs.
(D) Stray dogs are the main source of rabies worldwide.

Ans: (A)

Sol: Only option 'A' can be logically inferred from the information provided in the argument.

- 8) A flat is shared by four first year undergraduate students. They agreed to allow the oldest of them to enjoy some extra space in the flat. Manu is two months older than Sravan, who is three months younger than Trideep. Pavan is one month older than Sravan. Who should occupy the extra space in the flat?
- (A) Manu (B) Sravan (C) Trideep (D) Pavan

Ans: (C)

Sol: Manu age = Sravan age + 2 months

Manu age = Trideep age - 3 months

Pavan age = Sravan's age + 1 month

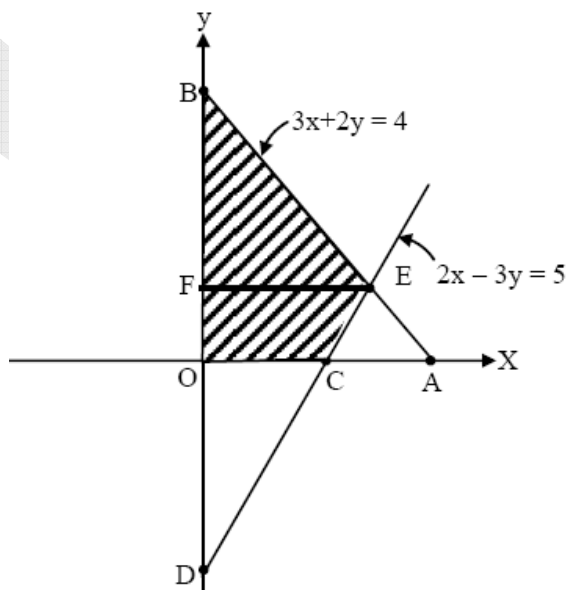
From this Trideep age > Manu > Pavan > Sravan

∴ Trideep can occupy the extra space in the flat.

- 9) Find the area bounded by the lines $3x+2y=14$, $2x-3y=5$ in the first quadrant.
- (A) 14.95 (B) 15.25 (C) 15.70 (D) 20.35

Ans: (B)

Sol:



GENERAL APTITUDE

$$A = \left[\frac{14}{3}, 0 \right] \quad B = [0, 7], \quad C = \left[\frac{5}{2}, 0 \right], \quad D = \left[0, \frac{-5}{3} \right], \quad E = [4, 1], \quad F = [0, 1]$$

Required area = Area of ΔOAB – Area of ΔCEA

$$= \left(\frac{1}{2} \times \frac{14}{3} \times 7 \right) - \left[\frac{1}{2} \times \left(\frac{14}{3} - \frac{5}{2} \right) \times 1 \right] = 15.25 \text{ sq. units}$$

Another method:

Required area = Area of ΔBFE + Area of $\Delta FEOC$

$$= \frac{1}{2} \times 4 \times 6 + \frac{1}{2} \times (4 + 2.5) \times 1 = 12 + 3.25 = 15.25 \text{ sq. units}$$

- 10) A straight line is fit to a data set (In x, y). This line intercepts the abscissa at In x = 0.1 and has a slope of -0.02. What is the value of y at x = 5 from the fit?

(A) -0.030

(B) -0.014

(C) 0.014

(D) 0.030

Ans: (A)

Sol: Straight line equation $y = mx + c$

$$m = \text{slope} = -0.02$$

set (log x, y)

If $\log x = X$, then set (x, y)

$$[x = 0.1 \quad y = 0]$$

$$y = mX + C$$

$$0 = -0.02 \times 0.1 + C$$

$$\therefore C = 0.002$$

$$y = mX + C$$

$$y = -0.02 \times \log x + C$$

$$@ x = 5$$

$$y = -0.02 \times \log 5 + 0.002$$

$$= -0.030$$

GENERAL APTITUDE

MECH PAPER – I

Q. 1 – Q. 5 carry one mark each.

- 1) Which of the following is **CORRECT** with respect to grammar and usage?

Mount Everest is _____.

- (A) the highest peak in the world
(B) highest peak in the world
(C) one of highest peak in the world
(D) one of the highest peak in the world

Ans: (A)

Sol: Before superlative article 'the' has to be used. "One of" the expression should take plural noun and so option 'C' and 'D' can't be the answer.

- 2) The policeman asked the victim of a theft, "What did you _____?"
(A) loose (B) lose (C) loss (D) louse

Ans: (B)

Sol: 'lose' is verb

- 3) Despite the new medicine's _____ in treating diabetes, it is not _____ widely.
(A) effectiveness --- prescribed (B) availability --- used
(C) prescription --- available (D) acceptance --- proscribed

Ans: (A)

Sol: 'effectiveness' is noun and 'prescribed' is verb. These words are apt and befitting with the word 'medicine'.

- 4) In a huge pile of apples and oranges, both ripe and unripe mixed together, 15% are unripe fruits. Of the unripe fruits, 45% are apples. Of the ripe ones, 66% are oranges. If the pile contains a total of 5692000 fruits, how many of them are apples?
(A) 2029198 (B) 2467482 (C) 2789080 (D) 3577422

Ans: (A)

Sol: Total no. of fruits = 5692000

Unripe type of apples = 45% of 15% of 5692000

$$= \frac{45}{100} \times \frac{15}{100} \times 5692000 = 384210$$

$$\text{Ripe type of apples} = \frac{34}{100} \times \frac{85}{100} \times 5692000 = 1644988$$

$$\therefore \text{Total no. of apples} = 384210 + 1644988 = 2029198$$

- 5) Michael lives 10 km away from where I live. Ahmed lives 5 km away and Susan lives 7 km away from where I live. Arun is farther away than Ahmed but closer than Susan from where I live. From the information provided here, what is one possible distance (in km) at which I live from Arun's place ?
(A) 3.00 (B) 4.99 (C) 6.02 (D) 7.01

Ans: (c)

Sol: From given data, the following diagram is possible

GENERAL APTITUDE

- 8) Leela is older than her cousin Pavithra. Pavithra's brother Shiva is older than Leela. When Pavithra and Shiva are visiting Leela, all three like to play chess. Pavithra wins more often than Leela does. Which one of the following statements must be **TRUE** based on the above?
- (A) When Shiva plays chess with Leela and Pavithra, he often loses.
 (B) Leela is the oldest of the three.
 (C) Shiva is a better chess player than Pavithra.
 (D) Pavithra is the youngest of the three.

Ans: (D)

Sol: From given data, the following arrangement is possible

Shiva

Leela

Pavithra

Among four alternatives, option D is TRUE.

- 9) If $q^{-a} = \frac{1}{r}$ and $r^{-b} = \frac{1}{s}$ and $S^{-c} = \frac{1}{q}$, the value of abc is _____
- (A) $(rqs)^{-1}$ (B) 0 (C) 1 (D) $r + q + s$

Ans: (C)

$$\text{Sol: } q^{-a} = \frac{1}{r} \Rightarrow \frac{1}{q^a} = \frac{1}{r} \Rightarrow q^a = r$$

$$r^{-b} = \frac{1}{s} \Rightarrow \frac{1}{r^b} = \frac{1}{s} \Rightarrow S = r^b$$

$$S^{-c} = \frac{1}{q} \Rightarrow \frac{1}{S^c} = \frac{1}{q} \Rightarrow S^c = q$$

$$q^a = r \Rightarrow (S^c)^a = r \Rightarrow S^{ac} = r$$

$$(S^{ac})^b = S \Rightarrow S^{abc} = S^1$$

$$\therefore abc = 1 \quad \therefore \text{Option 'C' is correct}$$

- 10) **P, Q, R** and **S** are working on a project. **Q** can finish the task in 25 days, working alone for 12 hours a day. **R** can finish the task in 50 days, working alone for 12 hours per day. **Q** worked 12 hours a day but took sick leave in the beginning for two days. **R** worked 18 hours a day on all days. What is the ratio of work done by **Q** and **R** after 7 days from the start of the project?

- (A) 10:11 (B) 11:10 (C) 20:21 (D) 21:20

Ans: (C)

Sol: Q can finish the task = 25 days, 12 hrs /day

$$= 300 \text{ hrs} \Rightarrow 1 \text{ hr} = \frac{1}{300} \text{ th}$$

R can finish the task = 50 days, 12 hrs/day = 50×12

$$= 600 \text{ hrs} \Rightarrow 1 \text{ hr} = \frac{1}{600} \text{ th}$$

Q Working hours $\Rightarrow (7 - 2) \times 12 = 60 \text{ hrs}$

R Working hours $\Rightarrow 7 \times 18 = 126 \text{ hrs}$

After 7 days, the ratio of Work done by Q and R

$$\begin{array}{ccc} Q & : & R \\ \frac{60}{300} & : & \frac{126}{600} \\ 20 & : & 21 \end{array}$$



GENERAL APTITUDE

MECH PAPER – II

Q. 1 – Q. 5 carry one mark each.

- 1) The volume of a sphere of diameter 1 unit is _____ than the volume of a cube of side 1 unit.
(A) least (B) less (C) lesser (D) low

Ans: (C)

Sol: 'Lesser than' is apt because the sentence should be in comparative degree.

- 2) The unruly crowd demanded that the accused be _____ without trial.
(A) hanged (B) hanging (C) hankering (D) hung

Ans: (A)

Sol: 'hanged' means death by hanging 'hung' is used only with things and not with people.

- 3) Choose the statement(s) where the underlined word is used correctly:
(i) A prone is a dried plum.
(ii) He was lying prone on the floor.
(iii) People who eat a lot of fat are prone to heart disease.
(A) (i) and (iii) only (B) (iii) only (C) (i) and (ii) only (D) (ii) and (iii) only

Ans: (D)

Sol: 'lying prone' means lie down flat. 'Prone to' means vulnerable to.

- 4) **Fact:** If it rains, then the field is wet.

Read the following statements:

- (i) It rains
(ii) The field is not wet
(iii) The field is wet
(iv) It did not rain

Which one of the options given below is **NOT** logically possible, based on the given fact?

- (A) If (iii), then (iv) (B) If (i), then (iii)
(C) If (i), then (ii) (D) If (ii), then (iv)

Ans: (C)

Sol: Statements i and ii are not logically possible based on the given fact.

- 5) A window is made up of a square portion and an equilateral triangle portion above it. The base of the triangular portion coincides with the upper side of the square. If the perimeter of the window is 6 m, the area of the window in m^2 is _____

- (A) 1.43 (B) 2.06 (C) 2.68 (D) 2.88

Ans: (B)

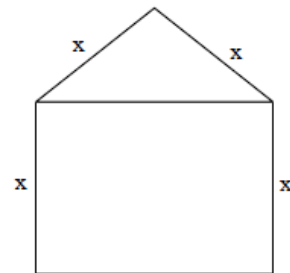
Sol: From the given data, the following window is possible

The perimeter of the window = $5x = 6$ m

$$x = \frac{6}{5} = 1.20 \text{ m}$$

Area of the window = Area of square + Area of Δ

$$\begin{aligned} &= 1.2 \times 1.2 + \frac{\sqrt{3}}{4} \times (1.2)^2 \\ &= 1.44 + 0.623 = 2.06 \text{ m}^2 \end{aligned}$$



GENERAL APTITUDE

Q. 6 – Q. 10 carry two marks each.

- 6) Students taking an exam are divided into two groups, **P** and **Q** such that each group has the same number of students. The performance of each of the students in a test was evaluated out of 200 marks. It was observed that the mean of group **P** was 105, while that of group **Q** was 85. The standard deviation of group **P** was 25, while that of group **Q** was 5. Assuming that the marks were distributed on a normal distribution, which of the following statements will have the highest probability of being **TRUE**?

- (A) No student in group **Q** scored less marks than any student in group **P**.
(B) No student in group **P** scored less marks than any student in group **Q**.
(C) Most students of group **Q** scored marks in a narrower range than students in group **P**.
(D) The median of the marks of group **P** is 100.

Ans: (c)

Sol: Group 'P'

Mean (μ) = 105

Standard deviation (σ_1) = 25

$\Pr(\mu - \sigma \leq x \leq \mu + \sigma) \approx 0.6827$

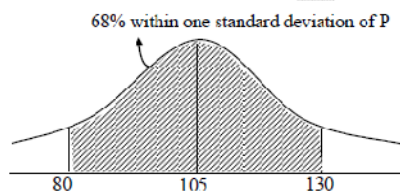
$\therefore$ 68% within one standard deviation

$\mu_1 - \sigma_1 = 105 - 25 = 80$

$\mu_1 + \sigma_1 = 105 + 25 = 130$

$\therefore$ range = 80 to 130

Distribution of P:



Group Q

Mean (μ_2) = 85

Standard deviation (σ_2) = 5

$\Pr(\mu - \sigma \leq x \leq \mu + \sigma) \approx 0.6827$

$\therefore$ 68% within one standard deviation

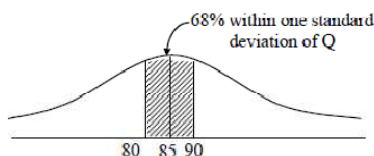
$\mu_2 - \sigma_2 = 85 - 5 = 80$

$\mu_2 + \sigma_2 = 85 + 5 = 90$

$\therefore$ range of Q in one standard deviation is 80 to 90

68% within one standard deviation of Q is narrower

Distribution of 'Q'



$\therefore$ 68% within one standard deviation of Q means most students of group Q.

$\therefore$ Most students of group 'Q' scored marks in a narrower range than students in group 'P'

GENERAL APTITUDE

- 7) A smart city integrates all modes of transport, uses clean energy and promotes sustainable use of resources. It also uses technology to ensure safety and security of the city, something which critics argue, will lead to a surveillance state.

Which of the following can be logically inferred from the above paragraph?

- (i) All smart cities encourage the formation of surveillance states.
 (ii) Surveillance is an integral part of a smart city.
 (iii) Sustainability and surveillance go hand in hand in a smart city.
 (iv) There is a perception that smart cities promote surveillance.
 (A) (i) and (iv) only (B) (ii) and (iii) only
 (C) (iv) only (D) (i) only

Ans: (B)

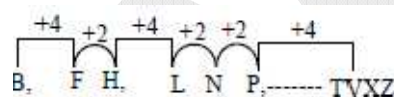
- 8) Find the missing sequence in the letter series.

B, FH, LNP, _ _ _ _ .

- (A) SUWY (B) TUVW (C) TVXZ (D) TWXZ

Ans: (C)

Sol: The following letter series is in the order of even letters series



- 9) The binary operation $\square$ is defined as $a \square b = a \times b + (a+b)$, where a and b are any two real numbers. The value of the identity element of this operation, defined as the number x such that $a \square x = a$, for any a , is .

- (A) 0 (B) 1 (C) 2 (D) 10

Ans: (A)

Sol: The binary operation $\square$ is defined

$$\Rightarrow a \square b = ab + (a+b)$$

$$a \square x = a$$

$\therefore$ From the equation 'b' is the variable

Option A: $x = 0$

$$a \square 0 = a \times 0 + (a + 0) = 0 + a = a$$

Option B: $x = 1$

$$a \square 1 = a \times 1 + (a + 1) = a + a + 1 = 2a + 1$$

Option C: $x = 2$

$$a \square 2 \Rightarrow a \times 2 + (a + 2) = 2a + a + 2 = 3a + 2$$

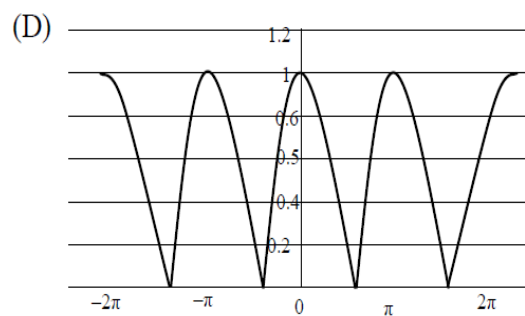
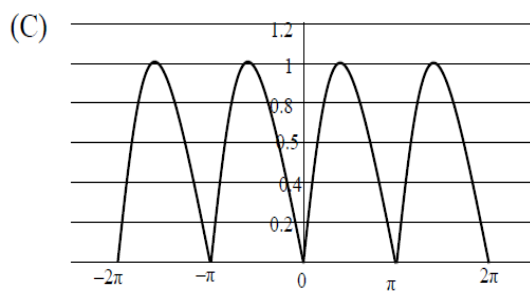
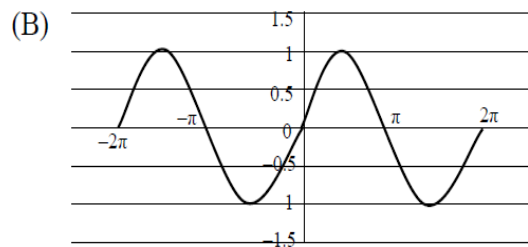
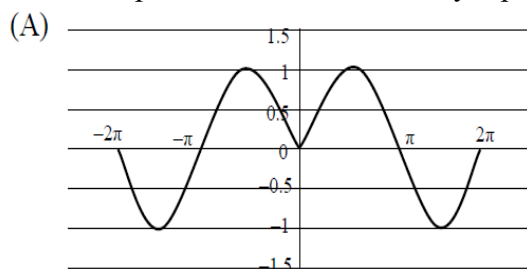
Option D: $x = 10$

$$a \square 10 \Rightarrow a \times 10 + (a + 10) = 10a + a + 10 = 11a + 10$$

$\therefore$ Option 'A' only True.

GENERAL APTITUDE

- 10) Which of the following curves represents the function $y = \ln(|e^{(\sin(|x|))}|)$ for $|x| < 2\pi$? Here, x represents the abscissa and y represents the ordinate.



Ans: (C)

GENERAL APTITUDE

MECH PAPER – III

Q. 1 – Q. 5 carry one mark each.

- 1) Based on the given statements, select the appropriate option with respect to grammar and usage.

Statements

- (i) The height of Mr. X is 6 feet.
- (ii) The height of Mr. Y is 5 feet.
- (A) Mr. X is longer than Mr. Y.
- (B) Mr. X is more elongated than Mr. Y.
- (C) Mr. X is taller than Mr. Y.
- (D) Mr. X is lengthier than Mr. Y.

Ans: (C)

Sol: In degrees of comparison Mr. X is taller than Mr. Y is apt.

Positive degree – tall

Comparative degree – taller

Superlative degree – tallest

- 2) The students _____ the teacher on teachers' day for twenty years of dedicated teaching.

- (A) facilitated (B) felicitated (C) fantasized (D) facillitated

Ans: (B)

Sol: Felicitate means honour.

- 3) After India's cricket world cup victory in 1985, Shrotria who was playing both tennis and cricket till then, decided to concentrate only on cricket. And the rest is history.

What does the underlined phrase mean in this context?

- (A) history will rest in peace (B) rest is recorded in history books
(C) rest is well known (D) rest is archaic

Ans: (C)

Sol: 'rest is history' is an idiomatic expression which means 'rest is well known.'

- 4) Given $(9 \text{ inches})^{1/2} = (0.25 \text{ yards})^{1/2}$
which one of the following statements is **TRUE**?

- (A) 3 inches = 0.5 yards (B) 9 inches = 1.5 yards
(C) 9 inches = 0.25 yards (D) 81 inches = 0.0625 yards

Ans: (C)

- 5) *S*, *M*, *E* and *F* are working in shifts in a team to finish a project. *M* works with twice the efficiency of others but for half as many days as *E* worked. *S* and *M* have 6 hour shifts in a day, whereas *E* and *F* have 12 hours shifts. What is the ratio of contribution of *M* to contribution of *E* in the project?

- (A) 1:1 (B) 1:2 (C) 1:4 (D) 2:1

Ans: (B)

Sol: $M \text{ efficiency} = 2 [\text{efficiency of } S, E, \text{ and } F]$

Contribution of M in the project

$$= x \text{ days} \times 6 \text{ hrs} \times 2$$

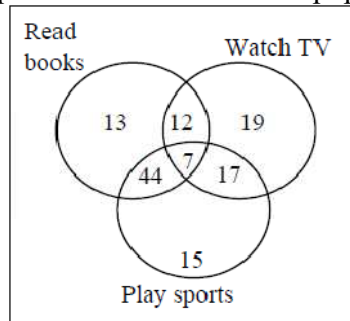
GENERAL APTITUDE

Contribution of E in the project
 $= 2x \text{ days} \times 12 \text{ hrs} \times 1$

Contribution of M : Contribution of E
 $x \times 6 \times 2 : 2x \times 12 \times 1$
 $1 : 2$

Q. 6 – Q.10 Carry two marks each

- 6) The Venn diagram shows the preference of the student population for leisure activities.



From the data given, the number of students who like to read books or play sports is

- (A) 44 (B) 51 (C) 79 (D) 108

Ans: (D)

Sol: Read books = $n(R) = 12 + 44 + 7 + 13 = 76$

Play sports = $n(S) = 44 + 7 + 17 + 15 = 83$

$n(R \cap S) = 44 + 7 = 51$

$n(R \cup S) = n(R) + n(S) - n(R \cap S) = 76 + 83 - 51 = 108$

- 7) Social science disciplines were in existence in an amorphous form until the colonial period when they were institutionalized. In varying degrees, they were intended to further the colonial interest. In the time of globalization and the economic rise postcolonial countries like India, conventional ways of knowledge production have become obsolete. Which of the following can be logically inferred from the above statements?

- (i) Social science disciplines have become obsolete
- (ii) Social science disciplines had a pre colonial origin
- (iii) Social science disciplines always promote colonialism.
- (iv) Social science must maintain disciplinary boundaries.

(A) (ii) only

(B) (i) and (iii) only

(C) (ii) and (iv) only

(D) (iii) and (iv) only

Ans: (A)

Sol: Until the colonial period means pre-colonial origin. Other options can't be inferred.

- 8) Two and a quarter hours back, when seen in a mirror, the reflection of a wall clock without number markings seemed to show 1:30. What is the actual current time shown by the clock?

(A) 8:15

(B) 11:15

(C) 12:15

(D) 12:45

Ans: (D)

GENERAL APTITUDE

- 10) A wire of length 340 mm is to be cut into two parts. One of the parts is to be made into a square and the other into a rectangle where sides are in the ratio of 1:2. What is the length of the side of the square (in mm) such that the combined area of the square and the rectangle is a **MINIMUM**?

(A) 30

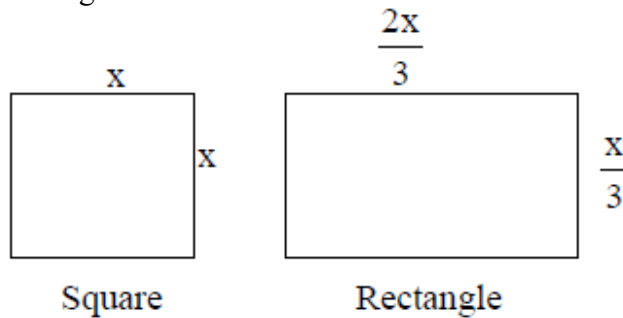
(B) 40

(C) 120

(D) 180

Ans: (B)

Sol: Length of the wire = 340 m



$$\text{Perimeter of rectangle} = 2 \left[\frac{x}{3} + \frac{2x}{3} \right] = 2x$$

$$\text{Perimeter of square} = 340 - 2x$$

$$\text{Side of square} = \frac{340 - 2x}{4}$$

$$\text{Total area} = \text{Area of square} + \text{Area of rectangle}$$

$$= \left[\frac{340 - 2x}{4} \right]^2 + \frac{x}{3} \times \frac{2x}{3}$$

$$= \left[\frac{340 - 2x}{4} \right]^2 + \frac{2x^2}{3}$$

$$\text{Combined area of square + rectangle} = \text{minimum}$$

$$f(x) = 0$$

$$f(x) = \left[\frac{340 - 2x}{4} \right]^2 + \frac{2}{9}x^2$$

$$f'(x) = \frac{4}{9}x - \frac{340 - 2x}{4} = 0$$

$$\frac{4}{9}x = \frac{1}{4}[340 - 2x] \Rightarrow x = 90$$

$$\text{Side of square} = \frac{340 - 2x}{4} = 40 \text{ mm}$$

GENERAL APTITUDE

CIVIL PAPER - I

Q. 1 – Q. 5 carry one mark each.

- 1) Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.
- (A) I will not leave the place until the minister does not meet me.
(B) I will not leave the place until the minister doesn't meet me.
(C) I will not leave the place until the minister meet me.
(D) I will not leave the place until the minister meets me.

Ans (D)

Soln: 'until' itself is negative so it can't take one more negative i.e., does not'. Hence, Option (D) is the right answer

- 2) A rewording of something written or spoken is a _____.
(A) paraphrase (B) paradox (C) paradigm (D) paraffin

Ans (A)

Soln: 'paraphrase' means a restatement of a text, passage or a rewording of something written or spoken.

- 3) Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world."

The sentence above is an example of a _____ statement.

- (A) figurative (B) collateral (C) literal (D) figurine

Ans (A)

Soln: 'Figurative' means representing by a figure or resemblance or expressing one thing in terms normally denoting another with which it may be regarded as analogous.

- 4) If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?

- (A) zentaga (B) tagafer (C) tagazen (D) relffer

Ans: (C)

Soln: From given codes

Relftage $\Rightarrow$ carefree

Otaga $\Rightarrow$ careful

Fertaga $\Rightarrow$ careless

From these codes, clearly know that "care" means "taga". From given alternatives, option 'C' is connect.

- 5) A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.

- (A) 56 (B) 64 (C) 72 (D) 96

Ans: (D)

Soln: From given data, 64 cubic blocks of one unit sizes are formed

No of faces of the

Cube is '6'

No of corners of the

Cube is '8'

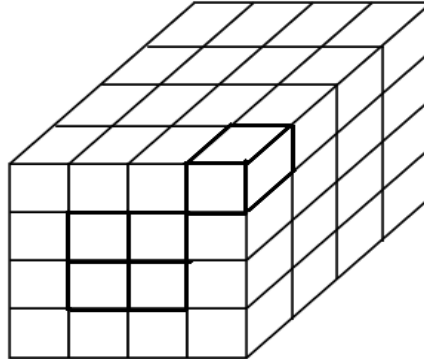
After removing one

Cubic block from



GENERAL APTITUDE

Each corner,
The resulting surface area of the body = $6 \times (4) = 96$ sq. Units.



Q. 6 – Q. 10 carry two marks each.

- 6) A shaving set company sells 4 different types of razors, Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter \ Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

- (A) Elegance (B) Executive (C) Smooth (D) Soft

Ans (B)

Elegance	Smooth	Soft	Executive
27300	20009	17602	9999
25222	19392	18445	8942
28976	22429	19544	10234
21012	18229	16595	10109
102510	80059	72186	39284

Rs.48

Rs.63

Rs.78

Rs.173

Total revenue

102510

$\times 48$

4920480

80059

$\times 63$

5043717

72186

$\times 78$

5630508

39284

$\times 173$

6796132

More revenue is on executive

- 7) Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is.

Which of the following can be logically inferred from the above sentences?

- (A) India is a country of exactly seventeen languages.
(B) Linguistic pluralism is the only indicator of a nation's diversity.
(C) Indian currency notes have sufficient space for all the Indian languages.
(D) Linguistic pluralism is strong evidence of India's diversity.

Ans: (D)

GENERAL APTITUDE

Soln: If seventeen languages were not an indication of the nation's diversity, nothing else is if nothing else is so the best inference is option 'D'

- 8) Consider the following statements relating to the level of poker play of four players P, Q, R and S.

- I. P always beats Q
- II. R always beats S
- III. S loses to P only sometimes
- IV. R always loses to Q

Which of the following can be logically inferred from the above statements?

- (i) P is likely to beat all the three other players
- (ii) S is the absolute worst player in the set

- (A) (i) only
- (C) (i) and (ii)

- (B) (ii) only
- (D) neither (i) nor (ii)

Ans: (A)

Soln: From the given data

Player P > Player Q

Player R > Player S

Players S < Player P (Sometimes only)

Player R < Player Q

∴ Player 'P' > Player 'Q' > Player 'R' > Players 'S'

∴ Logically, the statement (i) is definitely true, but statement (ii) is not

∴ Option 'A' is true

- 9) If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?

(A) $(x^3 + 8)$

(B) $(x - 1)$

(C) $(2x - 5)$

(D) $(x + 1)$

Ans: (B)

Soln: option (A)

$$X^3 + 8 = 0, X^3 = -8, X = -2$$

Sub $x = -2$, in $f(x) = 2x^7 + 3x - 5$

$$F(-2) = 2(-2)^7 + 3(-2) - 5$$

$$= -256 - 6 - 5 = -267$$

∴ This is not a factor of $f(x)$

Option (B)

$$X - 1 = 0, X = 1$$

Sub @ $x = 1$, in $f(x) = 2x^7 + 3x - 5$

$$F(1) = 2(1)^7 + 3(1) - 5 = 5 - 5 = 0$$

∴ this is a factor of $f(x)$

Option (C)

$$2x - f = 0, x = 5/2$$

Sub @ $x = 5/2$ in $f(x) = 2x^7 + 3x - 5$

$$f = \left(\frac{5}{2}\right) = 2 \left(\frac{5}{2}\right)^7 + 3 \left(\frac{5}{2}\right) - 5$$

$$= -2 - 3 - 5 = -10$$

∴ This is not a factor of $f(x)$

∴ Option 'B' is true

GENERAL APTITUDE

- 10) In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is ____.

(A) 40.00 (B) 46.02 (C) 60.01 (D) 92.02

Ans (B)

Soln: Load failure for cycles

80 100

40 10000

There is one failure 5000 cycles load must be between 80 and 40

∴ 46.02

GENERAL APTITUDE

CIVIL PAPER - II

Q. 1 – Q. 5 carry one mark each.

- 1) If I were you, I _____ that laptop. It's much too expensive.
(A) won't buy (B) shan't buy (C) wouldn't buy (D) would buy

Ans: (C)

Soln: In if clause (type2)'were' is in the past tense so the so main clause should be in the conditional clause. Therefore 'C' is the best answer.

- 2) He turned a deaf ear to my request. What does the underlined phrasal verb mean?
(A) ignored (B) appreciated (C) twisted (D) returned

Ans: (A)

Soln: 'turned a deaf ear' means ignored.

- 3) Choose the most appropriate set of words from the options given below to complete the following sentence. _____ is a will, _____ is a way.

(A) Wear, there, their (B) Were, their, there
(C) Where, there, there (D) Where, their, their

Ans: (C)

Soln: where there is a will there is a way. It is a quotation.

- 4) $(x\% \text{ of } y) + (y\% \text{ of } x)$ is equivalent to _____
(A) 2 % of xy (B) 2 % of $(xy/100)$ (C) $xy\%$ of 100 (D) 100 % of xy

Ans: (A)

Soln: $(x\% \text{ of } y) + (y\% \text{ of } x) = \frac{x}{100} \times y + \frac{y}{100} \times x$

$$= \frac{xy + xy}{100}$$

$$= \frac{2xy}{100}$$

$$= 2\% \text{ of } xy$$

- 5) The sum of the digits of a two digit number is 12. If the new number formed by reversing the digits is greater than the original number by 54, find the original number.

(A) 39 (B) 57 (C) 66 (D) 93

Ans: (A)

Soln: The new number formed by reversing the digits is greater than the original number is possible in options A and B only.

Option 'a':

Sum of two digits in the number = $3+9=12$

After reversing the two digits number = 93

The difference between the new number formed and original number = $93 - 39 = 54$

$\therefore$ option 'A' is correct.

Option 'b':

Original number = 57

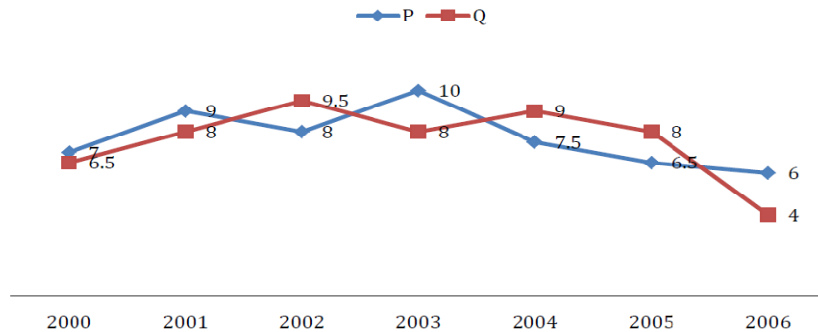
After reversing, the number is formed = 75

The difference between these two numbers = $75 - 57 = 18$

$\therefore$ it is not.

GENERAL APTITUDE

- 6) Two finance companies, P and Q, declared fixed annual rates of interest on the amounts invested with them. The rates of interest offered by these companies may differ from year to year. Year-wise annual rates of interest offered by these companies are shown by the line graph provided below.



If the amounts invested in the companies, P and Q, in 2006 are in the ratio 8:9, then the amounts received after one year as interests from companies P and Q would be in the ratio:

- (A) 2:3 (B) 3:4 (C) 6:7 (D) 4:3

Ans: (D)

Soln: The amounts invested in the companies of P and Q in 2006 = 8:9

The rate of interest of company 'P' in 2006 = 6%

The rate of interest of company 'Q' in 2006 = 4%

The amounts received after one year by P and Q companies in 2006 year

P	:	Q
6% of 8	:	4% of 9
$\frac{6}{100} \times 8$	:	$\frac{4}{100} \times 9$
4	:	3

- 7) Today, we consider Ashoka as a great ruler because of the copious evidence he left behind in the form of stone carved edicts. Historians tend to correlate greatness of a king at his time with the availability of evidence today. Which of the following can be logically inferred from the above sentences?
- (A) Emperors who do not leave significant sculpted evidence are completely forgotten.
 (B) Ashoka produced stone carved edicts to ensure that later historians will respect him.
 (C) Statues of kings are a reminder of their greatness.
 (D) A king's greatness, as we know him today, is interpreted by historians.

Ans: (D)

Soln: Today, historians correlate greatness of a king at his time with the availability of evidence. This statement leads to the best inference option 'd'

- 8) Fact 1: Humans are mammals.
 Fact 2: Some humans are engineers.
 Fact 3: Engineers build houses.
 If the above statements are facts, which of the following can be logically inferred?
- I. All mammals build houses.
 II. Engineers are mammals.
 III. Some humans are not engineers.
- (A) II only. (B) III only. (C) I, II and III. (D) I only.

GENERAL APTITUDE

Ans: (B)

Soln: From given facts, the following venn diagram is possible

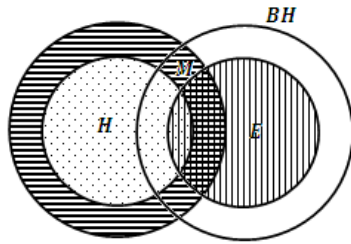
H= Humans

M= mammals

E = Engineers

BH = Build houses

∴ From above diagram, statement III is true



- 9) A square pyramid has a base perimeter x , and the slant height is half of the perimeter. What is the lateral surface area of the pyramid?

(A) x^2

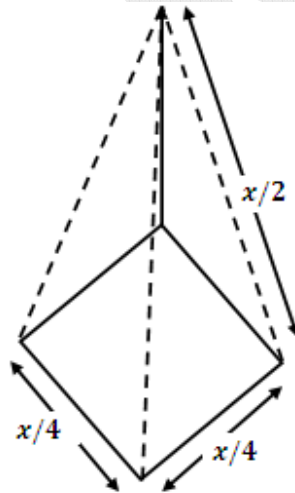
(B) $0.75 x^2$

(C) $0.50 x^2$

(D) $0.25 x^2$

Ans:(D)

Soln:



Base perimeter of square pyramid = $x = p$

Slant height = $\frac{x}{2} = l$

Lateral surface area of pyramid

$= \frac{1}{2} \times \text{Base perimeter (P)} \times \text{slant height (l)}$

$$\frac{x}{2} = \frac{1}{2} \times p \times l$$

$$= \frac{1}{2} \times x \times \frac{x}{2}$$

$$= \frac{x^2}{4}$$

$$= 0.25 x^2$$

GENERAL APTITUDE

- 10) Ananth takes 6 hours and Bharath takes 4 hours to read a book. Both started reading copies of the book at the same time. After how many hours is the number of pages **to be** read by Ananth, twice that **to be** read by Bharath? Assume Ananth and Bharath read all the pages with constant pace.

(A) 1 (B) 2 (C) 3 (D) 4

Ans: (C)

Soln: Ananth takes 6 hours to read a book

Bharath takes 4 hours to read a book

L.C.M = 12

The number of pages ready by Ananth and Bharath must be 12 (or) multiple of 12 only

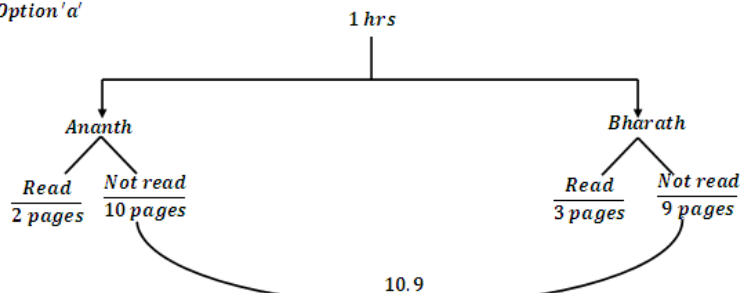
If Ananth read 12 number of pages in 6 hrs

$$\therefore 1 \text{ hr} = \frac{12}{6} = 2 \text{ pages}$$

If Bharath read 12 number of pages in 4 hrs

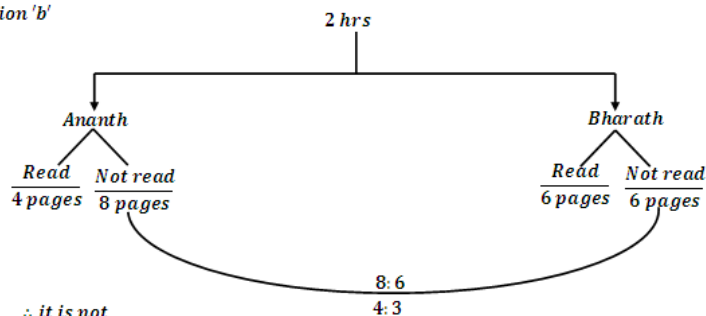
$$1 \text{ hr} = \frac{12}{4} = 3 \text{ pages}$$

Option 'a'



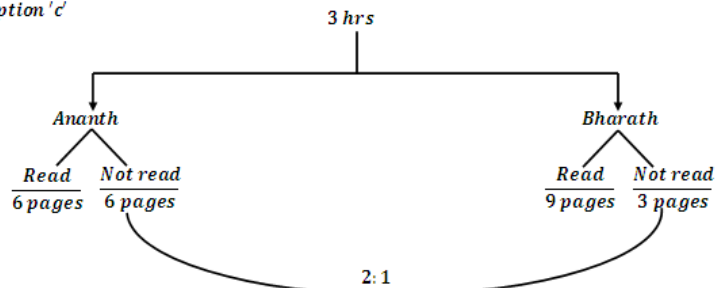
$\therefore$ it is not

Option 'b'



$\therefore$ it is not

Option 'c'



$\therefore$ After 3 hours is the number of pages to be read by Ananth, twice that to be ready by Bharath.

GENERAL APTITUDE

CSE PAPER – I

Q. 1 – Q. 5 carry one mark each.

- 1) Out of the following four sentences, select the most suitable sentence with respect to grammar and usage.

(A) I will not leave the place until the minister does not meet me.
(B) I will not leave the place until the minister doesn't meet me.
(C) I will not leave the place until the minister meet me.
(D) I will not leave the place until the minister meets me.

Ans (D)

Soln: 'until' itself is negative so it can't take one more negative i.e., does not'. Hence, Option (D) is the right answer.

- 2) A rewording of something written or spoken is a _____.
(A) paraphrase (B) paradox (C) paradigm (D) paraffin

Ans (A)

Soln: 'paraphrase' means a restatement of a text, passage or a rewording of something written or spoken.

- 3) Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I will move the world." The sentence above is an example of a _____ statement.
(A) figurative (B) collateral (C) literal (D) figurine

Ans (A)

Soln: 'Figurative' means representing by a figure or resemblance or expressing one thing in terms normally denoting another with which it may be regarded as analogous.

- 4) If 'relftaga' means carefree, 'otaga' means careful and 'fertaga' means careless, which of the following could mean 'aftercare'?
(A) zentaga (B) tagafer (C) tagazen (D) relffer

Ans: (C)

Soln: From given codes

Relftaga $\Rightarrow$ carefree

Otaga $\Rightarrow$ careful

Fertaga $\Rightarrow$ careless

From these codes, clearly know that "care" means "taga". From given alternatives, option 'C' is connect.

- 5) A cube is built using 64 cubic blocks of side one unit. After it is built, one cubic block is removed from every corner of the cube. The resulting surface area of the body (in square units) after the removal is _____.
(A) 56 (B) 64 (C) 72 (D) 96

Ans: (D)

Soln: form given data, 64 cubic blocks of one unit sizes are formed

No of faces of the

Cube is '6'

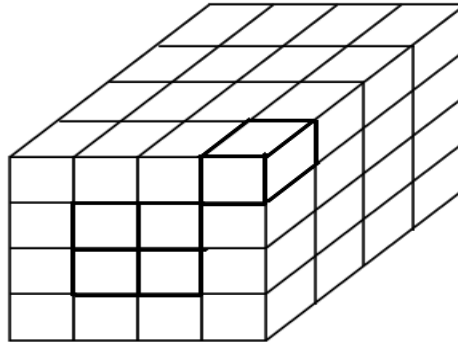
No of corners of the

Cub is '8'

GENERAL APTITUDE

After removing one
Cubic block from
Each corner,

The resulting surface area of the body = $6 \times (4) = 96$ sq. Units.



Q. 6 – Q. 10 carry two marks each.

- Q.6. A shaving set company sells 4 different types of razors, Elegance, Smooth, Soft and Executive. Elegance sells at Rs. 48, Smooth at Rs. 63, Soft at Rs. 78 and Executive at Rs. 173 per piece. The table below shows the numbers of each razor sold in each quarter of a year.

Quarter/Product	Elegance	Smooth	Soft	Executive
Q1	27300	20009	17602	9999
Q2	25222	19392	18445	8942
Q3	28976	22429	19544	10234
Q4	21012	18229	16595	10109

Which product contributes the greatest fraction to the revenue of the company in that year?

- (A) Elegance (B) Executive (C) Smooth (D) Soft

Ans: (B)

Soln: Total No. of razors Elegance type from all four quarters

$$= 27300 + 25222 + 28976 + 21012$$

$$= 102510$$

Total No. of razors of smooth type from all four quarters

$$= 20009 + 19392 + 22429 + 18229$$

$$= 80059$$

Total No. of razors of soft type from all four Quarts

$$= 17602 + 18445 + 19544 + 16595$$

$$= 72186$$

Total. No. of razors of Executive type from all four Quarters

$$= 9999 + 8942 + 10234 + 10109$$

$$= 39286$$

The revenue of the company in that year of 4 different types of razors

$$\text{Elegance} = 102510 \times 48 = 49,20,480$$

$$\text{Smooth} = 80059 \times 63 = 50,43,717$$

$$\text{Soft} = 72186 \times 78 = 56,30,508$$

$$\text{Executive} = 39284 \times 173 = 67,96,132$$

∴ The Executive of razors contributes the greatest revenue of the company in that year.

GENERAL APTITUDE

- 7) Indian currency notes show the denomination indicated in at least seventeen languages. If this is not an indication of the nation's diversity, nothing else is. Which of the following can be logically inferred from the above sentences?
- (A) India is a country of exactly seventeen languages.
(B) Linguistic pluralism is the only indicator of a nation's diversity.
(C) Indian currency notes have sufficient space for all the Indian languages.
(D) Linguistic pluralism is strong evidence of India's diversity.

Ans: (D)

Soln: If seventeen languages were not an indication of the nation's diversity, nothing else is. If nothing else is so the best inference is option 'D'

- 8) Consider the following statements relating to the level of poker play of four players P, Q, R and S.

- I. P always beats Q
II. R always beats S
III. S loses to P only sometimes
IV. R always loses to Q

Which of the following can be logically inferred from the above statements?

- (i) P is likely to beat all the three other players
(ii) S is the absolute worst player in the set

- (A) (i) only (B) (ii) only (C) (i) and (ii) (D) neither (i) nor (ii)

Ans: (A)

Soln: From the given data

Player P > player Q

Player R > player S

Player S < Player P (sometimes only)

Player R < Player Q

∴ Player 'P' > Player 'Q' > Player 'R' > Player 'S'

∴ Logically, the statement (i) is definitely true, but statement (ii) is not

∴ Option 'A' is true

- 9) If $f(x) = 2x^7 + 3x - 5$, which of the following is a factor of $f(x)$?
- (A) $(x^3 + 8)$ (B) $(x - 1)$ (C) $(2x - 5)$ (D) $(x + 1)$

Ans: (B)

Soln: $F(X) = 2x^7 + 3x - 5$ for $X = 1$ the equation is satisfied. The factor is $(X - 1)$

- 10) In a process, the number of cycles to failure decreases exponentially with an increase in load. At a load of 80 units, it takes 100 cycles for failure. When the load is halved, it takes 10000 cycles for failure. The load for which the failure will happen in 5000 cycles is ____.
- (A) 40.00 (B) 46.02 (C) 60.01 (D) 92.02

Ans: (B)

Soln: ↓ load cycles for failure ↑ exponentially

Eg : load = x

Cycles for failure = y

$$\frac{x}{2} = y^2 \quad \frac{x}{3} = y^3 \quad \frac{x}{4} = y^4$$

GENERAL APTITUDE

By elimination procedure from options.

Option (A)

Load = 40 units, it says that, the load is halved, it takes 10,000 cycles for failure

$$\frac{80}{2} = (100)^2$$

40 units = 10,000 cycles, it is not

Option (D)

Load = 92.02 units means it is more than 80 units, so it is not.

Option (C)

Load = 60.01 units

$$\frac{3}{4}(80 \text{ units}) = 60.01 \text{ units}$$

Form the given relation

$$\frac{3}{4}(80 \text{ units}) = (10)^{4/3} = (100)^1 \times (100)^{1/3} = 100 \times 4.64 = 464$$

It is not

∴ Option (B) only possible

∴ At the load of 46.02 units, the failure will happen in 5000 cycles.

GENERAL APTITUDE

CSE PAPER – II

Q. 1 – Q. 5 carry one mark each.

- 1) The man who is now Municipal Commissioner worked as _____.
- (A) the security guard at a university (B) a security guard at the university
(C) a security guard at university (D) the security guard at the university

Ans: (B)

Soln: Option (A) and (D) cannot be the answer because of the 'the security'

- 2) Nobody knows how the Indian cricket team is going to cope with the difficult and seamer-friendly wickets in Australia. Choose the option which is closest in meaning to the underlined phrase in the above sentence.
- (A) put up with (B) put in with (C) put down to (D) put up against

Ans: (A)

Soln: 'cope with' means put up with.

- 3) Find the odd one in the following group of words. mock, deride, praise, jeer
- (A) mock (B) deride (C) praise (D) jeer

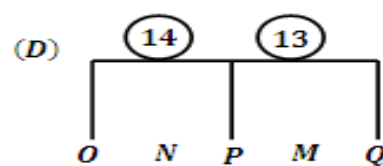
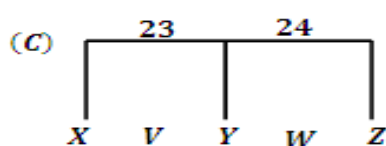
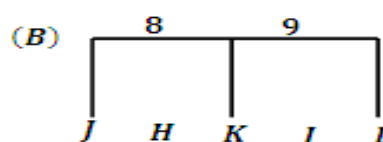
Ans: (C)

Sol: 'mock, deride and Jeer' are synonyms which means mockery. Therefore the odd is 'praise'

- 4) Pick the odd one from the following options.
- (A) CADBE (B) JHKIL (C) XZYWZ (D) ONPMQ

Ans: (D)

Soln:



In option A, B and C, the letters skipped between consecutive pair of letters in the English alphabet is in increasing order (i.e) A and B, H and I and V and W but in option 'D' N and M are present instead of M and N so, option 'D' is odd one from the group.

- 5) In a quadratic function, the value of the product of the roots (α, β) is 4. Find the value of
- $$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}}$$
- (A) n^4 (B) 4^n (C) 2^{2n-1} (D) 4^{n-1}

Ans: (B)

GENERAL APTITUDE

Soln:

$$\frac{\alpha^n + \beta^n}{\alpha^{-n} + \beta^{-n}} = \frac{\alpha^n + \beta^n}{\frac{1}{\alpha^n} + \frac{1}{\beta^n}}$$

$$\frac{\alpha^n + \beta^n}{\frac{\beta^n + \alpha^n}{\alpha^n \times \beta^n}} = \alpha^n \times \beta^n$$

$$\alpha^n \times \beta^n = (\alpha \times \beta)^n = (4)^n$$

Q. 6 – Q. 10 carry two marks each.

- 6) Among 150 faculty members in an institute, 55 are connected with each other through Facebook® and 85 are connected through WhatsApp®. 30 faculty members do not have Facebook® or WhatsApp® accounts. The number of faculty members connected only through Facebook® accounts is _____.
- (A) 35 (B) 45 (C) 65 (D) 90

Ans: (A)

Soln: Total faculty members = 150

The faculty members having facebook account = FB = 55

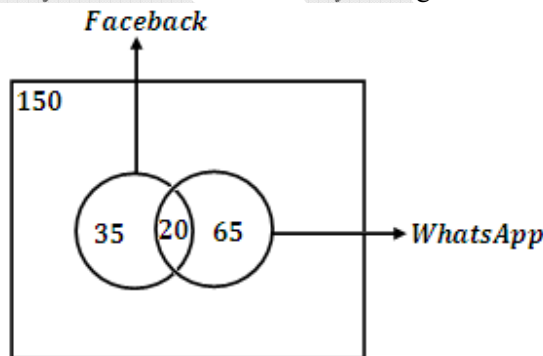
The faculty members having whatsapp = W = 85

The faculty members do not have face book (or) WhatsApp accounts = 30

The faculty members having any account = 150 – 30 = 120

The faculty members having both the accounts = (FB + W) – 120 = (55 + 85) – 120 = 20

∴ The number of faculty members connected only through Facebook accounts = 55 – 20 = 35



- 7) Computers were invented for performing only high-end useful computations. However, it is no understatement that they have taken over our world today. The internet, for example, is ubiquitous. Many believe that the internet itself is an unintended consequence of the original invention. With the advent of mobile computing on our phones, a whole new dimension is now enabled. One is left wondering if all these developments are good or, more importantly, required. Which of the statement(s) below is/are logically valid and can be inferred from the above paragraph?

(i) The author believes that computers are not good for us.

(ii) Mobile computers and the internet are both intended inventions

(A) (i) only (B) (ii) only (C) both (i) and (ii) (D) neither (i) nor (ii)

Ans: (B)

GENERAL APTITUDE

Soln: The first and second sentences tell us that computers are invented for computation and internet for intended invention. These sentences lead to option ii so option (B) is the right inference

- 8) All hill-stations have a lake.

Ooty has two lakes. Which of the statement(s) below is/are logically valid and can be inferred from the above sentences?

(i) Ooty is not a hill-station.

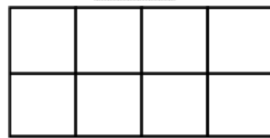
(ii) No hill-station can have more than one lake.

(A) (i) only (B) (ii) only (C) both (i) and (ii) (D) neither (i) nor (ii)

Ans: (D)

Soln: Statement (i) is not true because Ooty is a hill station due Ooty has two lakes statement (ii) is also not true, because in given statements., for hill station one lake is compulsory but not mentioned about number of lakes.

- 9) In a 2×4 rectangle grid shown below, each cell is a rectangle. How many rectangles can be observed in the grid?



(A) 21

(B) 27

(C) 30

(D) 36

Ans: (C)

Soln: In given 2×4 rectangle grid, the following type of rectangles are present

One figured rectangles = 8

Two figured rectangles = 10

Three figured rectangles = 4

Four figured rectangles = 5

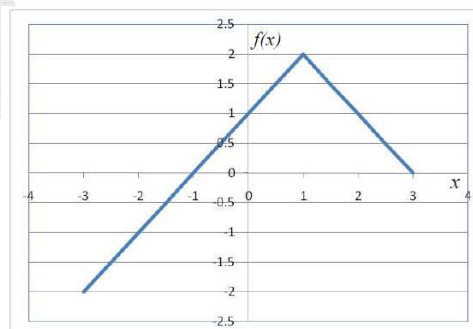
Six figured rectangles = 2

Eight figured rectangles = 1

Total No. of rectangles = 30

$\therefore$ The No. of rectangles observed in the given grid = 30

- 10)



Choose the correct expression for $f(x)$ given in the graph.

(A) $f(x) = 1 - |x - 1|$

(B) $f(x) = 1 + |x - 1|$

(C) $f(x) = 2 - |x - 1|$

(D) $f(x) = 2 + |x - 1|$

GENERAL APTITUDE

Ans : (C)

Soln: @ $x = 3$ from given graph $f(x)$ must be equals to zero

Option A

$$f(x) = 1 - |x - 1|$$

$$@ x = 3$$

$$f(x) = 1 - |3 - 1|$$
$$= 1 - 2 = -1$$

So, it is not

Option B

$$f(x) = 1 - |x - 1|$$

$$@ x = 3$$

$$f(x) = 1 + |3 - 1|$$
$$= 1 + 2 = 3$$

So, it is not

Option C

$$f(x) = 2 - |x - 1|$$

$$@ x = 3$$

$$f(x) = 2 - |3 - 1|$$
$$= 2 - 2 = 0$$

So, it is true

Option D

$$f(x) = 2 + |x - 1|$$

$$@ x = 3$$

$$f(x) = 2 + |3 - 1|$$
$$= 2 + 2 = 4$$

So, it is not

GENERAL APTITUDE

CHEMICAL PAPER – I

Q. 1 – Q. 5 carry one mark each.

- 1) The volume of a sphere of diameter 1 unit is ____ than the volume of a cube of side 1 unit.
(A) least (B) less (C) lesser (D) low

Ans: (B)

- 2) The unruly crowd demanded that the accused be _____ without trial.
(A) hanged (B) hanging (C) hankering (D) hung

Ans: (A)

- 3) Choose (The statement(s) where the underlined word is used correctly:
(i) A prone is a dried plum. (ii) He was lying prone on the floor.
(iii) People who eat a lot of fat are prone to heart disease.
(A) (i) and (iii) only (B) (iii) only (C) (i) and (ii) only (D) (ii) and (iii) only

Ans: (D)

- 4) **Fact:** If it rains, then the field is wet.
Read the following statements:
(i) It rains (ii) The field is not wet
(iii) The field is wet (iv) It did not rain
Which one of the options given below is **NOT** logically possible, based on the given fact?
(A) If (iii), then (iv). (B) If (i), then (iii).
(C) If (i), then (ii). (D) If (ii), then (iv).

Ans: (C)

- 5) A window is made up of a square portion and an equilateral triangle portion above it. The base of the triangular portion coincides with the upper side of the square. If the perimeter of the window is 6 in, the area of the window in m^2 is _____.
(A) 1.43 (B) 2.06 (C) 2.68 (D) 2.88

Ans: (B)

Q. 6 – Q. 10 carry two marks each.

- 6) Students taking an exam are divided into two groups, **P** and **Q** such that each group has the same number of students. The performance of each of the students in a test was evaluated out of 200 marks. It was observed that the mean of group **P** was 105, while that of group **Q** was 85. The standard deviation of group **P** was 25, while that of group **Q** was 5. Assuming that the marks were distributed on a normal distribution, which of the following statements will have the highest probability of being **TRUE**?
(A) No student in group **Q** scored less marks than any student in group **P**.
(B) No student in group **P** scored less marks than any student in group **Q**.
(C) Most students of group **Q** scored marks in a narrower range than students in group **P**.
(D) The median of the marks of group **P** is 100.

Ans: (C)

GENERAL APTITUDE

- 7) A smart city integrates all modes of transport, uses clean energy and promotes sustainable use of resources. It also uses technology to ensure safety and security of the city, something which critics argue, will lead to a surveillance state.

Which of the following can be logically inferred from the above paragraph?

- (i) All smart cities encourage the formation of surveillance states.
- (ii) Surveillance is an integral part of a smart city.
- (iii) Sustainability and surveillance go hand in hand in a smart city.
- (iv) There is a perception that smart cities promote surveillance.

(A) (i) and (iv) only

(B) (ii) and (iii) only

(C) (iv) only

(D) (i) only

Ans: (C)

- 8) Find the missing sequence in the letter series.

B, FH, LNP, _ _ _ _

(A) SUWY

(B) TUVW

(C) TVXZ

(D) TWXZ

Ans: (C)

- 9) The binary operation is defined as $a \square b = ab + (a + b)$, where a and b are any two real numbers. The value of the identity element of this operation, defined as the number x such that $a \square x$, for any a , is _____

(A) 0

(B) 1

(C) 2

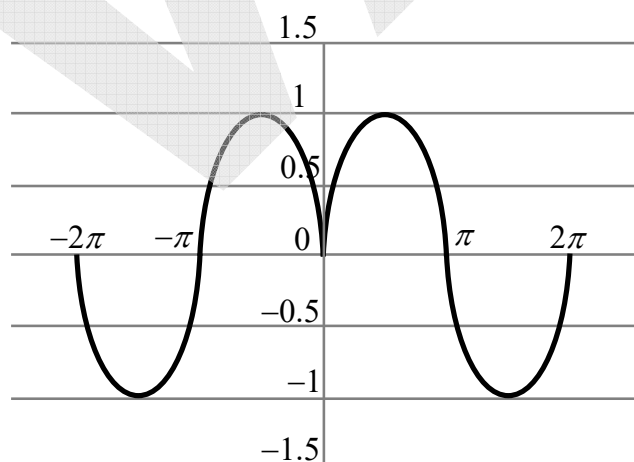
(D) 10

Ans: (A)

- 10) Which of the following curves represents the function $y = \ln \left(\left| e^{\left[\sin(|x|) \right]} \right| \right)$ for $|x| < 2\pi$?

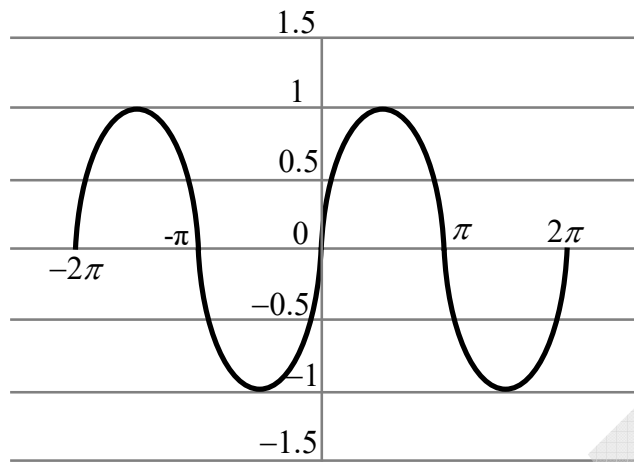
Here, x represents the abscissa and y represents the ordinate.

(A)

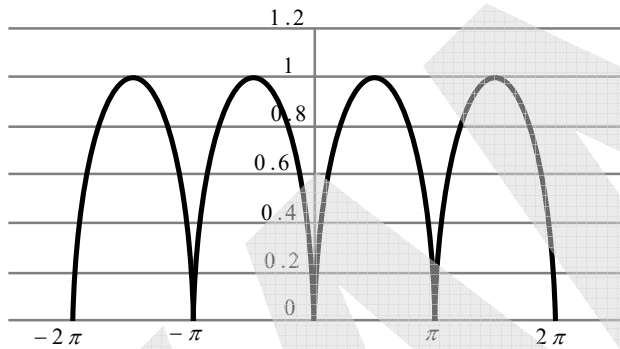


GENERAL APTITUDE

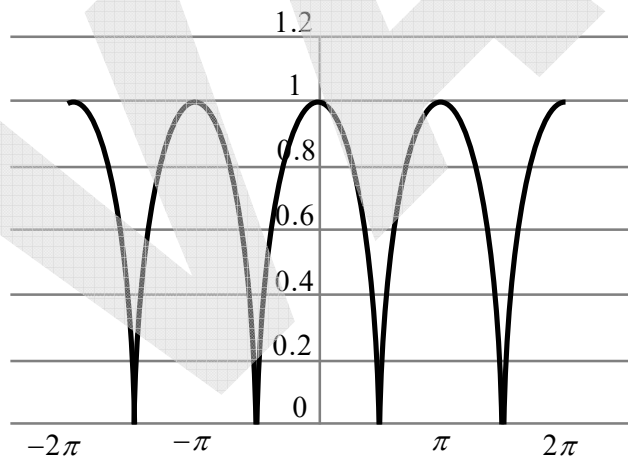
(B)



(C)



(D)



Ans: (C)

GENERAL APTITUDE

CIVIL-PAPER-1

1. Select the pair that best expresses a relationship similar to that expressed in the pair:
Children: Pediatrician (GATE-CE-15)
(A) Adult: Orthopaedist (B) Females: Gynaecologist
(C) Kidney: Nephrologist (D) Skin: Dermatologist

Answer: (B)

Exp: Community of people: Doctor

2. Extreme focus on syllabus and studying for test has become such a dominant concern of Indian students that this has closed their minds to anything _____ to the requirements of the exam (GATE-CE-15)
(A) related (B) extraneous (C) outside (D) useful

Answer: (B)

Exp: extraneous -irrelevant or unrelated to the subject being dealt with.

3. If ROAD is written as URDG, then SWAN should be written as: (GATE-CE-15)
(A) VXDQ (B) VZDQ (C) VZDP (D) UXDQ

Answer: (B)

Exp: R+3=U, O+3=R, A+3=D, D+3=G;
S+3=V, W+3=Z, A+3=D, N+3=Q

4. The Tamil version of _____ John Abraham-starrer Madras Café _____ cleared by the censor board with no cuts last week, but the film's distributors _____ no takers among the exhibitors for a release in Tamil Nadu _____ this Friday. (GATE-CE-15)
(A) Mr., was, found, on (B) a, was found, at
(C) the, was, found, on (D) a, being, find, at

Answer: (C)

Exp: John-Abraham starrer Madras Café talks about the movie not the person, so Mr. is ruled out.

'Find no takers' is not the correct phrase. At this Friday is incorrect. So, option C is correct.

5. A function $f(x)$ is linear and has a value of 29 at $x = -2$ and 39 at $x = 3$. Find its value at $x = 5$. (GATE-CE-15)
(A) 59 (B) 45 (C) 43 (D) 35

Answer: (C)

Exp: $f(x) = 2x + 33$

GENERAL APTITUDE

Q.No-6-10 Carry Two Marks Each

6. The head of a newly formed government desires to appoint five of the six selected members P,Q,R,S,T and U to portfolios of Home, Power, Defense, Telecom and Finance. U does not want any portfolio if S gets one of the five. R wants either Home or Finance or no portfolio. Q says that if S gets either Power or Telecom, then she must get the other one. T insists on a portfolio if P gets one.

Which is the valid distribution of portfolio?

(GATE-CE-15)

- (A) P-Home, Q-Power, R-Defense, S-Telecom, T-Finance
(B) R-Home, S-Power, P-Defense, Q-Telecom, T-Finance
(C) P-Home, Q-Power, T-Defense, S-Telecom, U-Finance
(D) Q-Home, U-Power, T-Defense, R-Telecom, P-Finance

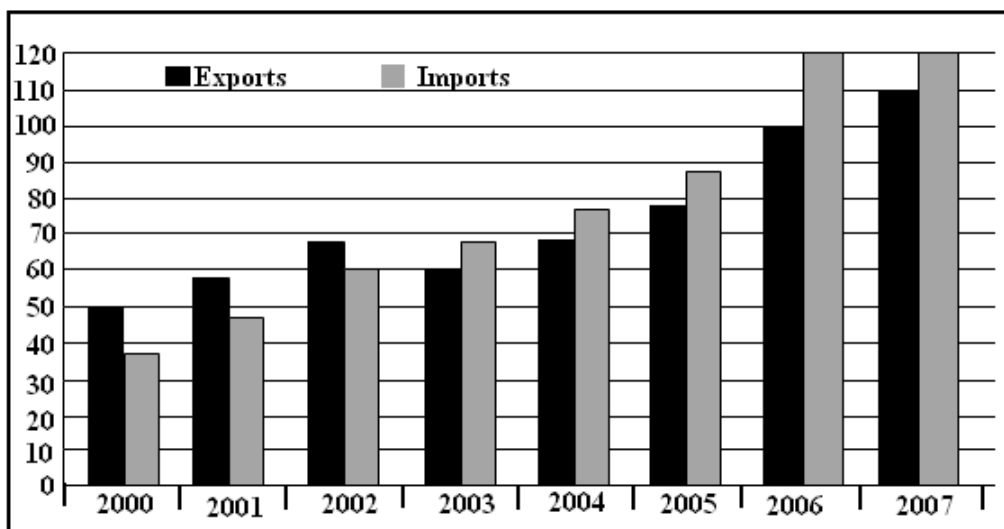
Answer: (B)

Exp: Since U does not want any portfolio, (C) and (D) are ruled out. R wants Home, or Finance or No portfolio, (A) is not valid. Hence option (B) is correct

7. The exports and imports (in crores of Rs.) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest?

(GATE-CE-15)

Answer: 2006



Exp: Increase in exports in 2006 = $\frac{100-70}{70} = 42.8\%$

Increase in imports in 2006 = $\frac{120-90}{90} = 33.3\%$

which is more than any other year

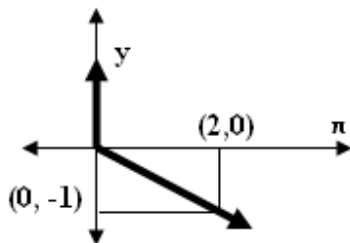
GENERAL APTITUDE

8. Most experts feel that in spite of possessing all the technical skills required to be a batsman of the highest order., he is unlikely to be so due to lack of requisite temperament. He was guilty of throwing away his wicket several times after working hard to lay a strong foundation. His critics pointed out that until he addressed to this problem, success at the highest level will continue to elude him. Which of the statement (s) below is/are logically valid and can be inferred from the above passage? **(GATE-CE-15)**

- (i) He was already a successful batsman at the highest level
 - (ii) He has to improve his temperament in order to become a great batsman
 - (iii) He failed to make many of his good starts count
 - (iv) Improving his technical skills will guarantee success
- (A) (iii) and (iv) (B) (ii) and (iii) (C) (i), (ii) and (iii) (D) (ii) only

Answer: (B)

9. Choose the most appropriate equation for the function drawn as a thick line, in the plot below. **(GATE-CE-15)**



- (A) $x = y - |y|$ (B) $x = -(y - |y|)$ (C) $x = y + |y|$ (D) $x = -(y + |y|)$

Answer: (B)

10. Alexander turned his attention towards India, since he had conquered Persia. Which one of the statements below is logically valid and can be inferred from the above sentence? **(GATE-CE-15)**

- (A) Alexander would not have turned his attention towards India had he not conquered Persia.
- (B) Alexander was not ready to rest on his laurels, and wanted to march to India
- (C) Alexander was completely in control of his army and could command it to move towards India.
- (D) Since Alexander's kingdom extended to Indian borders after the conquest of Persia, he was keen to move further.

Answer: (A)

GENERAL APTITUDE

CIVIL - PAPER - II

1. Choose the most appropriate word from the options given below to complete the following sentence The official answered _____ that the complaints of the citizen would be looked into. (GATE-CE-15)
- (A) respectably (B) respectfully
(C) reputably (D) respectively

Answer: (B)

2. Choose the statement where underlined word is used correctly (GATE-CE-15)
- (A) The minister insured the victims that everything would be all right.
(B) He ensured that the company will not have to bear any loss.
(C) The actor got himself ensured against any accident.
(D) The teacher insured students of good results

Answer: (B)

Exp: *insured*-the person, group, or organization whose life or property is covered by an insurance policy.

ensured- to secure or guarantee

3. Four cards are randomly selected from a pack of 52 cards. If the first two cards are kings, what is the probability that the third card is a king? (GATE-CE-15)
- (A) $4/52$ (B) $2/50$
(C) $1/52 \times (1/52)$ (D) $1/52 \times (1/52) \times (1/50)$

Answer: (B)

Exp: There are 4 kings in a pack of 52 cards.

If 2 cards are selected and both are kings, remaining cards will be 50 out of which 2 will be kings.

4. Which word is not a synonym for the word *vernacular*? (GATE-CE-15)
- (A) regional (B) indigeneous
(C) indigent (D) colloquial

Answer: (C)

Exp: *vernacular*- expressed or written in the native language of a place *indigent* -deficient in what is requisite

5. Mr. Vivek walks 6 meters North-East, then turns and walks 6 meters South- East, both at 60 degrees to East. He further moves 2 meters South and 4 meters West. What is the straight distance in meters between the point he started from and the point he finally reached? (GATE-CE-15)
- (A) $2\sqrt{2}$ (B) 2 (C) $\sqrt{2}$ (D) $1/\sqrt{2}$

Answer: (A)

GENERAL APTITUDE

Q. No. 6 - 10 Carry Two Marks Each

6. How many four digit numbers can be formed with the 10 digits 0, 1, 2, 9 if no number can start with 0 and if repetitions are not allowed? (GATE-CE-15)

Answer: 4536

Exp: In thousands place, 9 digits except 0 can be placed

In hundreds place, 9 digits can be placed (including 0, excluding the one used in thousands place)

In tens place, 8 digits can be placed (excluding the ones used in thousands and hundreds place)

In ones place, 7 digits can be placed (excluding the one used in thousands, hundreds and tens place)

Total number of combinations = $9 \times 9 \times 8 \times 7 = 4536$

7. The word similar in meaning to 'dreary' is (GATE-CE-15)
 (A) cheerful (B) dreamy (C) hard (D) dismal

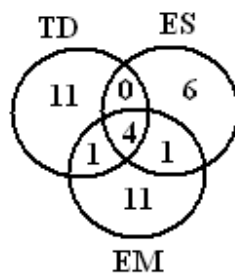
Answer: (D)

Exp: *dreary*- depressingly dull and bleak or repetitive.

8. There are 16 teachers who can teach Thermodynamics (TD), 11 who can teach Electrical Sciences (ES), and 5 who can teach both TD and Engineering Mechanics (EM). There are a total of 40 teachers, 6 cannot teach any of the three subjects, i.e. EM, ES or TD. 6 can teach only ES. 4 can teach all three subjects, i.e. EM, ES and TD. 4 can teach ES and TD. How many can teach both ES and EM but not TD? (GATE-CE-15)
 (A) 1 (B) 2 (C) 3 (D) 4

Answer: (A)

Exp:



9. Read the following table giving sales data of five types of batteries for years 2006 to 2012

Year	Type I	Type II	Type III	Type IV	Type V
2006	75	144	114	102	108
2007	90	126	102	84	126
2008	96	114	75	105	135
2009	105	90	150	90	75
2010	90	75	135	75	90
2011	105	60	165	45	120
2012	115	85	160	100	145

GENERAL APTITUDE

Out of the following, which type of battery achieved highest growth between the years 2006 and 2012? **(GATE-CE-15)**

- (A) Type V (B) Type III (C) Type II (D) Type I

Answer: (D)

Exp: Type-I achieved a growth of 53% in the period which is higher than any other type of battery

10. The given question is followed by two statements: select the most appropriate option that Capacity of a solution tank A is 70% of the capacity of tank B. How many gallons of solution are in tank A and tank B? **(GATE-CE-15)**

Statements:

- I. Tank A is 80% full and tank B is 40% full
II. Tank A if full contains 14,000 gallons of solution
(A) Statement I alone is sufficient
(B) Statement II alone is sufficient
(C) Either statement I or II alone is sufficient
(D) Both the statements I and II together are sufficient

Answer: (D)

Exp: Statement I can be used to solve the question if capacity of both tanks is already known
Statement-II can be used if it is known what quantity of each tank is full/empty.

Therefore, by using both statements

Let capacity of tank B is x

$$\frac{70}{100} \times = 14000$$
$$\Rightarrow 20000 \text{ gallons}$$

$$\text{Solution in tank A} = \frac{80}{100} \times 14000 = 11200 \text{ gallons}$$

$$\text{Solution in tank B} = \frac{40}{100} \times 20000 = 8000 \text{ gallons}$$

$$\text{Total Solution} = 11200 + 8000 = 19200 \text{ gallons}$$

GENERAL APTITUDE

GATE-CSE –PAPER -I

1. Didn't you buy _____ when you went shopping? (GATE-CSE-15)
(A) any paper (B) much paper (C) no paper (D) a few paper

Answer: (A)

2. Which of the following combinations is incorrect? (GATE-CSE-15)
(A) Acquiescence – Submission (B) Wheedle – Roundabout
(C) Flippancy – Lightness (D) Profligate – Extravagant

Answer: (B)

3. Given set A = {2, 3, 4, 5} and Set B = {11, 12, 13, 14, 15}, two numbers are randomly selected, one from each set. What is probability that the sum of the two numbers equals 16? (GATE-CSE-15)

(A) 0.20 (B) 0.25 (C) 0.30 (D) 0.33

Answer: (A) Exp:

4 x 5 = 20 Total mass

5,11

4,12

3,13 4 favorable

2,14

$$\therefore \frac{4}{20} = \frac{1}{5} = 0.2$$

4. Which of the following options is the closest in meaning to the sentence below? She enjoyed herself immensely at the party. (GATE-CSE-15)
(A) She had a terrible time at the party. (B) She had a horrible time at the party.
(C) She had a terrific time at the party (D) She had a terrifying time at the party

Answer: (C)

5. Based on the given statements, select the most appropriate option to solve the given question. If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor of the building? (GATE-CE-15)

Statements: (I) Each step is $\frac{3}{4}$ foot high. (II) Each step is 1 foot wide.

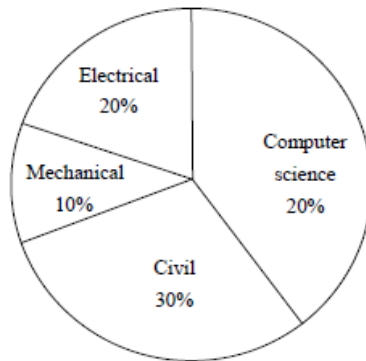
- (A) Statement I alone is sufficient, but statement II alone is not sufficient.
(B) Statement II alone is sufficient, but statement I alone is not sufficient.
(C) Both statements together are sufficient, but neither statement alone is sufficient.
(D) Statement I and II together are not sufficient.

Answer: (D)

GENERAL APTITUDE

Q.No-6-10 Carry Two Marks Each

6. The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is 5:4. There are 40 males in Electrical Engineering. What is the difference between numbers of female students in the Civil department and the female students in the Mechanical department? (GATE-CSE-15)



Answer: 32

Exp: Electrical male students = 40

$$\therefore \text{Electrical Female students} = \frac{4}{5} \times 40 = 32$$

$$\text{Total no. of Student} = 40 + 32 = 72$$

$$\text{Mechanical Strength is } 10\% = 20 + 16 = 36$$

$$\text{Civil Strength is } 30\% = 50 + 48 = 98$$

$$\text{Difference of civil female to mechanical female} = 48 - 16 = 32$$

7. Select the alternative meaning of the underlined part of the sentence. The chain snatchers took to their heels when the police party arrived. (GATE-CSE-15)
- (A) took shelter in a thick jungle (B) open indiscriminate fire
(C) took to flight (D) unconditionally surrendered

Answer: (C)

8. The probabilities that a student passes in Mathematics, Physics and Chemistry are m , p , and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m , p , c : (GATE-CSE-15)
- (I) $p + m + c = 27/20$
(II) $p + m + c = 13/20$
(III) $(p)(m)(c) = 1/10$
- (A) Only relation I is true
(B) Only relation II is true
(C) Relations II and III are true.
(D) Relations I and III are true.

Answer: (A)

GENERAL APTITUDE

Exp:

$$\begin{aligned} & P(\text{atleast two}) - p(\text{exat 2}) \\ &= 0.5 - 0.4 = 0.1 \\ & 0.75 = p + m + c + 0.1 - (0.5 + 0.11 \times 2) \\ & \therefore p + mc = 0.65 + 0.7 = 1.35 = \frac{27}{20} \end{aligned}$$

9. The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option. **(GATE-CSE-15)**

Statement:

There has been a significant drop in the water level in the lakes supplying water to the city.

Course of action:

- (I) The water supply authority should impose a partial cut in supply to tackle the situation.
(II) The government should appeal to all the residents through mass media for minimal use of water.
(III) The government should ban the water supply in lower areas.

- (A) Statements I and II follow. (B) Statements I and III follow
(C) Statements II and III follow. (D) All statements follow

Answer : (A)

10. The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking. **(GATE-CSE-15)**

D. No	Marks	Answered Correctly	Answered wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

What is the average of the marks obtained by the class in the examination?

- (A) 2.290 (B) 2.970 (C) 6.795 (D) 8.795

Answer: (B)

Exp:
$$\frac{21 \times 2 + 15 \times 3 + 11 \times 1 + 23 \times 2 + 31 \times 5}{21 + 15 + 11 + 23 + 31} = 2.970$$

GENERAL APTITUDE

GATE – CSE – PAPER -2

1. Based on the given statements, select the most appropriate option to solve the given question
What will be the total weight of 10 poles each of same weight? (GATE-CSE-15)

Statements

- (I) One fourth of the weight of a pole is 15kg.
(II) The total weight of these poles is 160 kg more than the total weight of two poles
(A) Statement I alone is not sufficient
(B) Statement II alone is not sufficient
(C) Either I or II alone is sufficient
(D) Both statements I and II together are not sufficient.

Answer: (C)

Exp: Let weight of each pole be x

I. Given, $\frac{x}{4} = 15$ kg

$\Rightarrow x = 60$ kg

All poles are of same weight

$\therefore$ Weight of 10 poles = $10x = 10 \times 60 = 600$ kg

II. Given, $10x = 2x + 160$

$\Rightarrow 8x = 160$ kg = $x = 20$ kg

$\therefore$ Weight of 10 poles = $10x = 10 \times 20 = 200$ kg

2. Consider a function $f(x) = 1 - |x|$ on $-1 \leq x \leq 1$. The value of x at which the function attains a maximum, and the maximum value of the function are. (GATE-CSE-15)
(A) 0, -1 (B) -1, 0 (C) 0, 1 (D) -1, 2

Answer: (C)

Exp:

For $-1 \leq x \leq 1$, $0 \leq |x| \leq 1$. $f(x)$ will be maximum when $|x|$ is minimum, i.e. $|x| = 0 \Rightarrow x = 0$

$\therefore$ Maximum value of $f(x) = 1 - 0 = 1$

3. A generic term that include various items of clothing such as a skirt, a pair of trousers and a shirt is (GATE-CSE-15)
(A) fabric (B) textile (C) fibre (D) apparel

Answer: (D)

Exp: apparel- clothing, especially outerwear; garments; attire; raiment

4. Choose the statement where underlined word is used correctly. (GATE-CSE-15)
(A) The industrialist load a personnel jet.
(B) I write my experience in my personnel diary.
(C) All personnel are being given the day off.
(D) Being religious is a personnel aspect.

Answer: (C)

Exp: personnel- people employed in an organization or engaged in an organized undertaking such as military service.

GENERAL APTITUDE

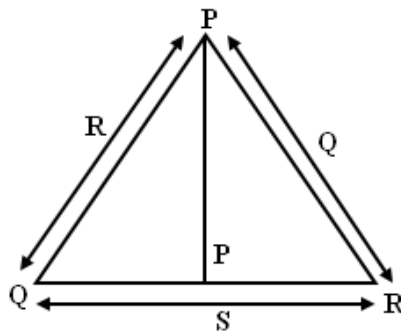
5. We _____ our friend's birthday and we _____ how to make it up to him. (GATE-CSE-15)

(A) Completely forgot - - - don't just know
(B) Forgot completely - - - don't just know
(C) Completely forgot - - - just don't know
(D) Forgot completely - - - just don't know

Answer: (C)

Q.No-6-10 Carry Two Marks Each

6. In a triangle PQR, PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60^\circ$. What is the length of PS? (GATE-CSE-15)



(A) $\frac{(q+r)}{qr}$

(B) $\frac{qr}{(q+r)}$

(C) $\sqrt{(q^2 + r^2)}$

(D) $\frac{(q+r)^2}{qr}$

Answer: (B)

7. Out of the following four sentences, select the most suitable sentence with respect to grammar and usage. (GATE-CE-15)

(A) Since the report lacked needed information, it was of no use to them.
(B) The report was useless to them because there were no needed information in it.
(C) Since the report did not contain the needed information, it was not real useful to them
(D) Since the report lacked needed information, it would not had been useful to them.

Answer: (A)

Exp: (B) there was no needed information

(C) not really useful

(D) would not have been

8. If the list of letters, P,R,S,T,U is an arithmetic sequence, which of the following are also in arithmetic sequence? (GATE-CSE-15)

I . 2P, 2R, 2S, 2T, 2U

II . P - 3, R - 3, S - 3, T - 3, U - 3

III. P^2 , R^2 , S^2 , T^2 , U^2

(A) I only

(B) I and II

(C) II and III

(D) I and III

Answer: (B)



GENERAL APTITUDE

Exp: Let us take the numbers as 3,4,5,6,7 $n=1$ and they are in arithmetic sequence

I. 6, 8,10,12,14 $n=2$ and they are in arithmetic sequence

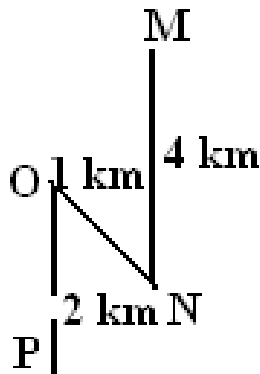
II. 0,1,2,3,4 $n=1$ and they are in arithmetic sequence

III. 9,16,25,36,49 $n=7,9,11,13$ they are not in an arithmetic sequence

9. Four branches of a company are located at M,N,O and P. M is north of N at a distance of 4km: P is south of O at a distance of 2 km: N is southeast of O by 1 km. What is the distance between M and P in km? (GATE-CE-15)
(A) 5.34 (B) 6.74 (C) 28.5 (D) 45.49

Answer: (A)

Exp:



GENERAL APTITUDE

ECE-GATE-PAPER -1

1. Choose the word most similar in meaning to the given word: (GATE-EC-15)

Educe

(A) Exert (B) Educate (C) Extract (D) Extend

Answer: (C)

2. If $\log_x (5/7) = -1/3$, then the value of x is (GATE-EC-15)

(A) 343/125 (B) 125/343 (C) -25/49 (D) -49/25

Answer: (A)

Exp: $\frac{5}{7} = x^{-1/3} \Rightarrow \frac{7}{5} = x^{1/3} \Rightarrow \left(\frac{7}{5}\right)^3 \Rightarrow x = 2.74$

3. Operators $\square$, $\diamond$ and $\rightarrow$ are defined by ; $a \square b = \frac{a-b}{a+b}$; $a \diamond b = \frac{a+b}{a-b}$; $a \rightarrow b = ab$. Find the value (GATE-EC-15)

$(66 \square 6) \rightarrow (66 \diamond 6)$

(A) -2 (B) -1 (C) 1 (D) 2

Answer: (C)

Exp:

$$66 \square 6 = \frac{66-6}{66+6} = \frac{60}{72} = \frac{5}{6}$$

$$66 \diamond 6 = \frac{66+6}{66-6} = \frac{72}{60} = \frac{6}{5}$$

$$(66 \square 6) \rightarrow (66 \diamond 6) = \frac{5}{6} \times \frac{6}{5} = 1$$

4. Choose the most appropriate word from the options given below to complete the following sentence. The principal presented the chief guest with a _____, as token of appreciation. (GATE-EC-15)

(A) momento (B) memento (C) momentum (D) moment

Answer: (B)

5. Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence: Frogs _____ . (GATE-EC-15)

(A) Croak (B) Roar (C) Hiss (D) Patter

Answer: (A)

Exp: Frogs make 'croak' sound.

Q. No. 6 – 10 Carry Two Marks Each

6. A cube of side 3 units is formed using a set of smaller cubes of side 1 unit. Find the proportion of the number of faces of the smaller cubes visible to those which are NOT visible. (GATE-EC-15)

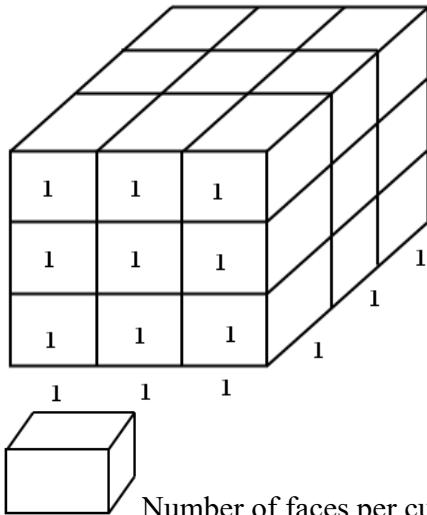
(A) 1 : 4 (B) 1 : 3 (C) 1 : 2 (D) 2 : 3

Answer: (C)



GENERAL APTITUDE

Exp:



Total number of cubes = $9 \times 3 = 27$

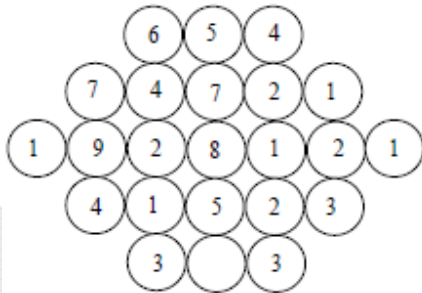
$\therefore$ Total number of faces = $27 \times 6 = 162$

$\therefore$ The number of non visible faces = $162 - 54 = 108$

$\therefore \frac{\text{Number of visible faces}}{\text{Number of non visible faces}} = \frac{54}{108} = \frac{1}{2}$

7. Fill in the missing value

(GATE-EC-15)



Answer : 3

Exp:

Middle number is the average of the numbers on both sides.

Average of 6 and 4 is 5

Average of (7+4) and (2+1) is 7

Average of (1+9+2) and (1+2+1) is 8

Average of (4+1) and (2+3) is 5

Therefore, Average of (3) and (3) is 3

GENERAL APTITUDE

8. Humpty Dumpty sits on a wall every day while having lunch. The wall sometimes breaks. A person sitting on the wall falls if the wall breaks. Which one of the statements below is logically valid and can be inferred from the above sentences? (GATE-EC-15)
- (A) Humpty Dumpty always falls while having lunch
 - (B) Humpty Dumpty does not fall sometimes while having lunch
 - (C) Humpty Dumpty never falls during dinner
 - (D) When Humpty Dumpty does not sit on the wall, the wall does not break

Answer: (B)

9. The following question presents a sentence, part of which is underlined. Beneath the sentence you find four ways of phrasing the underline part. Following the requirements of the standard written English, select the answer that produces the most effective sentence. Tuberculosis, together with its effects, ranks one of the leading causes of death in India. (GATE-EC-15)
- (A) ranks as one of the leading causes of death
 - (B) rank as one of the leading causes of death
 - (C) has the rank of one of the leading causes of death
 - (D) are one of the leading causes of death

Answer: (A)

10. Read the following paragraph and choose the correct statement. Climate change has reduced human security and threatened human well being. An ignored reality of human progress is that human security largely depends upon environmental security. But on the contrary, human progress seems contradictory to environmental security. To keep up both at the required level is a challenge to be addressed by one and all. One of the ways to curb the climate change may be suitable scientific innovations, while the other may be the Gandhian perspective on small scale progress with focus on sustainability. (GATE-EC-15)

- (A) Human progress and security are positively associated with environmental security.
- (B) Human progress is contradictory to environmental security.
- (C) Human security is contradictory to environmental security.
- (D) Human progress depends upon environmental security.

Answer : B

GENERAL APTITUDE

ECE-GATE-PAPER -II

1. What is the adverb for the given word below? Misogynous (GATE-EC-15)
(A) Misogynousness (B) Misogynity (C) Misogynously (D) Misogynous

Answer: (C)

2. Choose the appropriate word–phrase out of the four options given below, to complete the following sentence Dhoni, as well as the other team members of Indian team _____ present on the occasion (GATE-EC-15)
(A) Were (B) Was (C) Has (D) Have

Answer: (B)

3. Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is $\frac{1}{6}$ and that of Ramesh is $\frac{1}{8}$. What is the probability that only one of them will be selected? (GATE-EC-15)
(A) $\frac{47}{48}$ (B) $\frac{1}{4}$ (C) $\frac{13}{48}$ (D) $\frac{35}{48}$

Answer : B

Exp: $P(\text{Ram}) = \frac{1}{6}$; $p(\text{Ramesh}) = \frac{1}{8}$

$$P(\text{only at}) = p(\text{Ram}) \times p(\text{not Ramesh}) + p(\text{Ramesh}) \times p(\text{not Ram}) = \frac{1}{6} + \frac{7}{8} \times \frac{5}{6}$$

$$\Rightarrow \frac{12}{40} = \frac{1}{4}$$

4. Choose the word most similar in meaning to the given word: Awkward (GATE-EC-15)
(A) Inept (B) Graceful (C) Suitable (D) Dreadful

Answer: (A)

5. An electric bus has onboard instruments that report the total electricity consumed since the start of the trip as well as the total distance covered. During a single day of operation, the bus travels on stretches M, N, O and P, in that order. The cumulative distances traveled and the corresponding electricity consumption are shown in the Table below. (GATE-EC-15)

Stretch	Cummulative Distance (km)	Electricity used (kWh)
M	20	12
N	45	25
O	75	45
P	100	57

The stretch where the electricity consumption per km is minimum is

- (A) M (B) N (C) O (D) P

Answer: (B)

GENERAL APTITUDE

Exp:

Stretch	Cummulative Distance (km)	Electricity used (kWh)	Individual (km) Distance	Individual Electricity (kWh)
M	20	12	20	12
N	45	25	25	13
O	75	45	30	20
P	100	57	25	12

$$\text{For M} \Rightarrow 12/20 = 0.6$$

$$\text{N} \Rightarrow 13/25 = 0.555$$

$$\text{O} \Rightarrow 20/30 = 0.667$$

$$\text{P} \Rightarrow 12/25 = 0.48$$

6. Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows. Statements: I. All film stars are playback singers. II. All film directors are film stars. Conclusions: I. All film directors are playback singers. II. Some film stars are film directors.
- (A) Only conclusion I follows
(B) Only conclusion II follows
(C) Neither conclusion I nor II follows
(D) Both conclusions I and II follow

Answer: (D)

7. Lamenting the gradual sidelining of the arts in school curricula, a group of prominent artists wrote to the Chief Minister last year, asking him to allocate more funds to support arts education in schools. However, no such increase has been announced in this year's Budget. The artists expressed their deep anguish at their request not being approved, but many of them remain optimistic about funding in the future. Which of the statement(s) below is/are logically valid and can be inferred from the above statements? (GATE-EC-15)
- (i) The artists expected funding for the arts to increase this year
(ii) The Chief Minister was receptive to the idea of increasing funding for the arts
(iii) The Chief Minister is a prominent artist
(iv) Schools are giving less importance to arts education nowadays
- (A) (iii) and (iv) (B) (i) and (iv) (C) (i), (ii) and (iv) (D) (i) and (iii)

Answer: (B)

GENERAL APTITUDE

8. A tiger is 50 leaps of its own behind a deer. The tiger takes 5 leaps per minute to the deer's 4. If the tiger and the deer cover 8 metre and 5 metre per leap respectively. What distance in meters will be tiger have to run before it catches the deer? (GATE-EC-15)

Answer: 800

Exp: Tiger – 1 leap $\Rightarrow$ 8 meter

Speed = 5 leap/hr = 40m/min

Deer $\rightarrow$ 1 leap = 5 meter

Speed = 4hr = 20m/min

Let at time 't' the tiger catches the deer.

$\therefore$ Distance traveled by deer + initial distance between them

$50 \times 8 = 400\text{m} = \text{distance covered by tiger}$

$\Rightarrow 40 \times t = 400 + 20t$

$\Rightarrow t = \frac{400}{20} = 20 \text{ min}$

$\Rightarrow \text{total distance} \Rightarrow 400 + 20 \times t = 800 \text{ ms}$

9. If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ac$ lies in the interval (GATE-EC-15)
(A) $[1, 2/3]$ (B) $[-1/2, 1]$ (C) $[-1, 1/2]$ (D) $[2, -4]$

Answer: (B)

10. In the following sentence certain parts are underlined and marked P, Q and R. One of the parts may contain certain error or may not be acceptable in standard written communication. Select the part containing an error. Choose D as your answer if there is no error. The student corrected all the errors that the instructor marked on the answer book (GATE-EC-15)

(A) P (B) Q (C) R (D) No error

Answer: (B)

Exp: The is not required in 'Q'

GENERAL APTITUDE

GATE – EEE – PAPER –I

1. Which of the following combinations is incorrect? (GATE-EE-15)
(A) Acquiescence – Submission (B) Wheedle – Roundabout
(C) Flippancy – Lightness (D) Profligate – Extravagant

Answer: (B)

2. Given set A = {2, 3, 4, 5} and Set B = {11, 12, 13, 14, 15}, two numbers are randomly selected, one from each set. What is probability that the sum of the two numbers equals 16? (GATE-EE-15)
(A) 0.20 (B) 0.25 (C) 0.30 (D) 0.33

Answer: (A)

Exp:

4 x 5 = 20 Total mass

5,11

4,12

3,13 4 favorable

2,14

$$\therefore \frac{4}{20} = \frac{1}{5} = 0.2$$

3. Which of the following options is the closest in meaning to the sentence below? She enjoyed herself immensely at the party. (GATE-EE-15)
(A) She had a terrible time at the party. (B) She had a horrible time at the party.
(C) She had a terrific time at the party (D) She had a terrifying time at the party

Answer: (C)

4. Based on the given statements, select the most appropriate option to solve the given question. If two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor of the building? (GATE-EE-15)
Statements: (I) Each step is $\frac{3}{4}$ foot high. (II) Each step is 1 foot wide.
(A) Statement I alone is sufficient, but statement II alone is not sufficient.
(B) Statement II alone is sufficient, but statement I alone is not sufficient.
(C) Both statements together are sufficient, but neither statement alone is sufficient.
(D) Statement I and II together are not sufficient.

Answer: (D)

5. Didn't you buy _____ when you went shopping? (GATE-EE-15)
(A) any paper (B) much paper (C) no paper (D) a few paper
Answer: (A)



GENERAL APTITUDE

Q.No-6-10 Carry Two Marks Each

6. The given statement is followed by some courses of action. Assuming the statement to be true, decide the correct option. (GATE-EE-15)

Statement:

There has been a significant drop in the water level in the lakes supplying water to the city.

Course of action:

- (I) The water supply authority should impose a partial cut in supply to tackle the situation.
(II) The government should appeal to all the residents through mass media for minimal use of water.
(III) The government should ban the water supply in lower areas.

(A) Statements I and II follow

(B) Statements I and III follow

(C) Statements II and III follow

(D) All statements follow

Answer : (A)

7. The number of students in a class who have answered correctly, wrongly, or not attempted each question in an exam, are listed in the table below. The marks for each question are also listed. There is no negative or partial marking. (GATE-EE-15)

D. No	Marks	Answered Correctly	Answered wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	1	11	29	4
4	2	23	18	3
5	5	31	12	1

What is the average of the marks obtained by the class in the examination?

(A) 2.290

(B) 2.970

(C) 6.795

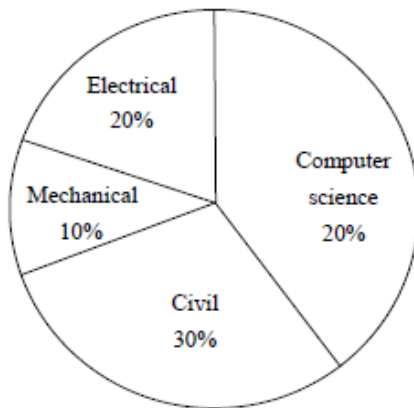
(D) 8.795

Answer: (B)

Exp:
$$\frac{21 \times 2 + 15 \times 3 + 11 \times 1 + 23 \times 2 + 31 \times 5}{44} = 6.795$$

8. The pie chart below has the breakup of the number of students from different departments in an engineering college for the year 2012. The proportion of male to female students in each department is 5:4. There are 40 males in Electrical Engineering. What is the difference between numbers of female students in the Civil department and the female students in the Mechanical department? (GATE-EE-15)

GENERAL APTITUDE



Answer: 32

Exp: Electrical male students = 40

$\therefore$ Electrical Female students $\frac{4}{5} \times 40 = 32$

Total no. of Student = 40 + 32 = 72

Mechanical Strength is 10% = 20 + 16 = 36

Civil Strength is 30% = 50 + 48 = 98

Difference of civil female to mechanical female = 48 - 16 = 32

9. Select the alternative meaning of the underlined part of the sentence. The chain snatchers took to their heels when the police party arrived. (GATE-EE-15)

(A) took shelter in a thick jungle

(B) open indiscriminate fire

(C) took to flight

(D) unconditionally surrendered

Answer: (C)

10. The probabilities that a student passes in Mathematics, Physics and Chemistry are m, p , and c respectively. Of these subjects, the student has 75% chance of passing in at least one, a 50% chance of passing in at least two and a 40% chance of passing in exactly two. Following relations are drawn in m, p, c :

(I) $p + m + c = 27/20$

(II) $p + m + c = 13/20$

(III) $(p)(m)(c) = 1/10$

(A) Only relation I is true

(B) Only relation II is true

(C) Relations II and III are true.

(D) Relations I and III are true.

Answer: (A)

Exp:

$P(\text{atleast two}) - p(\text{exat 2})$

$= 0.5 - 0.4 = 0.1$

$0.75 = p + m + c + 0.1 - (0.5 + 0.1 \times 2)$

$\therefore p + mc = 0.65 + 0.7 = 1.35 = \frac{27}{20}$

GENERAL APTITUDE

GATE – EEE – PAPER -2

1. A generic term that include various items of clothing such as a skirt, a pair of trousers and a Shirt is (GATE-EE-15)
(A) Fabric (B) textile (C) fibre (D) apparel

Answer: (D)

Exp: apparel- clothing, especially outerwear; garments; attire; raiment

2. Choose the statement where underlined word is used correctly. (GATE-EE-15)
(A) The industrialist load a personnel jet.
(B) I write my experience in my personnel diary.
(C) All personnel are being given the day off.
(D) Being religious is a personnel aspect.

Answer: (C)

Exp: personnel- people employed in an organization or engaged in an organized undertaking such as military service.

3. Based on the given statements, select the most appropriate option to solve the given question
What will be the total weight of 10 poles each of same weight? (GATE-EE-15)

Statements

- (I) One fourth of the weight of a pole is 15kg.
(II) The total weight of these poles is 160 kg more than the total weight of two poles
(A) Statement I alone is not sufficient
(B) Statement II alone is not sufficient
(C) Either I or II alone is sufficient
(D) Both statements I and II together are not sufficient.

Answer: (C)

Exp: Let weight of each pole be x

I. Given, $\frac{x}{4} = 15 \text{ kg}$

$\Rightarrow x = 60 \text{ kg}$

All poles are of same weight

$\therefore$ Weight of 10 poles = $10x = 10 \times 60 = 600 \text{ kg}$

II. Given, $10x = 2x + 160$

$\Rightarrow 8x = 160 \text{ kg} = x = 20 \text{ kg}$

$\therefore$ Weight of 10 poles = $10x = 10 \times 20 = 200 \text{ kg}$

4. Consider a function $f(x) = 1 - |x|$ on $-1 \leq x \leq 1$. The value of x at which the function attains a maximum, and the maximum value of the function are. (GATE-EE-15)
(A) 0, -1 (B) -1, 0 (C) 0, 1 (D) -1, 2

Answer: (C)



GENERAL APTITUDE

Exp:

For $-1 \leq x \leq 1$, $0 \leq |x| \leq 1$. $f(x)$ will be maximum when $|x|$ is minimum, i.e. $|x| = 0 \Rightarrow x = 0$
 $\therefore$ Maximum value of $f(x) = 1 - 0 = 1$

5. We _____ our friend's birthday and we _____ how to make it up to him. (GATE-EE-15)

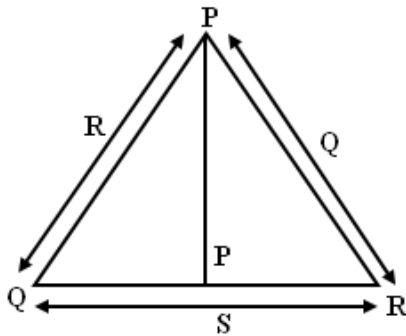
(A) Completely forgot - - - don't just know
 (B) Forgot completely - - - don't just know
 (C) Completely forgot - - - just don't know
 (D) Forgot completely - - - just don't know

Answer: (C)

Q.No-6-10 Carry Two Marks Each

6. In a triangle PQR, PS is the angle bisector of $\angle QPR$ and $\angle QPS = 60^\circ$. What is the length of PS? (GATE-EE-15)

Answer: (B)



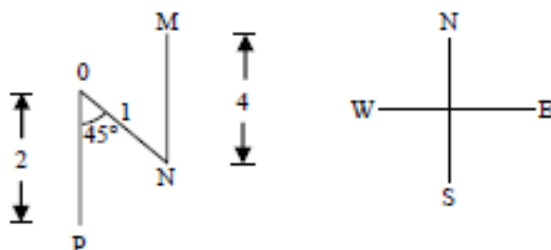
(A) $\frac{(q+r)}{qr}$ (B) $\frac{qr}{(q+r)}$ (C) $\sqrt{(q^2 + r^2)}$ (D) $\frac{(q+r)^2}{qr}$

7. Four branches of a company are located at M, N, O and P. M is north of N at a distance of 4 km; P is south of O at a distance of 2 km; N is southeast of O by 1 km. What is the distance between M and P in km? (GATE-EE-15)

(A) 5.34 (B) 6.74 (C) 28.5 (D) 45.49

Answer: (A)

Exp:



GENERAL APTITUDE

8. If the list of letters, P,R,S,T,U in an arithmetic sequence, which of the following are also in arithmetic sequence? (GATE-EE-15)

I . 2P, 2R, 2S, 2T, 2U

II . P - 3, R - 3, S - 3, T - 3, U - 3

III. P^2 , R^2 , S^2 , T^2 , U^2

(A) I only (B) I and II (C) II and III (D) I and III

Answer: (B)

Exp: Let us take the numbers as 3,4,5,6,7 $n=1$ and they are in arithmetic sequence

I. 6, 8,10,12,14 $n=2$ and they are in arithmetic sequence

II. 0,1,2,3,4 $n=1$ and they are in arithmetic sequence

III. 9,16,25,36,49 $n=7,9,11,13$ they are not in an arithmetic sequence

9. Out of the following four sentences, select the most suitable sentence with respect to grammar and usage. (GATE-EE-15)

(A) Since the report lacked needed information, it was of no use to them.

(B) The report was useless to them because there were no needed information in it.

(C) Since the report did not contain the needed information, it was not real useful to them

(D) Since the report lacked needed information, it would not had been useful to them.

Answer: (A)

Exp: (B) there **was** no needed information

(C) not **really** useful

(D) would not **have** been

10. If p,q,r,s are distinct integers such that:

$$f(p,q,r,s) = \max(p,q,r,s)$$

$$g(p,q,r,s) = \min(p,q,r,s)$$

$$h(p,q,r,s) = \text{remainder of } (p \times q)/(r \times s) \text{ if } (p \times q) > (r \times s) \text{ or remainder of } (r \times s)/(p \times q) \text{ if } (r \times s) > (p \times q)$$

$$\text{Also a function } fgh(p,q,r,s) = f(p,q,r,s) \times g(p,q,r,s) \times h(p,q,r,s)$$

Also the same operations are valid with two variable functions of the form $f(p,q)$

What is the value of $fg(h(2,5,7,3),4,6,8)$?

Ans : 8

GENERAL APTITUDE

GATE – MECH – PAPER –I

2. Ram and Ramesh appeared in an interview for two vacancies in the same department. The probability of Ram's selection is $1/6$ and that of Ramesh is $1/8$. What is the probability that only one of them will be selected? (GATE-ME-15)
- (A) $47/48$ (B) $1/4$ (C) $13/48$ (D) $35/48$

Answer : B

Exp: $P(\text{Ram}) = 1/6$; $p(\text{Ramesh}) = 1/8$

$$P(\text{only at}) = p(\text{Ram}) \times p(\text{not Ramesh}) + p(\text{Ramesh}) \times p(\text{not Ram}) = \frac{1}{6} + \frac{7}{8} \times \frac{1}{8} \times \frac{5}{6}$$
$$\Rightarrow \frac{12}{40} = 1/4$$

3. Choose the appropriate word-phrase out of the four options given below, to complete the following sentence Dhoni, as well as the other team members of Indian team _____ present on the occasion (GATE-ME-15)
- (A) Were (B) Was (C) Has (D) Have

Answer: (B)

4. An electric bus has onboard instruments that report the total electricity consumed since the start of the trip as well as the total distance covered. During a single day of operation, the bus travels on stretches M, N, O and P, in that order. The cumulative distances traveled and the corresponding electricity consumption are shown in the Table below. (GATE-ME-15)

Stretch	Cumulative Distance (km)	Electricity used (kWh)
M	20	12
N	45	25
O	75	45
P	100	57

The stretch where the electricity consumption per km is minimum is

- (A) M (B) N (C) O (D) P

Answer: (B)

Exp:

Stretch	Cummulative Distance (km)	Electricity used (kWh)	Individual (km) Distance	Individual Electricity (kWh)
M	20	12	20	12
N	45	25	25	13
O	75	45	30	20
P	100	57	25	12



GENERAL APTITUDE

$$\text{For M} \Rightarrow 12/20 = 0.6$$

$$\text{N} \Rightarrow 13/25 = 0.555$$

$$\text{O} \Rightarrow 20/30 = 0.667$$

$$\text{P} \Rightarrow 12/25 = 0.48$$

5. Choose the word most similar in meaning to the given word: Awkward (GATE-ME-15)
(A) Inept (B) Graceful (C) Suitable (D) Dreadful

Answer: (A)

6. In the following sentence certain parts are underlined and marked P, Q and R. One of the parts may contain certain error or may not be acceptable in standard written communication. Select the part containing an error. Choose D as your answer if there is no error. The student corrected all the errors that the instructor marked on the answer book

(A) P (B) Q (C) R (D) No error (GATE-ME-15)

Answer: (B)

Exp: The is not required in 'Q'

7. Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows. Statements: I. All film stars are playback singers. II. All film directors are film stars. Conclusions: I. All film directors are playback singers. II. Some film stars are film directors. (GATE-ME-15)
(A) Only conclusion I follows
(B) Only conclusion II follows
(C) Neither conclusion I nor II follows
(D) Both conclusions I and II follow

Answer: (D)

8. Lamenting the gradual sidelining of the arts in school curricula, a group of prominent artists wrote to the Chief Minister last year, asking him to allocate more funds to support arts education in schools. However, no such increase has been announced in this year's Budget. The artists expressed their deep anguish at their request not being approved, but many of them remain optimistic about funding in the future. Which of the statement(s) below is/are logically valid and can be inferred from the above statements? (GATE-ME-15)
(i) The artists expected funding for the arts to increase this year
(ii) The Chief Minister was receptive to the idea of increasing funding for the arts
(iii) The Chief Minister is a prominent artist
(iv) Schools are giving less importance to arts education nowadays
(A) (iii) and (iv) (B) (i) and (iv) (C) (i), (ii) and (iv) (D) (i) and (iii)

Answer: (B)



GENERAL APTITUDE

9. If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ac$ lies in the interval (GATE-ME-15)
(A) $[1, 2/3]$ (B) $[-1/2, 1]$ (C) $[-1, 1/2]$ (D) $[2, -4]$

Answer: (B)

10. A tiger is 50 leaps of its own behind a deer. The tiger takes 5 leaps per minute to the deer's 4. If the tiger and the deer cover 8 metre and 5 metre per leap respectively. What distance in metres will be tiger have to run before it catches the deer? (GATE-ME-15)

Answer: 800

Exp: Tiger – 1 leap $\Rightarrow$ 8 meter

Speed = 5 leap/hr = 40m/min

Deer $\rightarrow$ 1 leap = 5 meter

Speed = 4hr = 20m/min

Let at time 't' the tiger catches the deer.

$\therefore$ Distance traveled by deer + initial distance between them

$50 \times 8 = 400\text{m} = \text{distance covered by tiger}$

$\Rightarrow 40 \times t = 400 + 20t$

$\Rightarrow t = \frac{400}{20} = 20 \text{ min}$

$\Rightarrow \text{total distance} \Rightarrow 400 + 20 \times t = 800 \text{ ms}$

GENERAL APTITUDE

GATE-MECH-PAPER-II

1. Find the missing sequence in the letter series below: (GATE-ME-15)

A, CD, GHI,?, UVWXY

- (A) LMN (B) MNO (C) MNOP (D) NOPQ

Answer: (C)

Exp:

A, CD, GHI, MNOP, UVWXY

Logic : 1 LEFT

2 Letters Left

3 Letters Left

Next '4' Letters left

2. Choose the correct verb to fill in the blank below: Let us _____. (GATE-ME-15)

- (A) Introvert (B) alternate (C) atheist (D) altruist

Answer: (B)

3. Choose the most appropriate word from the options given below to complete the following sentence? If the athlete had wanted to come first in the race, he _____ several hours every day. (GATE-ME-15)

- (A) Should practice (B) Should have practiced
(C) Practised (D) Should be practicing

Answer: (B)

Exp: For condition regarding something which already happened, should have practiced is the correct choice.

4. Choose the most suitable one word substitute for the following expression Connotation of a road or way (GATE-ME-15)

- (A) Pertinacious (B) Viaticum (C) Clandestine (D) Ravenous

Answer: (B)

Exp: Viaticum and connotation both are associated with the feeling of spirituality.

5. If $x > y > 1$, which of the following must be true? (GATE-ME-15)

- (i) $\ln x > \ln y$ (ii) $e^x > e^y$ (iii) $Y^x > X^y$ (iv) $\cos z > \cos y$
(A) (i) and (ii) (B) (i) and (iii)
(C) (iii) and (iv) (D) (ii) and (iv)

Answer : (A)

Exp: For whole numbers, greater the value greater will be its log. Same logic for power of e.

GENERAL APTITUDE

Q. No. 6 – 10 Carry Two Marks Each

6. From a circular sheet of paper of radius 30cm, a sector of 10% area is removed. If the remaining part is used to make a conical surface, then the ratio of the radius and height of the cone is _____. (GATE-ME-15)

Answer: 2.06

Exp: 90% of area of sheet = Cross sectional area of cone

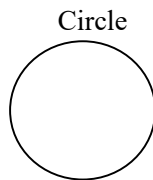
$$\Rightarrow 0.9 \times \pi \times 30 \times 30 = \pi \times r_1 \times l$$

$$\Rightarrow 27 \text{ cm} = r_1$$

$$\therefore \text{Height of the cone} = \sqrt{30^2 - 27^2}$$

$$= 13.08 \text{ cm}$$

$$\text{Then } r/h = 27/13.08 = 2.06$$

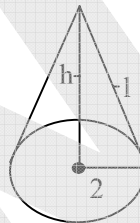


Circle

$$A = \pi r^2$$

$$l^2 = h^2 + r^2$$

$$h = \sqrt{l^2 - r^2}$$



7. In the following question, the first and the last sentence of the passage are in order and numbered 1 and 6. The rest of the passage is split into 4 parts and numbered as 2,3,4, and 5. These 4 parts are not arranged in proper order. Read the sentences and arrange them in a logical sequence to make a passage and choose the correct sequence from the given options. (GATE-ME-15)

1. One Diwali, the family rises early in the morning.
2. The whole family, including the young and the old enjoy doing this,
3. Children let off fireworks later in the night with their friends.
4. At sunset, the lamps are lit and the family performs various rituals
5. Father, mother, and children visit relatives and exchange gifts and sweets.
6. Houses look so pretty with lighted lamps all around.

(A) 2, 5, 3, 4 (B) 5, 2, 4, 3 (C) 3, 5, 4, 2 (D) 4, 5, 2, 3

Answer: (B)

8. Ms X will be in Bagdogra from 01/05/2014 to 20/05/2014 and from 22/05/2014 to 31/05/2014. On the morning of 21/05/2014, she will reach Kochi via Mumbai Which one of the statements below is logically valid and can be inferred from the above sentences? (GATE-ME-15)

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- (A) Ms. X will be in Kochi for one day, only in May
- (B) Ms. X will be in Kochi for only one day in May
- (C) Ms. X will be only in Kochi for one day in May
- (D) Only Ms. X will be in Kochi for one day in May.

Answer: (B)

9. $\log \tan 1^\circ + \log \tan 2^\circ + \dots + \log \tan 89^\circ$ is **(GATE-ME-15)**
(A) 1 (B) $1/\sqrt{2}$ (C) 0 (D) -1

Answer: (C)

Exp:

$$\begin{aligned}\log \tan 1^\circ + \log \tan 89^\circ &= \log (\tan 1^\circ \times \tan 89^\circ) \\ &= \log (\tan 1^\circ \times \cot 1^\circ) \\ &= \log 1 \\ &= 0\end{aligned}$$

10. Ram and Shyam shared a secret and promised to each other that it would remain between them. Ram expressed himself in one of the following ways as given in the choices below. Identify the correct way as per standard English. **(GATE-ME-15)**
- (A) It would remain between you and me.
 - (B) It would remain between I and you
 - (C) It would remain between you and I
 - (D) It would remain with me.

Answer: (A)



GENERAL APTITUDE

GATE-CHEMICAL-PAPER

1. Choose the word most similar in meaning to the given word: (GATE-CH-15)

Educe

(A) Exert

(B) Educate

(C) Extract

(D) Extend

Answer: (C)

2. If $\log_x (5/7) = -1/3$, then the value of x is (GATE-CH-15)

(A) 343/125

(B) 125/343

(C) -25/49

(D) -49/25

Answer: (A)

Exp:

$$\frac{5}{7} = x^{-1/3} \Rightarrow \frac{7}{5} = x^{1/3} \Rightarrow \left(\frac{7}{5}\right)^3 \Rightarrow x = 2.74$$

3. Operators $\square$, $\diamond$ and $\rightarrow$ are defined by ; $a \square b = \frac{a-b}{a+b}$; $a \diamond b = \frac{a+b}{a-b}$; $a \rightarrow b = ab$. Find the value . (GATE-CH-15)

$(66 \square 6) \rightarrow (66 \diamond 6)$

(A) -2

(B) -1

(C) 1

(D) 2

Answer: (C)

Exp:

$$66 \square 6 = \frac{66-6}{66+6} = \frac{60}{72} = \frac{5}{6}$$

$$66 \diamond 6 = \frac{66+6}{66-6} = \frac{72}{60} = \frac{6}{5}$$

$$(66 \square 6) \rightarrow (66 \diamond 6) = \frac{5}{6} \times \frac{6}{5} = 1$$

4. Choose the most appropriate word from the options given below to complete the following sentence. The principal presented the chief guest with a _____, as token of appreciation. (GATE-CH-15)

(A) momento

(B) memento

(C) momentum

(D) moment

Answer: (B)

5. Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence: Frogs _____. (GATE-CH-15)

(A) Croak

(B) Roar

(C) Hiss

(D) Patter

Answer: (A)

Exp: Frogs make 'croak' sound.



GENERAL APTITUDE

Q. No. 6 – 10 Carry Two Marks Each

6. A cube of side 3 units is formed using a set of smaller cubes of side 1 unit. Find the proportion of the number of faces of the smaller cubes visible to those which are NOT visible. (GATE-CH-15)

(A) 1 : 4

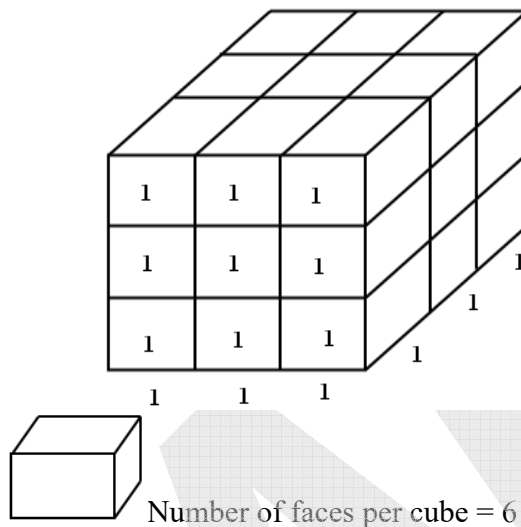
(B) 1 : 3

(C) 1 : 2

(D) 2 : 3

Answer: (C)

Exp:



Total number of cubes = $9 \times 3 = 27$

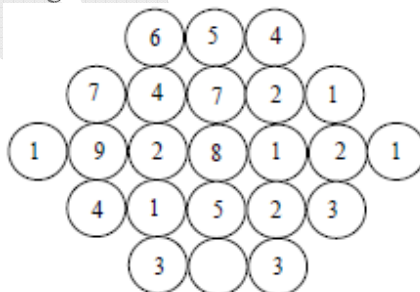
$\therefore$ Total number of faces = $27 \times 6 = 162$

$\therefore$ The number of non visible faces = $162 - 54 = 108$

$\therefore \frac{\text{Number of visible faces}}{\text{Number of non visible faces}} = \frac{54}{108} = \frac{1}{2}$

7. Fill in the missing value

(GATE-CH-15)



Answer : 3

Exp:

Middle number is the average of the numbers on both sides.

Average of 6 and 4 is 5

GENERAL APTITUDE

Average of $(7+4)$ and $(2+1)$ is 7

Average of $(1+9+2)$ and $(1+2+1)$ is 8

Average of $(4+1)$ and $(2+3)$ is 5

Therefore, Average of (3) and (3) is 3

8. Humpty Dumpty sits on a wall every day while having lunch. The wall sometimes breaks. A person sitting on the wall falls if the wall breaks. Which one of the statements below is logically valid and can be inferred from the above sentences? **(GATE-CH-15)**

(A) Humpty Dumpty always falls while having lunch
(B) Humpty Dumpty does not fall sometimes while having lunch
(C) Humpty Dumpty never falls during dinner
(D) When Humpty Dumpty does not sit on the wall, the wall does not break

Answer: (B)

9. The following question presents a sentence, part of which is underlined. Beneath the sentence you find four ways of phrasing the underlined part. Following the requirements of the standard written English, select the answer that produces the most effective sentence. Tuberculosis, together with its effects, ranks one of the leading causes of death in India. **(GATE-CH-15)**

(A) ranks as one of the leading causes of death
(B) rank as one of the leading causes of death
(C) has the rank of one of the leading causes of death
(D) are one of the leading causes of death

Answer: (A)

10. Read the following paragraph and choose the correct statement. Climate change has reduced human security and threatened human well being. An ignored reality of human progress is that human security largely depends upon environmental security. But on the contrary, human progress seems contradictory to environmental security. To keep up both at the required level is a challenge to be addressed by one and all. One of the ways to curb the climate change may be suitable scientific innovations, while the other may be the Gandhian perspective on small scale progress with focus on sustainability. **(GATE-CH-15)**

(A) Human progress and security are positively associated with environmental security.
(B) Human progress is contradictory to environmental security.
(C) Human security is contradictory to environmental security.
(D) Human progress depends upon environmental security.

Answer : B

GENERAL APTITUDE

GATE-INSTRUMENTATION-PAPER-I

1. Five teams have to compete in a league, with every team playing every other team exactly once before going to the next round. How many matches will have to be held to complete the league round of matches? (GATE-IN-15)
(A) 20 (B) 10 (C) 8 (D) 5

Answer: (B)

Exp: For a match to be played, we need 2 teams

No. of matches = No. of ways of selections 2 teams out of 5 = $c_2^5 = 10$

2. Fill in the blank with the correct idiom/phrase. That boy from the town was a _____ in the sleepy village. (GATE-IN-15)
(A) dog out of herd (B) sheep from the heap (C) fish out of water (D) bird from the flock

Answer: (C)

Exp: From the statement, it appears that boy found it tough to adapt to a very different situation.

3. Choose the statement where underlined word is used correctly. (GATE-IN-15)
(A) When the teacher eludes to different authors, he is being elusive.
(B) When the thief keeps eluding the police, he is being elusive.
(C) Matters that are difficult to understand, identify or remember are allusive.
(D) Mirages can be allusive, but a better way to express them is illusory.

Ans: (B)

Exp: Elusive: Difficult to answer.

4. Tanya is older than Eric. Cliff is older than Tanya. Eric is older than Cliff. If the first two statements are true, then the third statement is: (GATE-IN-15)
(A) True (B) False (C) Uncertain (D) Data insufficient

Ans: (B)

5. Choose the appropriate word/phrase, out of the four options given below, to complete the following sentence:
Apparent lifelessness _____ dormant life. (GATE-IN-15)
(A) harbours (B) leads to (C) supports (D) affects

Answer: (A)

Exp: Apparent: looks like

dormant: hidden

Harbour: give shelter

Effect (verb): results in

GENERAL APTITUDE

6. A coin tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occurs in three tosses. (GATE-IN-15)

Based on the above information, which one of the following statements is TRUE?

- (A) X and Y are not independent (B) Y and Z are dependent
(C) Y and Z are independent (D) X and Z are independent

Answer: (B)

Exp: Let y as tail occurred in third toss
and z as two tails in third toss which can be {TTH, THT, HTT}
 $y = \{TTH, TTT\}$
 $\therefore$ both y and z are dependent.

7. Given below are two statements followed by two conclusions. Assuming these statements to be true, decide which one logically follows. (GATE-IN-15)

Statements: I. No manager is a leader.

II. All leaders are executives.

Conclusions: I. No manager is an executive.

II. No executive is a manager.

- (A) Only conclusion I follows. (B) Only conclusion II follows.
(C) Neither conclusion I nor II follows. (D) Both conclusions I and II follow.

Answer: (C)

Exp: Therefore concluding diagram can be
It can be manager that is manager can be executive also.
Some executives are also leaders that is not a manager

8. In the given figure angle Q is a right angle, PS:QS=3:1, RT:QT=5:2 and PU:UR=1:1. If area of triangle QTS is 20 cm^2 , then the area of triangle PQR in cm^2 is _____. (GATE-IN-15)

Answer: 280

Exp: Let are triangle PQR be 'A'

$$\frac{SQ}{PQ} = \frac{1}{1+3} = \frac{1}{4} \quad \frac{QT}{QR} = \frac{2}{2+5} = \frac{2}{7}$$

$$\begin{aligned} \therefore \text{Area of } \Delta^{\text{le}} \text{QTS} &= \frac{1}{2} \times SQ \times QT \\ &= \frac{1}{2} \times \left(\frac{1}{4}PQ\right) \times \left(\frac{2}{7}QR\right) \\ &= \frac{1}{4} \times \frac{2}{7} \times \left(\frac{1}{2} \times PQ \times QR\right) \\ &= \frac{1}{14} \times \text{Area of } \Delta^{\text{le}} \text{PQR} \end{aligned}$$

$$\begin{aligned} \text{Given } 20 \text{ cm}^2 &= \frac{1}{14} \times A \\ \therefore A &= 14 \times 20 = 280 \text{ cm}^2 \end{aligned}$$

GENERAL APTITUDE

9. Select the appropriate option in place of underlined part of the sentence. (GATE-IN-15)
Increased productivity necessary reflects greater efforts made by the employees.

(A) Increase in productivity necessary (B) Increase productivity is necessary
(C) Increase in productivity necessarily (D) No improvement required

Answer: (C)

10. Right triangle PQR is to be constructed in the xy-plane so that the right angle is at P and line PR is parallel to the x-axis. The x and y coordinates of P, Q, and R are to be integers that satisfy the inequalities: $-4 \leq x \leq 5$ and $6 \leq y \leq 16$. How many different triangles could be constructed with these properties? (GATE-IN-15)

(A) 110 (B) 1,100 (C) 9,900 (D) 10,000

Answer: (C)

Exp:

$$X_1 \rightarrow 4 \leq X_1 \leq 5$$

$$Y_1 \rightarrow 6 \leq Y_1 \leq 16$$

$$X_2 \rightarrow 9 \text{ chairs } (\because X_1 \neq X_2)$$

$$Y_2 \rightarrow 10 \text{ chairs } (\because Y_1 \neq Y_2)$$

$$\therefore \text{Total triangles} = 10 \times 11 \times 9 \times 10 = 9900$$

GENERAL APTITUDE

GATE-CSE-PAPER-3

1. Extreme focus on syllabus and studying for test has become such a dominant concern of Indian students that this has closed their minds to anything _____ to the requirements of the exam (GATE-CS-15)
(A) related (B) extraneous (C) outside (D) useful

Answer: (B)

Exp: extraneous -irrelevant or unrelated to the subject being dealt with.

2. A function $f(x)$ is linear and has a value of 29 at $x = -2$ and 39 at $x = 3$. Find its value at $x = 5$. (GATE-CS-15)
(A) 59 (B) 45 (C) 43 (D) 35

Answer: (C)

Exp: $f(x) = 2x + 33$

3. The Tamil version of _____ John Abraham-starrer Madras Café _____ cleared by the censor board with no cuts last week, but the film's distributors _____ no takers among the exhibitors for a release in Tamil Nadu _____ this Friday. (GATE-CS-15)
(A) Mr., was, found, on (B) a, was found, at
(C) the, was, found, on (D) a, being, find, at

Answer: (C)

Exp: John-Abraham starrer Madras Café talks about the movie not the person, so Mr. is ruled out.

'Find no takers' is not the correct phrase. At this Friday is incorrect. So, option C is correct.

4. If ROAD is written as URDG, then SWAN should be written as: (GATE-CS-15)
(A) VXDQ (B) VZDQ (C) VZDP (D) UXDQ

Answer: (B)

Exp: $R+3=U$, $O+3=R$, $A+3=D$, $D+3=G$;
 $S+3=V$, $W+3=Z$, $A+3=D$, $N+3=Q$

Logic: Every letter increased by 3

5. Select the pair that best expresses a relationship similar to that expressed in the pair:
Children: Pediatrician (GATE-CS-15)
(A) Adult: Orthopaedist (B) Females: Gynaecologist
(C) Kidney: Nephrologist (D) Skin: Dermatologist

Answer: (B)

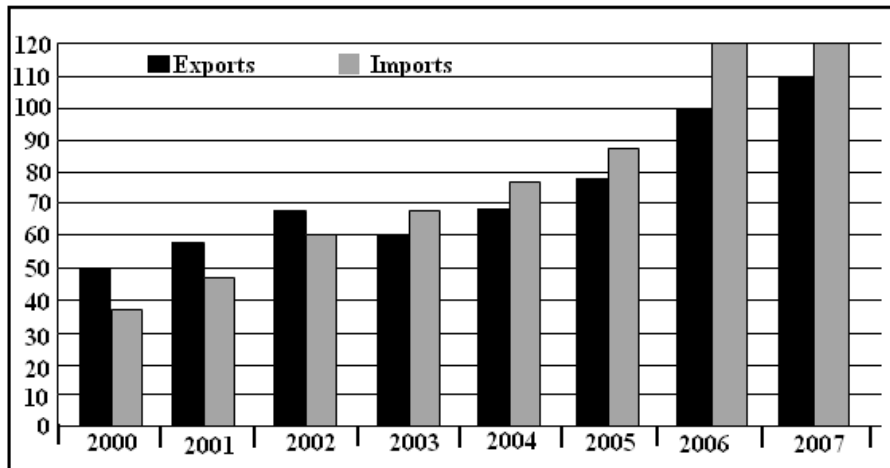
Exp: Community of people: Doctor

GENERAL APTITUDE

Q.No-6-10 Carry Two Marks Each

6. The exports and imports (in crores of Rs.) of a country from the year 2000 to 2007 are given in the following bar chart. In which year is the combined percentage increase in imports and exports the highest? (GATE-CS-15)

Answer: 2006



Exp: Increase in exports in 2006 = $\frac{100-70}{70} = 42.8\%$

Increase in imports in 2006 = $\frac{120-90}{90} = 33.3\%$

Which is more than any other year

7. The head of a newly formed government desires to appoint five of the six selected members P,Q,R,S,T and U to portfolios of Home, Power, Defense, Telecom and Finance. U does not want any portfolio if S gets one of the five. R wants either Home or Finance or no portfolio. Q says that if S gets either Power or Telecom, then she must get the other one. T insists on a portfolio if P gets one. (GATE-CS-15)

Which is the valid distribution of portfolio?

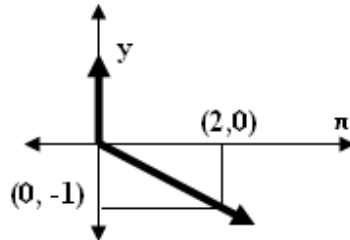
- (A) P-Home, Q-Power, R-Defense, S-Telecom, T-Finance
- (B) R-Home, S-Power, P-Defense, Q-Telecom, T-Finance
- (C) P-Home, Q-Power, T-Defense, S-Telecom, U-Finance
- (D) Q-Home, U-Power, T-Defense, R-Telecom, P-Finance

Answer: (B)

Exp: Since U does not want any portfolio, (C) and (D) are ruled out. R wants Home, or Finance or No portfolio, (A) is not valid. Hence option (B) is correct

GENERAL APTITUDE

8. Choose the most appropriate equation for the function drawn as a thick line, in the plot below. (GATE-CS-15)



- (A) $x = y - |y|$ (B) $x = -(y - |y|)$ (C) $x = y + |y|$ (D) $x = -(y + |y|)$

Answer: (B)

9. Most experts feel that in spite of possessing all the technical skills required to be a batsman of the highest order., he is unlikely to be so due to lack of requisite temperament. He was guilty of throwing away his wicket several times after working hard to lay a strong foundation. His critics pointed out that until he addressed to this problem, success at the highest level will continue to elude him. Which of the statement (s) below is/are logically valid and can be inferred from the above passage? (GATE-CS-15)

- (i) He was already a successful batsman at the highest level
- (ii) He has to improve his temperament in order to become a great batsman
- (iii) He failed to make many of his good starts count
- (iv) Improving his technical skills will guarantee success

- (A) (iii) and (iv) (B) (ii) and (iii) (C) (i), (ii) and (iii) (D) (ii) only

Answer: (B)

10. Alexander turned his attention towards India, since he had conquered Persia. Which one of the statements below is logically valid and can be inferred from the above sentence? (GATE-CS-15)

- (A) Alexander would not have turned his attention towards India had he not conquered Persia.
- (B) Alexander was not ready to rest on his laurels, and wanted to march to India
- (C) Alexander was completely in control of his army and could command it to move towards India.
- (D) Since Alexander's kingdom extended to Indian borders after the conquest of Persia, he was keen to move further.

Answer: (A)

GENERAL APTITUDE

GATE-CIVIL-2014 PAPER-1

Q. No. 1 – 5 Carry One Mark Each

- Q.1 A student is required to demonstrate a high level of comprehension of the subject, especially in the social sciences. (GATE-CE-14)

The word closest in meaning to comprehension is

- (A) understanding (B) meaning (C) concentration (D) stability

Answer: (A)

- Q.2 Choose the most appropriate word from the options given below to complete the following sentence. (GATE-CE-14)

One of his biggest _____ was his ability to forgive.

- (A) vice (B) virtues (C) choices (D) strength

Answer: (B)

- Q.3 Rajan was not happy that Sajan decided to do the project on his own. On observing his unhappiness, Sajan explained to Rajan that he preferred to work independently. Which one of the statements below is logically valid and can be inferred from the above sentences? (GATE-CE-14)

- (A) Rajan has decided to work only in a group.
(B) Rajan and Sajan were formed into a group against their wishes.
(C) Sajan had decided to give in to Rajan's request to work with him.
(D) Rajan had believed that Sajan and he would be working together.

Answer: (D)

- Q.4 If $y = 5x^2 + 3$, then the tangent at $x = 0, y = 3$ (GATE-CE-14)

- (A) passes through $x = 0, y = 0$ (B) has a slope of +1
(C) is parallel to the x-axis (D) has a slope of -1

Answer: (C)

Exp: $y = 5x^2 + 3 \Rightarrow \frac{dy}{dx} = 10x$

Slope of tangent = $\left(\frac{dy}{dx}\right)_{x=0, y=3} = 10 \times 0 = 0$

Slope = 0 $\Rightarrow$ tangent is parallel to x-axis

- Q.5 A foundry has a fixed daily cost of Rs 50,000 whenever it operates and a variable cost of Rs 800 Q, where Q is the daily production in tonnes. What is the cost of production in Rs per tonne for a daily production of 100 tonnes? (GATE-CE-14)

Answer: 1300 to 1300

GENERAL APTITUDE

Exp: Fixed cost = Rs. 50,000

Variable cost = Rs. 800Q

Q = daily production in tonnes

For 100 tonnes of production daily, total cost of production = 50,000 + 800 × 100 = 1,30,000

So, cost of production per tonne of daily production

$$= \frac{1,30,000}{100} = \text{Rs. } 1300$$

Q. No. 6 – 10 Carry One Mark Each

Q.6 Find the odd one in the following group: ALRVX, EPVZB, ITZDF, OYEIK

(GATE-CE-14)

(A) ALRVX

(B) EPVZB

(C) ITZDF

(D) OYEIK

Answer: (D)

Exp: ALRVX → only one vowel

EPVZB → only one vowel

ITZDF → only one vowel

OYEIK → three vowels

Q.7 Anuj, Bhola, Chandan, Dilip, Eswar and Faisal live on different floors in a six - storeyed building (the ground floor is numbered 1, the floor above it 2, and so on). Anuj lives on an even-numbered floor. Bhola does not live on an odd numbered floor. Chandan does not live on any of the floors below Faisal's floor. Dilip does not live on floor number 2. Eswar does not live on a floor immediately above or immediately below Bhola. Faisal lives three floors above Dilip. Which of the following floor-person combinations is correct? (GATE-CE-14)

	Anuj	Bhola	Chandan	Dilip	Eswar	Faisal
(A)	6	2	5	1	3	4
(B)	2	6	5	1	3	4
(C)	4	2	6	3	1	5
(D)	2	4	6	1	3	5

Answer: (B)

Exp: (a) Anuj: Even numbered floor (2,4,6)

(b) Bhola: Even numbered floor (2,4,6)

(c) Chandan lives on the floor above that of Faisal.

(d) Dilip: not on 2nd floor.

(e) Eswar: does not live immediately above or immediately below Bhola

From the options its clear, that only option (B) satisfies condition (e).

So, correct Ans is (B).

GENERAL APTITUDE

- Q.8 The smallest angle of a triangle is equal to two thirds of the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is 3:4:5:6. The largest angle of the triangle is twice its smallest angle. What is the sum, in degrees, of the second largest angle of the triangle and the largest angle of the quadrilateral? (GATE-CE-14)

Answer: 180 to 180

Exp: Let the angles of quadrilateral are $3x, 4x, 5x, 6x$

$$\text{So, } 3x + 4x + 5x + 6x = 360$$

$$x = 20$$

$$\text{Smallest angle of quadrilateral} = 3 \times 20 = 60^\circ$$

$$\text{Smallest angle of triangle} = \frac{2}{3} \times 60^\circ = 40^\circ$$

$$\text{Largest angle of triangle} = 2 \times 40^\circ = 80^\circ$$

$$\text{Three angles of triangle are } 40^\circ, 60^\circ, 80^\circ$$

$$\text{Largest angle of quadrilateral is } 120^\circ$$

$$\text{Sum (2}^{\text{nd}} \text{ largest angle of triangle + largest angle of quadrilateral)} \\ = 60^\circ + 120^\circ = 180^\circ.$$

- Q.9 One percent of the people of country X are taller than 6 ft. Two percent of the people of country Y are taller than 6 ft. There are thrice as many people in country X as in country Y. Taking both countries together, what is the percentage of people taller than 6 ft? (GATE-CE-14)

(A) 3.0

(B) 2.5

(C) 1.5

(D) 1.25

Answer: (D)

Exp: Let number of people in country y = 100

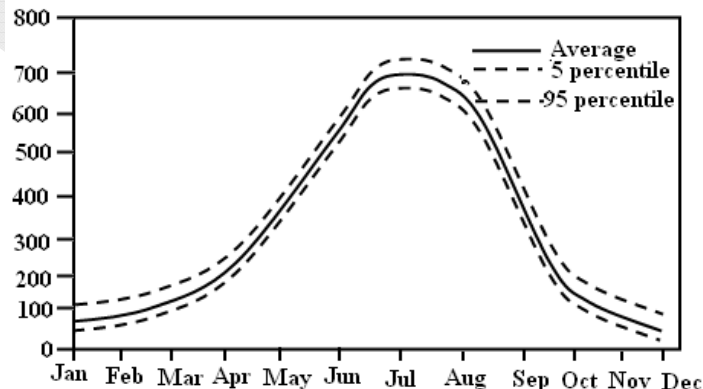
$$\text{So, number of people in country x} = 300$$

Total number of people taller than 6ft in both the countries

$$= 300 \times \frac{1}{100} + 100 \times \frac{2}{100} = 5$$

$$\% \text{ of people taller than 6ft in both the countries} = \frac{5}{400} \times 100 = 1.25\%$$

- Q.10 The monthly rainfall chart based on 50 years of rainfall in Agra is shown in the following figure. Which of the following are true? (k percentile is the value such that k percent of the data fall below that value)



GENERAL APTITUDE

- (i) On average, it rains more in July than in December
 - (ii) Every year, the amount of rainfall in August is more than that in January
 - (iii) July rainfall can be estimated with better confidence than February rainfall
 - (iv) In August, there is at least 500 mm of rainfall
- (A) (i) and (ii) (B) (i) and (iii) (C) (ii) and (iii) (D) (iii) and (iv)

Answer: (B)

Exp: In the question the monthly average rainfall chart for 50 years has been given.
Let us check the options.

- (i) On average, it rains more in July than in December $\Rightarrow$ correct.
 - (ii) Every year, the amount of rainfall in August is more than that in January.
 $\Rightarrow$ may not be correct because average rainfall is given in the question.
 - (iii) July rainfall can be estimated with better confidence than February rainfall.
 $\Rightarrow$ From chart it is clear the gap between 5 percentile and 95 percentile from average is higher in February than that in July $\Rightarrow$ correct.
 - (iv) In August at least 500 mm rainfall
 $\Rightarrow$ May not be correct, because its 50 year average.
- So correct option (B) (i) and (iii).

GENERAL APTITUDE

GATE-CIVIL-2014 PAPER-II

- Q.1 Choose the most appropriate word from the options given below to complete the following sentence. (GATE-CE-14)

A person suffering from Alzheimer's disease _____ short-term memory loss.

- (A) experienced (B) unexperienced
(C) is experiencing (D) experiences

Answer: (D)

- Q.2. Choose the most appropriate word from the options given below to complete the following sentence. _____ is the key to their happiness; they are satisfied with what they have. (GATE-CE-14)

- (A) Contentment (B) Ambition
(C) Perseverance (D) Hunger

Answer: (A)

- Q.3. Which of the following options is the closest in meaning to the sentence below? "As a woman, I have no country." (GATE-CE-14)

- (A) Women have no country
(B) Women are not citizens of any country.
(C) Women's solidarity knows no national boundaries
(D) Women of all countries have equal legal rights.

Answer: (C)

- Q.4 In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is _____ years. (GATE-CE-14)

Answer: 25 to 25

Exp:

$$P = 0.04 = \frac{4}{100}$$

For 1 earth quake

$$\frac{100}{4} P = 1 \text{ earth quake}$$

25 years

Reverse Probability

- Q.5 The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate? (GATE-CE-14)
- (A) 3-4 years (B) 4-5 years (C) 5-6 years (D) 6-7 years

Answer: (A)



GENERAL APTITUDE

Exp:

$$\frac{20}{140} \times 8$$

After 1 year

$$P = 6$$

$$2 \text{ years} = 7.2$$

$$\text{After 3} = \frac{20}{100} \times 1.2$$
$$= 8.65$$

$$\text{After 4 years} = \frac{20}{100} \times 8.65 = 10$$

Time will be in between 3–4 years.

Q. No. 6 – 10 Carry One Mark Each

- Q6. In a group of four children, Som is younger to Riaz. Shiv is elder to Ansu. Ansu is youngest in the group. Which of the following statements is/are required to find the eldest child in the group?

Statements :

1. Shiv is younger to Riaz.

2. Shiv is elder to Som.

(A) Statement 1 by itself determines the eldest child.

(B) Statement 2 by itself determines the eldest child.

(C) Statement 1 and 2 are both required to determine the eldest child.

(D) Statement 1 and 2 are not sufficient to determine the eldest child.

Answer: (A)

- Q7. Moving into a world of big data will require us to change our thinking about the merits of exactitude. To apply the conventional mindset of measurement to the digital, connected world of the twenty-first century is to miss a crucial point. As mentioned earlier, the obsession with exactness is an artefact of the information-deprived analog era. When data was sparse, every data point was critical, and thus great care was taken to avoid letting any point bias the analysis. *From "BIG DATA" Viktor Mayer-Schonberger and Kenneth Cukier*
The main point of the paragraph is: **(GATE-CE-14)**

(A) The twenty-first century is a digital world

(B) Big data is obsessed with exactness

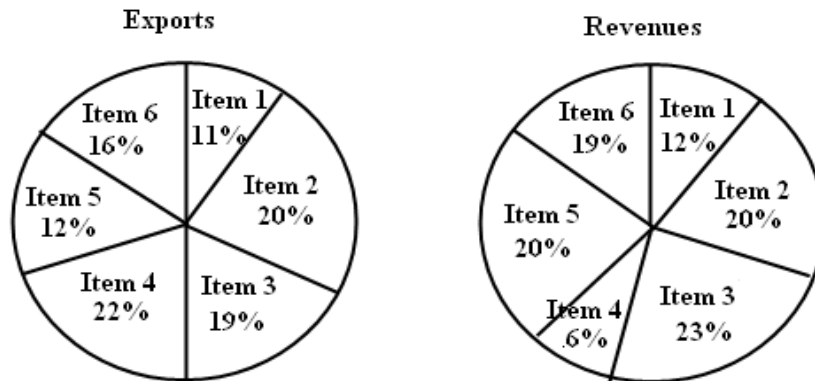
(C) Exactitude is not critical in dealing with big data

(D) Sparse data leads to a bias in the analysis

Answer: (C)

GENERAL APTITUDE

- Q.8 The total exports and revenues from the exports of a country are given in the two pie charts below. The pie chart for exports shows the quantity of each item as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 5 lakh tonnes and the total revenues are 250 crore rupees. What is the ratio of the revenue generated through export of Item 1 per kilogram to the revenue generated through export of Item 4 per kilogram? (GATE-CE-14)



- (A) 1:2 (B) 2:1 (C) 1:4 (D) 4:1

Answer: (D)

Exp:

revenue generated through export of item 1 Kg

$$\Rightarrow \frac{\text{Item}}{\text{Quantity}} = \frac{11}{100} \times 5 = \frac{11}{20} \text{ (lakhs tons)}$$

$$\text{Revenue gen Item 1} = \frac{12}{100} \times 6 \times 250 \times (C) = \frac{30cr}{11} \times 20 \dots \dots \dots (1)$$

$$\begin{aligned} \text{Revenue Item 4} &= \frac{6}{100} \times 250 (C) \\ &= \frac{15 cr}{22} \times 20 Lt \dots \dots \dots (2) \end{aligned}$$

$$\frac{1:2}{\frac{30}{11} \times \frac{20 \times 22}{15 \times 20}} = 4:1$$

- Q.9 X is 1 km northeast of Y. Y is 1 km southeast of Z. W is 1 km west of Z. P is 1 km south of W. Q is 1 km east of P. What is the distance between X and Q in km? (GATE-CE-14)
- (A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

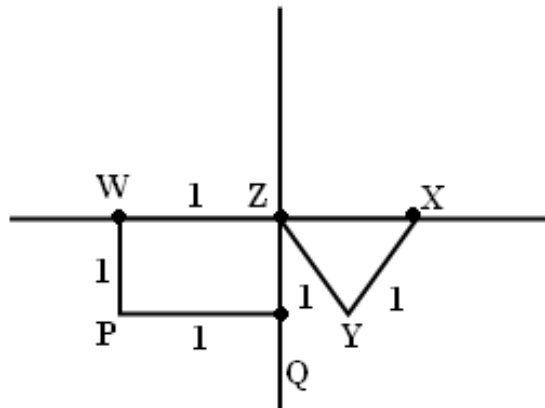
Answer : (C)

Exp: From the fig: $zx = \sqrt{2}$. [Pythagoras theorem] $zQ = 1$ Given

$\Rightarrow$ Considering ZQX, which is right angle, is

$$\begin{aligned} \Rightarrow Qx^2 &= ZQ^2 + Zx^2 \\ &= \sqrt{1+2} \\ &= \sqrt{3} \end{aligned}$$

GENERAL APTITUDE



Q.10. 10% of the population in a town is HIV+. A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV+ individuals 95% of the time, and HIV- individuals 89% of the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is _____

(GATE-CE-14)

Answer: 0.48 to 0.49

Exp: Let total population = 100

HIV + patients = 10

For the patient to be +Ve, should be either +Ve and test is showing +Ve or the patient should

be - Ve but test is showing +Ve

$$\Rightarrow \frac{0.1 \times 0.95}{0.1 \times 0.95 + 0.9 \times 0.11}$$

GENERAL APTITUDE

GATE-CSE-2014 PAPER-1

1. Which of the following options is the closest in meaning to the phrase underlined in the sentence below? It is fascinating to see life forms cope with varied environmental conditions. (GATE-CS-14)
(A) Adopt to (B) Adapt to (C) Adept in (D) Accept with

Answer: (B)

2. Choose the most appropriate word from the options given below to complete the following sentence. He could not understand the judges awarding her the first prize, because he thought that her performance was quite _____. (GATE-CS-14)
(A) Superb (B) Medium (C) Mediocre (D) Exhilarating

Answer: (C)

3. In a press meet on the recent scam, the minister said, "The buck stops here". What did the minister convey by the statement? (GATE-CS-14)
(A) He wants all the money (B) He will return the money
(C) He will assume final responsibility (D) He will resist all enquiries

Answer: (C)

4. If $(z+1/z)^2=98$, compute $(z^2+1/z^2) \left(z+\frac{1}{z}\right)^2 = z^2 + \frac{1}{z^2} + 2 \cdot z \cdot \frac{10}{z}$

Answer: (96)

Exp: Expanding

$$z^2 + \frac{1}{z^2} + 2 \cdot z \cdot \frac{1}{z} = 98 \Rightarrow z^2 + \frac{1}{z^2} = 96$$

5. The roots of $ax^2 + bx + c = 0$ are real and positive a, b and c are real. Then $ax^2 + b|x| + c = 0$ Has (GATE-CS-14)
(A) No roots (B) 2 real roots (C) 3 real roots (D) 4 real roots

Answer: (D)

Exp: $ax^2+bx+c=0$

for roots to be real & +ve

$$b^2-4ac>0$$

This will have 2 real positive roots.

$$Ax^2 + b|x| + c = 0$$

This can be written as;

$$Ax^2 + bx + c$$

$$\text{Discriminant } b^2 - 4ac > 0$$



GENERAL APTITUDE

$$Ax^2 - bx + c$$

$$(-b)^2 - 4ac$$

$$\Rightarrow b^2 - 4ac$$

Is also > 0 . This will have real roots

$\Rightarrow$ This will have 4 real roots

Q.No. 6 – 10 Carry One Mark Each

6. The Palghat Gap (or Palakkad Gap), a region about 30 km wide in the southern part of the Western Ghats in India, is lower than the hilly terrain to its north and south. The exact reasons for the formation of this gap are not clear. It results in the neighbouring regions of Tamil Nadu getting more rainfall from the South West monsoon and the neighbouring regions of Kerala having higher summer temperatures. What can be inferred from this passage?
- (A) The Palghat gap is caused by high rainfall and high temperatures in southern Tamil Nadu and Kerala
- (B) The regions in Tamil Nadu and Kerala that are near the Palghat Gap are low-lying
- (C) The low terrain of the Palghat Gap has a significant impact on weather patterns in neighbouring parts of Tamil Nadu and Kerala
- (D) Higher summer temperatures result in higher rainfall near the Palghat Gap area

Answer: (C)

7. Geneticists say that they are very close to confirming the genetic roots of psychiatric illnesses such as depression and schizophrenia, and consequently, that doctors will be able to eradicate these diseases through early identification and gene therapy.

(GATE-CS-14)

On which of the following assumptions does the statement above rely?

- (A) Strategies are now available for eliminating psychiatric illnesses
- (B) Certain psychiatric illnesses have a genetic basis
- (C) All human diseases can be traced back to genes and how they are expressed
- (D) In the future, genetics will become the only relevant field for identifying psychiatric illnesses

Answer: (B)

8. Round-trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is Rs 100, a group of 5 tourists purchasing round-trip tickets will be charged Rs _____

(GATE-CS-14)

Answer: (850)

Exp: One way fare = 100

Two way fare per person = 200

5 persons = 1000/-

Total discount applicable = 10 + 5 = 15%



GENERAL APTITUDE

$$\text{Discount amount} = \frac{15}{100} \times 1000 = 150$$

$$\text{Amount to be paid} = 1000 - 150 = 850$$

9. In a survey, 300 respondents were asked whether they own a vehicle or not. If yes, they were further asked to mention whether they own a car or scooter or both. Their responses are tabulated below. What percent of respondents do not own a scooter? (GATE-CS-14)

		Men	Women
Own vehicle	Car	40	34
	Scooter	30	20
	Both	60	46
Do not own vehicle		20	50

Answer: (48)

Exp: Total respondents=300

Those who don't have scooter

$$\Rightarrow \text{Men} = 40 + 20 = 60$$

$$\text{Women} = 34 + 50 = \frac{84}{144}$$

$$\% = \frac{144}{300} \times 100 = 48\%$$

10. When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines? (GATE-CS-14)

Answer: (6)

GENERAL APTITUDE

GATE-CSE-2014 PAPER-II

1. Choose the most appropriate phrase from the options given below to complete the following sentence. (GATE-CS-14)

India is a post-colonial country because

- (A) it was a former British colony
- (B) Indian Information Technology professionals have colonized the world
- (C) India does not follow any colonial practices
- (D) India has helped other countries gain freedom

Answer: (A)

2. Who _____ was coming to see us this evening? (GATE-CS-14)
(A) you said (B) did you say (C) did you say that (D) had you said

Answer: (B)

3. Match the columns. (GATE-CS-14)

Column 1	Column 2
(1) eradicate	(P) misrepresent
(2) distort	(Q) soak completely
(3) saturate	(R) use
(4) utilize	(S) destroy utterly

(A) 1:S, 2:P, 3:Q, 4:R

(B) 1:P, 2:Q, 3:R, 4:S

(C) 1:Q, 2:R, 3:S, 4:P

(D) 1:S, 2:P, 3:R, 4:Q

Answer: (A)

4. What is the average of all multiples of 10 from 2 to 198? (GATE-CS-14)
(A) 90 (B) 100 (C) 110 (D) 120

Answer: (B)

Exp:

$$10 + 190 \rightarrow 200$$

$$20 - 180 \rightarrow$$

$$:$$

$$\Rightarrow \frac{[(200) \times 9 + 100]}{19} = \frac{1900}{19} = 100$$

$$90 - 110$$

$$100$$



GENERAL APTITUDE

5. The value of $\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$ is (GATE-CS-14)
- a) 3.464 b) 3.932 c) 4.000 d) 4.444

Answer: (c)

Exp:

$$\begin{aligned}
 \text{Let } & \sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}} = y \\
 \Rightarrow & \sqrt{12 + y} = y \\
 \Rightarrow & 12 + y = y^2 \\
 \Rightarrow & (y + 3)(y - 4) = 0 \\
 \Rightarrow & y = -3, y = 4
 \end{aligned}$$

$\sqrt{12 + \sqrt{12 + \sqrt{12 + \dots}}}$

In given problem base written as product of consecutive numbers & in given question “+” symbol is there so biggest number answer.

Formula :

$$\begin{aligned}
 \sqrt{a + \sqrt{a + \dots}} &= \frac{1 + \sqrt{1 + 4a}}{2} \\
 &= \frac{1 + \sqrt{1 + 4(12)}}{2} \\
 &= \frac{1 + 7}{2} = \frac{8}{2} = 4
 \end{aligned}$$

Q. No. 6 – 10 Carry Two Marks Each

6. The old city of Koenigsberg, which had a German majority population before World War 2, is now called Kaliningrad. After the events of the war, Kaliningrad is now a Russian territory and has a predominantly Russian population. It is bordered by the Baltic Sea on the north and the countries of Poland to the south and west and Lithuania to the east respectively. Which of the statements below can be inferred from this passage? (GATE-CS-14)
- (A) Kaliningrad was historically Russian in its ethnic make up
 (B) Kaliningrad is a part of Russia despite it not being contiguous with the rest of Russia
 (C) Koenigsberg was renamed Kaliningrad, as that was its original Russian name
 (D) Poland and Lithuania are on the route from Kaliningrad to the rest of Russia

Answer: (B)

GENERAL APTITUDE

7. The number of people diagnosed with dengue fever (contracted from the bite of a mosquito) in north India is twice the number diagnosed last year. Municipal authorities have concluded that measures to control the mosquito population have failed in this region. **(GATE-CS-14)** Which one of the following statements, if true, does not contradict this conclusion?
- (A) A high proportion of the affected population has returned from neighbouring countries where dengue is prevalent
- (B) More cases of dengue are now reported because of an increase in the Municipal Office's administrative efficiency
- (C) Many more cases of dengue are being diagnosed this year since the introduction of a new and effective diagnostic test
- (D) The number of people with malarial fever (also contracted from mosquito bites) has increased this year

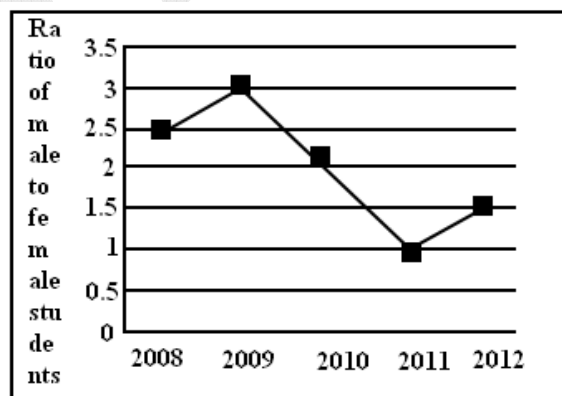
Answer: (D)

8. If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include **(GATE-CS-14)**
- (A) 2, 4 (B) 2, 14 (C) 4, 52 (D) 14, 52

Answer: (D)

Exp: $x^2 - 2x + 3 = 11$
 $(x-4)(x+2) = 0 \Rightarrow x = 4, x = -2$
Values of $|-x^3 + x^2 - x|$
For $x = 4$
Value = 52
for $x = -2$
Value = 14

9. The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009? **(GATE-CS-14)**



Answer: (140)

GENERAL APTITUDE

Exp:

$$\frac{m}{f} = 3$$

$$M = 6f$$

$$\frac{m}{f} = 2.5 \text{ m} = 2.5f$$

$$\frac{m}{2f} = 3$$

$$= \frac{m - m}{m}$$

$$\% \uparrow = \frac{3.5f}{2.5f} \times 100$$

$$= \frac{7}{8} = 1.4$$

$$\% \uparrow = 140\%$$

10. At what time between 6 *a.m.* and 7 *a.m.* will the minute hand and hour hand of a clock make an angle closest to 60° ? **(GATE-CS-14)**
 (A) 6: 22 *a. m.* (B) 6:27 *a.m.* (C) 6: 38 *a.m.* (D) 6:45 *a.m.*

Answer: (A)

Exp: Angle by minute's hand

$$60 \text{ min} \rightarrow 360^\circ$$

$$1 \text{ min} \rightarrow \frac{360}{60} = 6^\circ$$

$$8 \text{ min} \rightarrow 48^\circ$$

Angle $\rightarrow 48^\circ$ with number '6'

Angle by hours hand

$$60 \text{ min} = 30^\circ$$

$$22 \text{ min} \rightarrow \frac{30}{60} \times 22$$

$$\text{Total Angle} = 48 + 11 = 59^\circ$$



GENERAL APTITUDE

GATE-CSE-2014 PAPER-III

1. While trying to collect an envelope from under the table, Mr. X fell down and
I II III
was losing consciousness. (GATE-CS-14)
IV
- Which one of the above underlined parts of the sentence is NOT appropriate?
(A) I (B) II (C) III (D) IV

Answer: (D)

2. If she _____ how to calibrate the instrument, she _____ done the experiment. (GATE-CS-14)
(A) knows, will have (B) knew, had
(C) had known, could have (D) should have known, would have

Answer: (C)

3. Choose the word that is opposite in meaning to the word “coherent”.
(A) sticky (B) well-connected (C) rambling (D) friendly

Answer: (C)

4. Which number does not belong in the series below?
2, 5, 10, 17, 26, 37, 50, 64
(A) 17 (B) 37 (C) 64 (D) 26

Answer: (C)

Explanation : Logic : n^2+1 , so 64 not satisfied
Remaining everything
 n^2+1 .

5. The table below has question-wise data on the performance of students in an examination. The marks for each question are also listed. There is no negative or partial marking in the Examination.

Q.No	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

What is the average of the marks obtained by the class in the examination?

- (A) 1.34 (B) 1.74 (C) 3.02 (D) 3.91

Answer: (C)



GENERAL APTITUDE

Exp: Total question

$$44 \times 2 = 88$$

$$44 \times 3 = 132$$

$$\frac{144}{132} = \frac{88}{308}$$

$$\text{Total marks obtained} = (21 \times 2) + (15 \times 3) + (23 \times 2) = 133$$

$$\text{Total Number of students} = 44$$

$$\text{Average} = \frac{133}{44} = 3.02$$

Q. No. 6 – 10 Carry One Mark Each

6. A dance programme is scheduled for 10.00 a.m. Some students are participating in the programme and they need to come an hour earlier than the start of the event. These students should be accompanied by a parent. Other students and parents should come in time for the programme. The instruction you think that is appropriate for this is **(GATE-CS-14)**
- (A) Students should come at 9.00 a.m. and parents should come at 10.00 a.m.
(B) Participating students should come at 9.00 a.m. accompanied by a parent, and other parents and students should come by 10.00 a.m.
(C) Students who are not participating should come by 10.00 a.m. and they should not bring their parents. Participating students should come at 9.00 a.m.
(D) Participating students should come before 9.00 a.m. Parents who accompany them should come at 9.00 a.m. All others should come at 10.00 a.m.

Answer: (B)

7. By the beginning of the 20th century, several hypotheses were being proposed, suggesting a paradigm shift in our understanding of the universe. However, the clinching evidence was provided by experimental measurements of the position of a star which was directly behind our sun. **(GATE-CS-14)**
- Which of the following inference(s) may be drawn from the above passage?
- (i) Our understanding of the universe changes based on the positions of stars
(ii) Paradigm shifts usually occur at the beginning of centuries
(iii) Stars are important objects in the universe
(iv) Experimental evidence was important in confirming this paradigm shift
- (A) (i), (ii) and (iv) (B) (iii) only (C) (i) and (iv) (D) (iv) only

Answer: (D)

8. The Gross Domestic Product (GDP) in Rupees grew at 7% during 2012-2013. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period 2012-2013 the exchange rate for the USD increased from Rs. 50/ USD to Rs. 60/ USD. India's GDP in USD during the period 2012 - 2013. **(GATE-CS-14)**
- (A) increased by 5 % (B) decreased by 13%
(C) decreased by 20% (D) decreased by 11%

Answer: (D)



GENERAL APTITUDE

Exp: Per 100 Rs final value 107 Rs

$$\Rightarrow \text{per } \frac{100}{50} \text{ Dollars final value } \frac{107}{60}$$

For 100 dollars _____ ?

$$= \frac{100 \times 50}{100} \times \frac{107}{60} = 89.16$$

Decreased by 11%.

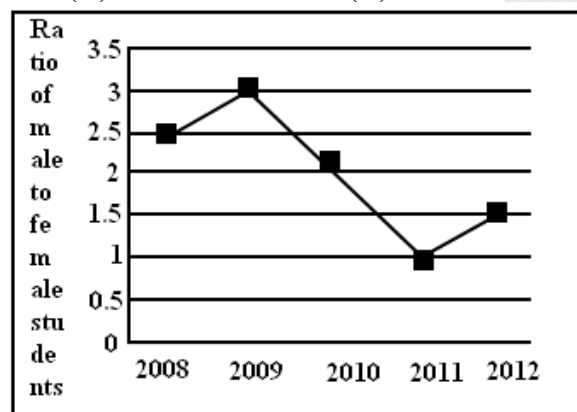
9. The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students in 2011 and 2012 is equal, what is the ratio of male students in 2012 to male students in 2011? **(GATE-CS-14)**

(A) 1:1

(B) 2:1

(C) 1.5:1

(D) 2.5:1



Answer: (C)

Exp: Take number of female students in 2011=100

$\therefore$ Number of male in 2011=100

No. of female in 2012=100

No. of male in 2012=150

$$\text{Ratio} = \frac{150}{100} = 15:10 = 3:2$$

10. Consider the equation: $(7526)_8 - (Y)_8 = (4364)_8$, where (X) N stands for X to the base N. Find Y. **(GATE-CS-14)**

(A) 1634

(B) 1737

(C) 3142

(D) 3162

Answer: (C)

Logic:- octal system

$$\begin{array}{r} 7526(8) \\ -4364(8) \\ \hline 3142(8) \end{array}$$

Otherwise everything change into decimal system, finally change in to octal

GENERAL APTITUDE

GATE-ECE-2014 PAPER-I

1. Choose the most appropriate phrase from the options given below to complete the following sentence. The aircraft _____ take off as soon as its flight plan was filed. (GATE-EC-14)

(A) Is allowed to (B) will be allowed to
(C) Was allowed to (D) has been allowed to

Answer: (C)

2. Read the statements:
All women are entrepreneurs.
Some women are doctors
Which of the following conclusions can be logically inferred from the above statements? (GATE-EC-14)

(A) All women are doctors (B) All doctors are entrepreneurs
(C) All entrepreneurs are women (D) Some entrepreneurs are doctors

Answer: (D)

3. Choose the most appropriate word from the options given below to complete the following Sentence. Many ancient cultures attributed disease to supernatural causes. However, modern science has largely helped _____ such notions. (GATE-EC-14)
- (A) impel (B) dispel (C) propel (D) repel

Answer: (B)

4. The statistics of runs scored in a series by four batsmen are provided in the following table, Who is the most consistent batsman of these four?

Batsman	Average	Standard deviation
K	31.2	5.21
L	46.0	6.35
M	54.4	6.22
N	17.9	5.90

(A) K (B) L (C) M (D) N

Answer: (A)

Exp: If the standard deviation is less, there will be less deviation or batsman is more consistent

GENERAL APTITUDE

5. What is the next number in the series?

12 35 81 173 357 _____

Answer: 725

Exp:



Q. No. 6 – 10 Carry One Mark Each

6. Find the odd one from the following group:

W,E,K,O

I,Q,W,A

F,N,T,X

N,V,B,D

(A) W,E,K,O

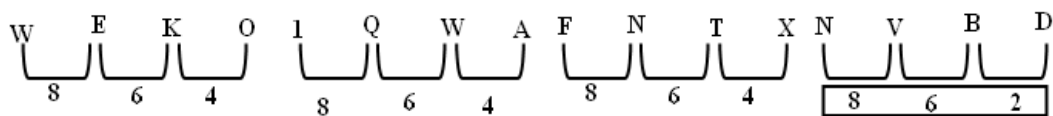
(B) I,Q,W,A

(B) F,N,T,X

(D) N,V,B,D

Answer: (D)

Exp:



Difference of position: D

7. For submitting tax returns, all resident males with annual income below Rs 10 lakh should fill up Form P and all resident females with income below Rs 8 lakh should fill up Form All people with incomes above Rs 10 lakh should fill up Form R, except non residents with income above Rs 15 lakhs, who should fill up Form S. All others should fill Form T. An example of a person who should fill Form T is

(A) a resident male with annual income Rs 9 lakh

(B) a resident female with annual income Rs 9 lakh

(C) a non-resident male with annual income Rs 16 lakh

(D) a non-resident female with annual income Rs 16 lakh

Answer: (B)

Exp: Resident female in between 8 to 10 lakhs haven't been mentioned.

8. A train that is 280 metres long, travelling at a uniform speed, crosses a platform in 60 seconds and passes a man standing on the platform in 20 seconds. What is the length of the platform in metres?

Answer: 560

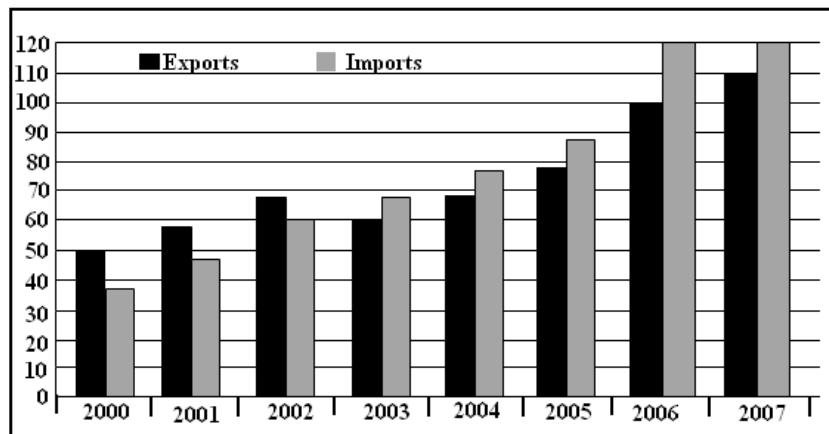


GENERAL APTITUDE

Exp: For a train to cross a person, it takes 20 seconds for its 280m.

So, for second 60 seconds. Total distance travelled should be 840. Including 280 train length so length of plates = 840-280=560

9. The exports and imports (in crores of Rs.) of a country from 2000 to 2007 are given in the following bar chart. If the trade deficit is defined as excess of imports over exports, in which year is the trade deficit $\frac{1}{5}$ th of the exports?



- (A) 2005 (B) 2004 (C) 2007 (D) 2006

Answer: (D)

Exp:

$$2004, \frac{\text{imports} - \text{exports}}{\text{exports}} = \frac{10}{70} = \frac{1}{7}$$

$$2005, \frac{26}{76} = \frac{2}{7}$$

$$2006, \frac{20}{100} = \frac{1}{5}$$

$$2007, \frac{10}{100} = \frac{1}{10}$$

10. You are given three coins: one has heads on both faces, the second has tails on both faces and the third has a head on one face and a tail on the other. You choose a coin at random and toss it, and it comes up heads. The probability that the other face is tails is

(GATE-EC-14)

- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Answer: (B)

GENERAL APTITUDE

GATE-ECE-2014 PAPER-II

1. Choose the most appropriate word from the options given below to complete the following sentence. Communication and interpersonal skills are _____ important in their own ways. (GATE-EC-14)
- (A) each (B) both (C) all (D) either

Answer: (B)

2. Which of the options given below best completes the following sentence?
She will feel much better if she _____. (GATE-EC-14)
- (A) will get some rest (B) gets some rest
(C) will be getting some rest (D) is getting some rest

Answer: (B)

3. Choose the most appropriate pair of words from the options given below to complete the following sentence. (GATE-EC-14)
- She could not _____ the thought of _____ the election to her bitter rival.
- (A) bear, loosing (B) bare, loosing (C) bear, losing (D) bare, losing

Answer: (C)

4. A regular die has six sides with numbers 1 to 6 marked on its sides. If a very large number of throws show the following frequencies of occurrence: 1@0.167; 2@0.167; 3@0.152; 4@0.166; 5@0.168; 6@0.180. We call this die (GATE-EC-14)
- (A) irregular (B) biased (C) Gaussian (D) insufficient

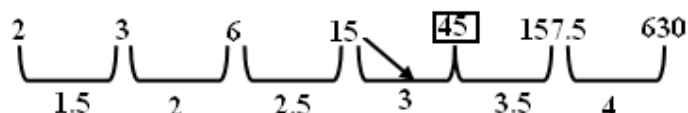
Answer: (B)

Exp: For a very large number of throws, the frequency should be same for unbiased throw. As it not same, then the die is biased.

5. Fill the missing number in the series. (GATE-EC-14)
- 2, 3, 6, 15, _____, 157.5, 630

Answer: 45

Exp:



$\frac{\text{2nd number}}{\text{1st number}}$ is in increasing order as shown above

Logic : Multiplication

Multiplication increased by .5 Decimal 5

GENERAL APTITUDE

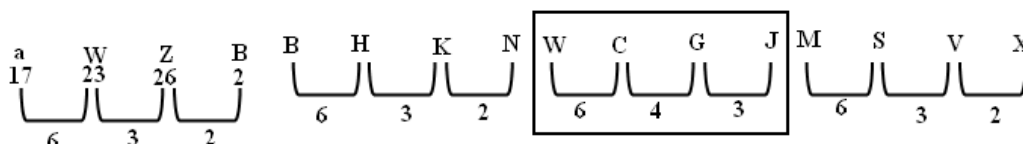
Q. No. 6 – 10 Carry One Mark Each

6. Find the odd one in the following group

- Q,W,Z,B B,H,K,M W,C,G,J M,S,V,X
 (A) Q,W,Z,B (B) B,H,K,M (C) W,C,G,J (D) M,S,V,X

Answer: (C)

Exp:



7. Lights of four colors (red, blue, green, yellow) are hung on a ladder. On every step of the ladder there are two lights. If one of the lights is red, the other light on that step will always be blue. If one of the lights on a step is green, the other light on that step will always be yellow. Which of the following statements is not necessarily correct? **(GATE-EC-14)**

- (A) The number of red lights is equal to the number of blue lights
 (B) The number of green lights is equal to the number of yellow lights
 (C) The sum of the red and green lights is equal to the sum of the yellow and blue lights
 (D) The sum of the red and blue lights is equal to the sum of the green and yellow lights

Answer: (D)

8. The sum of eight consecutive odd numbers is 656. The average of four consecutive even numbers is 87. What is the sum of the smallest odd number and second largest even number? **(GATE-EC-14)**

Answer: 163

Exp: Eight consecutive odd number = 656

$$a-6, a-4, a-2, a, a+2, a+4, a+6$$

$$a+8=656$$

$$a=81$$

$$\text{Smallest } m=75 \dots (1)$$

Average consecutive even numbers

$$\Rightarrow \frac{a-2 + a + a+2 + a+4}{4} = 87$$

$$\Rightarrow a = 86$$

$$\text{Second largest number} = 88$$

$$1+2=163$$

$$\text{Sum of '8' consecutive odd numbers} = 656$$

$$\text{So Avg} = \frac{656}{8} = 82$$

Here not terms even. So we got '2' middle number i.e. $82-1.82+1$

$$75, 77, 79, 81, 83, 85, 87, 89$$

$$\text{Smallest odd} = 75$$

GENERAL APTITUDE

II. Average of '4' consecutive even = 87, so middle two numbers 87-1, 87+1, 84, 86, 88, 90

Sum of smallest odd numbers and second largest even = $75 + 88 = 163$

Concept : if Avg = K

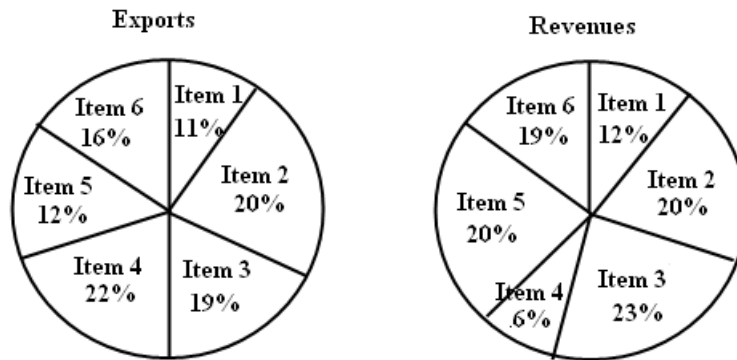
Average A.P., if no of Terms odd = then middle no is called Average.

If no of terms = Even

We got '2' middle number. i.e. K-1, K+1

9. The total exports and revenues from the exports of a country are given in the two charts shown below. The pie chart for exports shows the quantity of each item exported as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 500 thousand tonnes and the total revenues are 250 crore rupees. Which item among the following has generated the maximum revenue per kg?

(GATE-EC-14)



(A) Item 2

(B) Item 3

(C) Item 6

(D) Item 5

Answer: (D)

Exp:

Item: 2

$$0.5 \times 10^4 = 5 \times 10^3 \text{ t} = \text{Item 2}$$

Item: 3

$$1.2 = \text{Item 3}$$

Item : 5

Item : 6

$$\frac{20}{100} \times 250 \times 10^7$$

$$\frac{20}{100} \times 500 \times 10^3$$

$$\frac{23 \times 250 \times 10^7}{19 \times 500 \times 10^3}$$

$$\frac{20}{12} = \frac{5}{3} = 1.6 \Rightarrow 1.6 = \text{Item 5}$$

$$\frac{19}{16} = 1.18 = \text{Item 6}$$

GENERAL APTITUDE

10. It takes 30 minutes to empty a half-full tank by draining it at a constant rate. It is decided to simultaneously pump water into the half-full tank while draining it. What is the rate at which water has to be pumped in so that it gets fully filled in 10 minutes? (GATE-EC-14)

(A) 4 times the draining rate (B) 3 times the draining rate
(C) 2.5 times the draining rate (D) 2 times the draining rate

Answer: (A)

Exp: half $V_{\text{half}} = 30(s)$ drawing rate = s

Total volume = $60 S$ tank

$(s^1)(10) - (s)10 = 30s$

$S^1(s) s = 3s$

$s^1 = 4s, s^1 = 4 \text{ drawing rate}$



GENERAL APTITUDE

GATE-ECE-2014 PAPER-III

1. Which of the following options is the closest in meaning to the word underlined in the Sentence below? In a democracy, everybody has the freedom to disagree with the government. (GATE-EC-14)
(A) Dissent (B) Descent (C) Decent (D) Decadent

Answer: A

Exp: Dissent is to disagree

2. After the discussion, Tom said to me, 'Please revert!' He expects me to _____. (GATE-EC-14)
(A) Retract (B) Get back to him
(C) Move in reverse (D) Retreat

Answer: B

Exp: Revert means set back

3. While receiving the award, the scientist said, "I feel vindicated". Which of the following is closest in meaning to the word 'vindicated'? (GATE-EC-14)
(A) Punished (B) Substantiated (C) Appreciated (D) Chastened

Answer: B

Exp: Vindicate has 2 meanings

1. Clear of blain
2. Substantiate, justify

4. Let $f(x,y) = x^n y^m = P$. If x is doubled and y is halved, the new value of f is (GATE-EC-14)
(A) $2^{n-m} P$ (B) $2^{m-n} P$ (C) $2(n-m) P$ (D) $2(m-n) P$

Answer: A

Exp:

$$(2x)^n \times \left(\frac{y}{2}\right)^m = 2^{n-m} \times x^n y^m$$

5. In a sequence of 12 consecutive odd numbers, the sum of the first 5 numbers is 425. What is the sum of the last 5 numbers in the sequence? (GATE-EC-14)

Answer: 495

Exp: Let consecutive odd numbers be $a-10, a-8, a-6, a-4, a-2, a, \dots, a+12$

Sum of 1st 5 number $= 5a - 30 = 425 \Rightarrow a = 91$

Last 5 numbers $= (a+4) + (a+6) + \dots + (a+12)$

$= (95+97+99+101+103) = 495$

GENERAL APTITUDE

Q. No. 6 – 10 Carry Two Marks Each

6. Find the next term in the sequence: 13M, 17Q, 19S, ____ (GATE-EC-14)
(A) 21W (B) 21V (C) 23W (D) 23V

Answer: C

Exp:

13	M
17(13 4)	Q(M 4)
19(17 2)	S(Q 2)
23(19 4)	W(s 4)
$\Rightarrow 23W$	

7. If 'KCLFTSB' stands for 'best of luck' and 'SHSWDG' stands for 'good wishes', which of the following indicates 'ace the exam'? (GATE-EC-14)
(A) MCHTX (B) MXHTC (C) XMHCT (D) XMHTC

Answer: B

Exp: KCLFTSB

Reverse order:

BCS TOF LUCK

Ace the exam

Reverse order should be

MAXE EHT ECA

Looking at the options we have M X H T C

SHSWDG

Reverse order:

GO OD W I S HES

8. Industrial consumption of power doubled from 2000-2001 to 2010-2011. Find the annual rate of increase in percent assuming it to be uniform over the years. (GATE-EC-14)
(A) 5.6 (B) 7.2 (C) 10.0 (D) 12.2

Answer: B

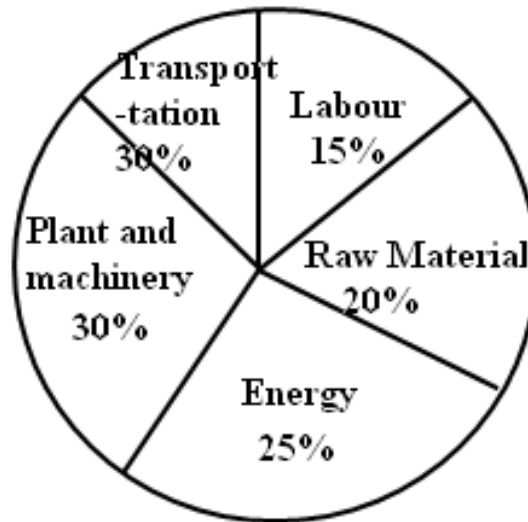
Exp:

$$A = P \left(1 + \frac{r}{100} \right)^n$$
$$A = 2P$$
$$2 = \left(1 + \frac{r}{100} \right)^{10}$$
$$\therefore r = 7.2$$



GENERAL APTITUDE

9. A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is Rs. 4,50,000. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. What is the percentage increase in total cost for the company in 2013? (GATE-EC-14)



Answer: 22%

Exp: Let total cost in 2012 is 100

Raw material increases in 2013 to $1.3 \times 20 = 26$

Other Expenses increased in 2013 to $1.2 \times 80 = 96$

Total Cost in 2013 = $96 + 26 = 122$

Total Cost increased by 22%

Hint: Labour cost (i.e, 4,50,000) in 2012 is redundant data.

10. A five digit number is formed using the digits 1,3,5,7 and 9 without repeating any of them. What is the sum of all such possible five digit numbers? (GATE-EC-14)
- (A) 6666660 (B) 6666600 (C) 6666666 (D) 6666606

Answer: B

Exp: 1 appears in units place in 4! Ways

Similarly all other positions in 4! Ways

Same for other digits.

Sum of all the numbers = $(11111) \times 4! (1+3+5+7+9) = 6666600$

GENERAL APTITUDE

GATE-EIE-2014 PAPER-I

1. Choose the most appropriate word from the options given below to complete the following sentence. (GATE-IN-14)

A person suffering from Alzheimer's disease _____ short-term memory loss.

- (A) experienced (B) has experienced
(C) is experiencing (D) experiences

Answer: D

2. Choose the most appropriate word from the options given below to complete the following sentence. _____ is the key to their happiness; they are satisfied with what they have. (GATE-IN-14)

- (A) Contentment (B) Ambition (C) Perseverance (D) Hunger

Answer: A

3. Which of the following options is the closest in meaning to the sentence below? "As a woman, I have no country." (GATE-IN-14)

- (A) Women have no country.
(B) Women are not citizens of any country.
(C) Women's solidarity knows no national boundaries.
(D) Women of all countries have equal legal rights.

Answer: C

4. In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is _____ years. (GATE-IN-14)

Answer: 25 years.

Exp:

$$P = 0.04 = \frac{4}{100}$$

For 1 earth quake

$$\frac{100}{4} P = 1 \text{ earth quake}$$

25 years

Reverse probability

5. The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate? (GATE-IN-14)
- (A) 3-4 years (B) 4-5 years (C) 5-6 years (D) 6-7 years

Answer: A

GENERAL APTITUDE

Exp:

$$= \frac{20}{140} \times 8$$

After 1 year = P = 6

2 years = 7.2

$$\text{After 3} = \frac{20}{100} \times 1.2 = 8.65$$

$$\text{After 4 years} = \frac{20}{100} \times 8.65 = 10$$

Time will be in between 3–4 years.

Q. 6 – 10 Carry Two Marks Each.

6. In a group of four children, Som is younger to Riaz. Shiv is elder to Ansu. Ansu is youngest in the group. Which of the following statements is/are required to find the eldest child in the group?

(GATE-IN-14)

Statements: 1. Shiv is younger to Riaz.

2. Shiv is elder to Som.

- (A) Statement 1 by itself determines the eldest child.
(B) Statement 2 by itself determines the eldest child.
(C) Statements 1 and 2 are both required to determine the eldest child.
(D) Statements 1 and 2 are not sufficient to determine the eldest child.

Answer: A

7. Moving into a world of big data will require us to change our thinking about the merits of exactitude. To apply the conventional mindset of measurement to the digital, connected world of the twenty-first century is to miss a crucial point. As mentioned earlier, the obsession with exactness is an artefact of the information-deprived analog era. When data was sparse, every data point was critical, and thus great care was taken to avoid letting any point bias the analysis. From “BIG DATA” Viktor Mayer-Schonberger and Kenneth Cukier The main point of the paragraph is:

(GATE-IN-14)

- (A) The twenty-first century is a digital world
(B) Big data is obsessed with exactness
(C) Exactitude is not critical in dealing with big data
(D) Sparse data leads to a bias in the analysis

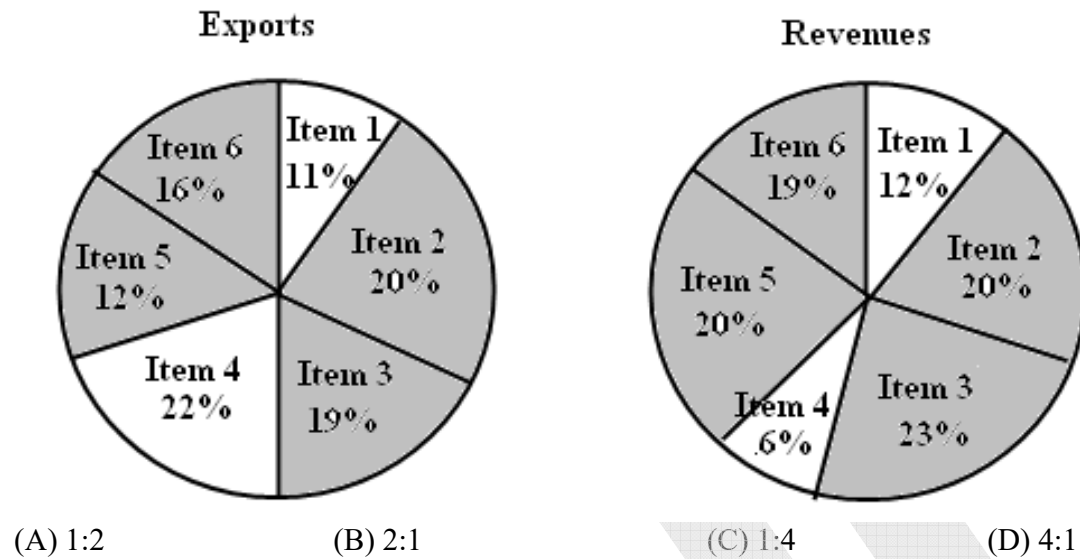
Answer: C

8. The total exports and revenues from the exports of a country are given in the two pie charts below. The pie chart for exports shows the quantity of each item as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 5 lakh tonnes and the total revenues are 250 crore rupees. What is the ratio of the revenue generated through export of Item 1 per kilogram to the revenue generated through export of Item 4 per kilogram?

(GATE-IN-14)



GENERAL APTITUDE



Answer: D

Exp:

revenue generated through export of item 1 Kg

$$\Rightarrow \frac{\text{Item}}{\text{quantity}} = \frac{11}{100} \times 5 = \frac{11}{20} \text{ (lakhs tons)}$$

$$\text{Revenue gen Item 4} = \frac{6}{100} \times 250 \times (C) = \frac{30 \text{ cr}}{11} \times 20 \quad \dots \dots \dots (1)$$

$$\begin{aligned} \text{Revenue gen Item 4} &= \frac{6}{100} \times 250 (C) \\ &= \frac{15 \text{ cr}}{22} \times 20 \text{ Lt} \quad \dots \dots \dots (2) \end{aligned}$$

$$\frac{1}{11} \times \frac{20 \times 22}{15 \times 20} = 4:1$$

9. X is 1 km northeast of Y. Y is 1 km southeast of Z. W is 1 km west of Z. P is 1 km south of W. Q is 1 km east of P. What is the distance between X and Q in km? **(GATE-IN-14)**
- (A) 1 (B) $\sqrt{2}$ (C) $\sqrt{3}$ (D) 2

Answer: C

Exp: From the fig: $ZX = \sqrt{2}$. [Pythagoras theorem $ZQ = 1$ Given]

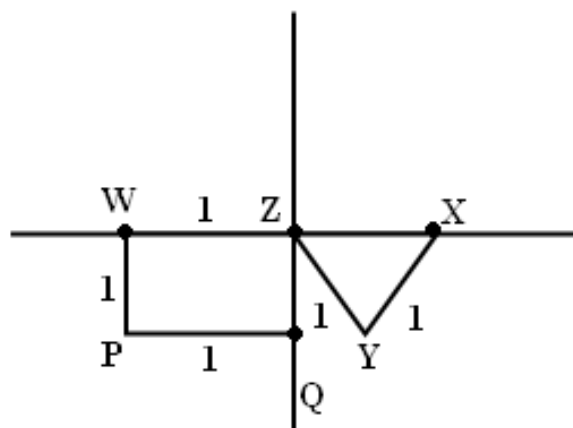
$\Rightarrow$ Considering ZQX , which is right angle, is

$$\Rightarrow QX^2 = ZQ^2 + ZX^2$$

$$= \sqrt{1 + 2}$$

$$= \sqrt{3}$$

GENERAL APTITUDE



10. 10% of the population in a town is HIV+. A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV+ individuals 95% of the time, and HIV- individuals 89% of the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is _____.

(GATE-IN-14)

Answer: 0.48 to 0.49

Exp: Let total population = 100

HIV + patients = 10

For the patient to be +Ve, should be either +VE and test is showing the or the patient should be - Ve but rest is showing +Ve

$$\Rightarrow \frac{0.1 \times 0.95}{0.1 \times 0.95 + 0.9 \times 0.11} = 0.489$$

GENERAL APTITUDE

GATE-MECH-2014 PAPER-I

1. "India is a country of rich heritage and cultural diversity." Which one of the following facts best supports the claim made in the above sentence? (GATE-ME-14)
- (A) India is a union of 28 states and 7 union territories.
(B) India has a population of over 1.1 billion.
(C) India is home to 22 official languages and thousands of dialects.
(D) The Indian cricket team draws players from over ten states.

Answer: C

Exp: Diversity is shown in terms of difference language

2. The value of one U.S. dollar is 65 Indian Rupees today, compared to 60 last year. The Indian Rupee has _____. (GATE-ME-14)
- (A) Depressed (B) Depreciated (C) Appreciated (D) Stabilized

Answer: B

3. 'Advice' is _____. (GATE-ME-14)
- (A) a verb (B) a noun
(C) an adjective (D) both a verb and a noun

Answer: B

4. The next term in the series 81, 54, 36, 24 ... is _____. (GATE-ME-14)

Answer: 16

Exp:

$$81 - 54 = 27; 27 \times \frac{2}{3} = 18$$

$$54 - 36 = 18; 18 \times \frac{2}{3} = 12$$

$$36 - 24 = 12; 12 \times \frac{2}{3} = 8$$

$$\therefore 24 - 8 = 16$$

Logic :
multiplication logic
is $\times \frac{2}{3}$

5. In which of the following options will the expression $P < M$ be definitely true? (GATE-ME-14)
- (A) $M < R > P > S$ (B) $M > S < P < F$
(C) $Q < M < F = P$ (D) $P = A < R < M$

Answer: D

GENERAL APTITUDE

Q. No. 6 – 10 Carry Two Marks Each

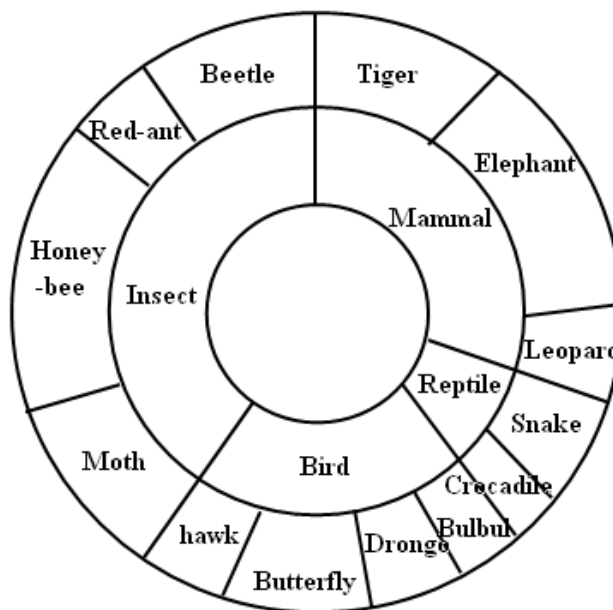
6. Find the next term in the sequence: 7G, 11K, 13M, ____ (GATE-ME-14)

(A) 15Q (B) 17Q (C) 15P (D) 17P

Answer: B

Logic : [Prime Number] . letter.

7. The multi-level hierarchical pie chart shows the population of animals in a reserve forest. The correct conclusions from this information are: (GATE-ME-14)



- (i) Butterflies are birds
(ii) There are more tigers in this forest than red ants
(iii) All reptiles in this forest are either snakes or crocodiles
(iv) Elephants are the largest mammals in this forest
(A) (i) and (ii) only (B) (i), (ii), (iii) and (iv)
(C) (i), (iii) and (iv) only (D) (i), (ii) and (iii) only

Answer: D

Exp: It is not mentioned that elephant is the largest animal

8. A man can row at 8 km per hour in still water. If it takes him thrice as long to row upstream, as to row downstream, then find the stream velocity in km per hour. (GATE-ME-14)

Answer: 4

Exp: Speed of man=8; Left distance =d

$$\text{Time taken} = \frac{d}{8}$$

GENERAL APTITUDE

Upstream:

Speed of stream = s

$\Rightarrow$ speed upstream $S' = (8 - s)$

$$t' = \left(\frac{d}{8 - s} \right)$$

Downstream:

Given speed downstream = $t'' = \frac{d}{8 + s}$

$$\Rightarrow 3t' = t'' \Rightarrow \frac{3d}{8 - s} = \frac{d}{8 + s}$$

$$\Rightarrow \frac{3d}{8 - s} = \frac{d}{8 + s} \Rightarrow s = 4 \text{ km/hr}$$

9. A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is Rs. 4,50,000. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. If the company registered a profit of Rs. 10 lakhs in 2012, at what price (in Rs.) was each air purifier sold? (GATE-ME-14)

Answer: 20,000

Exp: Total expenditure = $\frac{15}{100}x = 4,50,000$

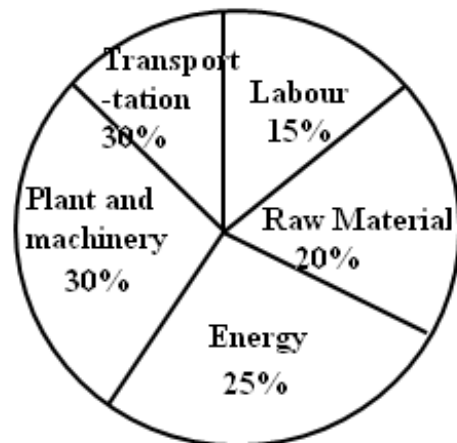
$$x = 3 \times 10^6$$

Profit = 10 lakhs

So, total selling price = 40,00,000 ... (1)

Total purifiers = 200 ... (2)

S.P of each purifier = (1)/(2) = 20,000



10. A batch of one hundred bulbs is inspected by testing four randomly chosen bulbs. The batch is rejected if even one of the bulbs is defective. A batch typically has five defective bulbs. The probability that the current batch is accepted is _____ (GATE-ME-14)

Answer: 0.8145

Exp: Probability for one bulb to be non defective is $\frac{95}{100}$

$\therefore$ Probabilities that none of the bulbs is defectives $\left(\frac{95}{100} \right)^4 = 0.8145$

GENERAL APTITUDE

ECE, EEE - 2013

Q. 01 Q.05 carry one mark each

01. They were requested not to quarrel with others (GATE-EC/EE-13)
Which one of the following options is the closest in meaning to the word quarrel ?

(a) make out (b) call out (c) dig out (d) fall out

ANS: D

02. In the summer of 2012, in New Delhi, the mean temperature of Monday to Wednesday was 41°C and of Tuesday to Thursday was 43°C . If the temperature on Thursday was 15°C higher than that of Monday, then the temperature in $^{\circ}\text{C}$ on Thursday was (GATE-EC/EE-13)

(a) 40 (b) 43 (c) 46 (d) 49

ANS: (c)

$$\frac{M + \text{Tue} + W}{3} = 41$$
$$\Rightarrow M + \text{Tue} + W = 123$$

$$\frac{\text{Tue} + M + \text{Th}}{3} = 43$$
$$\Rightarrow \text{Tue} + W + \text{Th} = 129$$

From eq.(ii) – eq.(i)

$$\text{Th} - M = 6$$

Also $\text{Th} = 1.15M$

$$1.15M - M = 6$$

$$\Rightarrow 0.15M = 6$$

$$\Rightarrow M = 40$$

$$\text{Th} = 1.15M$$

$$\Rightarrow \text{Th} = 1.15 \times 40$$

$$\Rightarrow \text{Th} = 46$$

03. Complete the sentence: (GATE-EC/EE-13)
Dare _____ Mistake

(a) Commit (b) to commit (c) committed (d) committing

ANS: A

GENERAL APTITUDE

04. Choose the grammatically CORRECT sentence

(GATE-EC/EE-13)

- (a) Two and two add four
- (b) Two and two become four
- (c) Two and two are four
- (d) Two and two make four

ANS: (d)

This is famous idiom in English.

05. Statement: You can always give me a ring whenever you need

(GATE-EC/EE-13)

Which one of the following is the best inference from the above statement?

- (a) Because I have a nice caller tune
- (b) Because I have a better telephone facility
- (c) Because a friend in need is a friend indeed
- (d) Because you need not pay towards the telephone bills when you give me a ring

ANS: (c)

You can always give me a ring whenever you need implies that I am available even at a single call (whenever you need me). Ring has been used as a metaphor here and not literally.

Q. 06 to Q. 10 carry two marks each

06. What is the chance that a leap year selected at random, will contain 53 Saturdays?

(GATE-EC/EE-13)

- (a) $2/7$
- (b) $3/7$
- (c) $1/7$
- (d) $5/7$

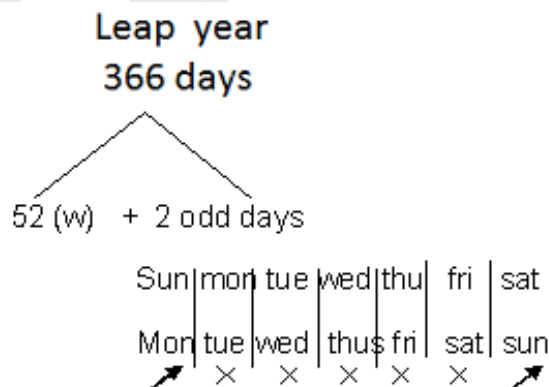
Ans: a

Sol: in a leap year 366 days

I.e 52 weeks and 2 odd days

Probability getting 53 Sundays = $2/7$

Concept wise:



We need 53 sun, we had 52 sun ,so we need another Sunday out of 7 pairs ,2 pairs accepted ,

Probability = $2/7$

GENERAL APTITUDE

07. Statement: There were different streams of freedom movements in colonial India carried out by the moderates, liberals, radicals, socialists and so on (GATE-EC/EE-13)
Which one of the following is the best inference from the above statement?
(a) The emergence of nationalism in colonial India led to our Independence
(b) Nationalism in India emerged in the context of colonialism
(c) Nationalism in India is homogeneous
(d) Nationalism in India is heterogeneous

Ans: B

08. The set of p for which the roots of the equation $3x^2 + 2x + p(p-1) = 0$ are of opposite sign is (GATE-EC/EE-13)
(a) $(-\infty, 0)$ (b) $(0, 1)$ (c) $(1, \infty)$ (d) $(0, \infty)$

Ans: B

Sol: since properties of roots ($ax^2+bx+c=0$)

If roots are negative, then product of roots will be negative
Product of the roots = c/a , result < 0 (that means negative)

Given $3x^2+2x+p(p-1)=0$

$$\text{Product of the roots} = \frac{p(p-1)}{3} < 0$$

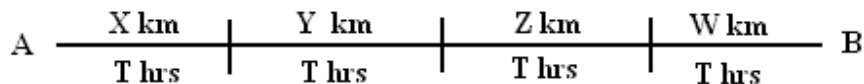
$p(p-1) < 0$. p should be in between 0 to 1 only

That means $0 < p < 1$, simply $p \in (0, 1)$

09. A car travels 8 km in the quarter of an hour, 6 km in the second quarter and 16 km in the third quarter. The average speed of the car in km per hour over the entire journey is (GATE-EC/EE-13)
(a) 30 (B) 36 (C) 40 (D) 24

Ans: C

Sol:



If time is same then avg speed = $\frac{\text{Sum of speeds}}{\text{No of speeds}}$

A car travelled 8 km/quarter ie 32 km/hr

Next quarter 6km, ie 24 km/hr

Next quarter 16km ie 64 km/hr

GENERAL APTITUDE

Here time is equal, then avg speed = $\frac{32+24+64}{3} = 40 \text{ km/hr}$

Another method:

$$\begin{aligned} \text{Avg speed} &= \frac{\text{total distance}}{\text{total time}} \\ &= \frac{8+6+16}{\frac{1}{4}+\frac{1}{4}+\frac{1}{4}} = 40 \text{ km/hr} \end{aligned}$$

10. Find the sum to n terms of the series $10 + 84 + 734 + \dots$

(GATE-EC/EE-13)

- (a) $\frac{9(9^n+1)}{10} + 1$ (b) $\frac{9(9^n-1)}{8} + 1$ (c) $\frac{9(9^n-1)}{8} + n$ (d) $\frac{9(9^n-1)}{8} + n^2$

Ans : d

Sol:

$$\begin{aligned} &10+84+734+\dots+n \text{ terms} \\ &(9+1)+(9^2+3)+(9^3+5)+\dots+n \text{ terms} \\ &(9+9^2+9^3+9^4+\dots+9^n)+(1+3+5+7+\dots \text{odd numbers}) \end{aligned}$$

$$\begin{aligned} &\frac{9(9^n-1)}{9-1} + \text{sum of odd numbers} \\ &\frac{9(9^n-1)}{8} + n^2 \end{aligned}$$

In geometric progression sum of n terms = $\frac{a(r^n-1)}{r-1}$

“r” means common ratio

GENERAL APTITUDE

MECH, CSE -2013

General Aptitude (GA) Questions Q. 01 to Q.05 Carry one mark each.

01. Which one of the following options is the closest in meaning to the word given below?
Nadir
(a) Highest (b) Lowest (c) Medium (d) Integration (GATE-ME/CS-13)

ANS: B

Nadir which is opposite to the word Zenith (the highest point) means the lowest point.

02. Complete the sentence: (GATE-ME/CS-13)
Universalism is to particularism as diffuseness is to _____
(a) Specificity (b) Neutrality (c) generality (d) adaptation

ANS: (a)

Specificity

Universalism and Particularize are opposite words. Similarly .We require a word that indicates an opposite to diffuseness. Diffuseness refers to spreading or causing to spread in all directions .Specificity is an appropriate opposite.

03. What will be the maximum sum of 44, 42, 40,? (GATE-ME/CS-13)
(a) 502 (b) 504 (c) 506 (d) 500

ANS: C

For sum to be maximum, we will consider only positive terms

$$44 + 42 + \dots + 2$$

$$\text{Or } 2 + 4 + 6 \dots + 42 + 44$$

$$2[1 + 2 + 3 \dots + 22]$$

$$= \frac{2[22 \times 23]}{2} = 506$$

Alternate:

Sum of even Natural Numbers

$$2 + 4 + 6 \dots + 44 = n(n + 1)$$

$$22 \times 23 = 506$$

04. Were you a bird, you _____ in the sky (GATE-ME/CS-13)
(a) would fly (b) shall fly (c) should fly (d) shall have flown

ANS: (a)

Would fly

This is a sentence based on Conditional Sentences. In the first clause imagination (Were) has been Used so the main clause should comprise of would +1 form of verb usage.



GENERAL APTITUDE

05. Choose the grammatically **INCORRECT** sentence:

(GATE-ME/CS-13)

- (a) He is Asian origin
- (b) The belonged to Africa
- (c) She is an European
- (d) They migrated from India to Australia

ANS: (c)

She is an European :

The usage of articles and an is not decided by the first alphabet of the word but the sound / pronunciation of the word. European though starts with 'E' which is a vowel but it does not yield a vowel sound. When read aloud it leaves the sound of 'Y' which is a consonant sound. So the correct sentence should be, 'She is a European'.

Q.06 to Q.10 carry two marks each

06. Find the sum of the expression

(GATE-ME/CS-13)

$$\frac{1}{\sqrt{1} + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \dots + \frac{1}{\sqrt{80} + \sqrt{81}}$$

- (a) 7
- (b) 8
- (c) 9
- (d) 10

ANS: (b)

The question can be written as

$$\frac{1}{\sqrt{2} + \sqrt{1}} + \frac{1}{\sqrt{3} + \sqrt{2}} + \frac{1}{\sqrt{4} + \sqrt{3}} + \dots + \frac{1}{\sqrt{81} + \sqrt{80}}$$

Rationalize each term by multiplying it with its conjugate in both numerator and denominator.

$$\left[\frac{1}{\sqrt{2} + \sqrt{1}} \times \frac{\sqrt{2} - \sqrt{1}}{\sqrt{2} - \sqrt{1}} \right] \times \left[\frac{1}{\sqrt{3} + \sqrt{2}} \times \frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} - \sqrt{2}} \right] \times \dots \times \left[\frac{1}{\sqrt{81} + \sqrt{80}} \times \frac{\sqrt{81} - \sqrt{80}}{\sqrt{81} - \sqrt{80}} \right]$$

Which can be resolved as

$$[\sqrt{2} - \sqrt{1}] + [\sqrt{3} - \sqrt{2}] + \dots + [\sqrt{81} - \sqrt{80}]$$

(as denominator in each term will become 1) now cancelling like terms we will be left with

$$\sqrt{81} - \sqrt{1} = (9 - 1) = 8$$

1.49 (b)

$$\text{Now cost} = 13200 \left[1 + \frac{1}{5} \right] \left[1 - \frac{1}{24} \right]$$

$$= 13200 \left[\frac{6}{5} \right] \times \left[\frac{23}{24} \right] = 15,180$$

GENERAL APTITUDE

07. Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7 ?

(GATE-ME/CS-13)

- (a) 13/90 (b) 12/90 (c) 78/90 (d) 77/90

ANS: (d)

There are Ninety (90) two digit numbers between 1 and 100 (typically 10 to 99).

Among these numbers which are divisible by 7 are (10, 21 --- 98) only 13 numbers.

$$\left[\because \left[\frac{100}{7} \right] - 1 = 13 \right]$$

So the required probability that the number is not divisible by

$$1 - \frac{13}{90} = \frac{77}{90}$$

08. After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

(GATE-ME/CS-13)

Which one of the following assertions is best supported by the above information?

- (a) Failure is the pillar of success
(b) Honesty is the best policy
(c) Life begins and ends with adventures
(d) No adversity justifies giving up hope

ANS: (d)

No adversity justifies giving up hope options (b) and (c) can be clearly discarded since Honesty and adventure making have no relationship with the given argument Option (a) May appear close. But remember every failure may not result into success as it happens in the case of spider.

09. A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

(GATE-ME/CS-13)

- (a) 36 (b) 30 (c) 24 (d) 18

ANS: (c)

$$\begin{aligned} \text{Average speed} &= \frac{\text{Total distance}}{\text{Total time}} \\ &= \frac{D}{\frac{D/2}{60} + \frac{D/4}{30} + \frac{D/4}{10}} = 24 \text{ kmph} \end{aligned}$$



GENERAL APTITUDE

10. The current erection cost of a structure is Rs. 13,200. If the labour wages per day increase by $\frac{1}{5}$ of the current wages and the working hours decrease by $\frac{1}{24}$ of the current period, then the new cost of erection in Rs is **(GATE-ME/CS-13)**

(a) 16,500 (b) 15,180 (c) 11,000 (d) 10,120

ANS: B

Now cost

$$\begin{aligned} &= 13200 \left[1 + \frac{1}{5} \right] \left[1 - \frac{1}{24} \right] \\ &= 13200 \left[\frac{6}{5} \right] \times \left[\frac{23}{24} \right] = 15,180 \end{aligned}$$

GENERAL APTITUDE

CIVIL – 2013

General Aptitude (GA) Questions Q. 01 – Q. 05 carry one mark each.

Q.01. A number is as much greater than 75 as it is smaller than 117. The number is:
(GATE-CE-13)

- (A) 91 (B) 93 (C) 89 (D) 96

ANS: D

Q.02 The professor ordered to the students to go out of the class. (GATE-CE-13)

- I II III IV
Which of the above underlined parts of the sentence is grammatically incorrect?
(A) I (B) II (C) III (D) IV

ANS: B

Q.03. Which of the following options is the closest in meaning to the word given below:
Primeval (GATE-CE-13)
(A) Modern (B) Historic (C) Primitive (D) Antique

ANS: (c)

Primeval pertains to the first age or ages especially of the world .Ex: Primeval forms of life.

Q.04. Friendship, no matter how _____ it is, has its limitations. (GATE-CE-13)
(A) cordial (B) intimate (C) secret (D) pleasant

ANS: (b)

Friends No matter how intimate it is has its limitation. Intimate refers to close personal relations Ex .an intimate friend . It is also characterized by or involving warm friendship or personally Close or familiar association or feeling.

Q.05. Select the pair that best expresses a relationship similar to that expressed in the pair:
Medicine: Health (GATE-CE-13)
(A) Science: Experiment (B) Wealth: Peace
(C) Education: Knowledge (D) Money: Happiness

ANS: (c)

Medicine leads to good health. Similarly Education leads to Knowledge. Science does not lead to experiment .Wealth may not necessarily lead to peace. Also money may not also lead to happiness all the time.

GENERAL APTITUDE

Q. 06 to Q. 10 carry two marks each.

Q.06. X and Y are two positive real numbers such that $2X + Y \leq 6$ and $X + 2Y \leq 8$. For which of the following values of (X, Y) the function $f(X, Y) = 3X + 6Y$ will give maximum value?

(GATE-CE-13)

- (A) $(4/3, 10/3)$ (B) $(8/3, 20/3)$ (C) $(8/3, 10/3)$ (D) $(4/3, 20/3)$

ANS: (a)

$$2x + y \leq 6$$

$$X + 2y \leq 8$$

$$X \leq 1.33$$

$$Y \leq 3.33$$

$\therefore$ option (b) and (c) is rejected because $X \nless 1.33$.

Option (d) is rejected $X \nless 10/3$

Q.07. If $|4X - 7| = 5$ then the values of $2|X| - |-X|$ is:

(GATE-CE-13)

- (A) 2, $1/3$ (B) $1/2$, 3 (C) $3/2$, 9 (D) $2/3$, 9

ANS: (b)

$$|4x - 7| = 5$$

$$4x - 7 = 5$$

$$\text{And } 4x - 7 = -5$$

$$\Rightarrow x = 3$$

$$\text{And } x = 0.5$$

$$\therefore 2|X| - |-X| = 2 \times 3 - 3 = 3$$

$$\text{And } 2|X| - |-X| = 2 \times \frac{1}{2} - \frac{1}{2} = \frac{1}{2}$$

$$\Rightarrow \frac{1}{2}, 3$$

Q.08. Following table provides figures (in rupees) on annual expenditure of a firm for two years 2010 and 2011.

(GATE-CE-13)

Category	2010	2011
Raw material	5200	6240
Power & fuel	7000	9450
Salary & wages	9000	12600
Plant & machinery	20000	25000
Advertising	15000	19500
Research & Development	22000	26400

GENERAL APTITUDE

In 2011, which of the following two categories have registered increase by same percentage?

- (A) Raw material and Salary & wages
- (B) Salary & wages and Advertising
- (C) Power & fuel and Advertising
- (D) Raw material and Research & Development

ANS: (d)

$$\% \text{ of raw material are salary and wages} = \frac{6240-5200}{5200} \times 100 = 20\%$$

$$\begin{aligned} \% \text{ of raw material and research and development} \\ = \frac{26400-22000}{22000} \times 100 = 20\% \end{aligned}$$

Q.09. A firm is selling its product at Rs. 60 per unit. The total cost of production is Rs. 100 and firm is earning total profit of Rs. 500. Later, the total cost increased by 30%. By what percentage the price should be increased to maintained the same profit level.

(GATE-CE-13)

- (A) 5 (B) 10 (C) 15 (D) 30

ANS: A

Q.10. Abhishek is elder to Savar.
Savar is younger to Anshul.

(GATE-CE-13)

Which of the given conclusions is logically valid and is inferred from the above statements?

- (A) Abhishek is elder to Anshul (B) Anshul is elder to Abhishek
(C) Abhishek and Anshul are of the same age (D) No conclusion follows

ANS: D

GENERAL APTITUDE

ECE, EEE - 2012

General Aptitude (GA) Questions Q. 01 - Q. 05 carry one mark each.

01. If $(1.001)^{1259} = 3.52$ and $(1.001)^{2062} = 7.85$, then $(1.001)^{3321} =$ (GATE-EC/EE-13)
(a) 2.23 (b) 4.33 (c) 11.37 (d) 27.64

ANS: (d)

$$\begin{aligned} (1.001)^{1259} &= 3.52 \text{ and } (1.001)^{2062} = 7.85 \\ (1.001)^{1259} \times (1.001)^{2062} &= (1.001)^{3321} \text{ (as } a^m \times a^n = a^{m+n}) \\ \downarrow \quad \quad \downarrow & \\ 3.52 \quad \times \quad 7.85 &= (1.001)^{3321} \\ \text{Hence } (1.001)^{3321} &= 27.632 \approx 27.64 \end{aligned}$$

02. Choose the most appropriate alternative from the options given below to complete the following sentence: (GATE-EC/EE-13)
If the tired soldier wanted to lie down, he ----- the matters out on the balcony.
(a) should take (b) shall take
(c) should have taken (d) will have taken

ANS: (c)

If the tired soldier wanted to lie down he should have taken the mattress out on the balcony.

03. Choose the most appropriate word from the options given below to complete the following Sentence: (GATE-EC/EE-13)
Given the seriousness of the situation that he had to face, his ----- was impressive.
(a) beggary (b) nomenclature (c) jealousy (d) nonchalance

ANS: (d)

Given the seriousness of the situation that he had to face his nonchalance was impressive
Nonchalance refers to casual /non – serious /indifferent attitude .The author wants to convey that in spite of the seriousness of the situation he was very casual.

No other word fits in the blank beggary is word that is related to extreme poverty (Beggars).Nomenclature refers to the art of naming .

04. Which one of the following options is the closest in meaning to the word given below? (GATE-EC/EE-13)
Latitude
(a) Eligibility (b) Freedom (c) Coercion (d) Meticulousness

ANS: (d)

Latitude refers to freedom of action freedom of expression from restrictions etc. For

GENERAL APTITUDE

Example He allowed his children a fair amount of latitude. Coercion refers to force which is an opposite of the word latitude. Meticulousness refers to being extremely careful and conscientious

05. One of the parts (A, B, C, D) in the sentence given below contains an ERROR. Which one of the following is **INCORRECT**? **(GATE-EC/EE-13)**

I requested that he should be given the driving test today instead of tomorrow.

- (a) Requested that
- (b) Should be given
- (c) The driving test
- (d) Instead of tomorrow

ANS: (a)

Requested that part should be modified the sentence is in reported speech so should be given part is correct .Instead of Request words like suggested recommended can be introduced.

Q. 06- Q. 10 carry two marks each.

06. One of the legacies of the Roman legions was discipline. In the legions, military law prevailed and discipline was brutal. Discipline on the battlefield kept units obedient, intact and fighting, even when the odds and conditions were against them.

(GATE-EC/EE-13)

Which one of the following statements best sums up the meaning of the above passage?

- (a) Thorough regimentation was the main reason for the efficiency of the Roman legions even in adverse circumstances
- (b) The legions were treated inhumanly as if the men were animals.
- (c) Discipline was the armies' inheritance from their seniors.
- (d) The harsh discipline to which the legions were subjected to led to the odds and conditions being against them.

ANS: (a)

Thorough regimentation was the main reason for the efficiency of the Roman legions even in Adverse circumstances .As the author says discipline on the battlefield kept units obedient Intact and fighting even when the odds and conditions were against them. So choice (A) Emerges as the best answer option. We have been asked to summarize the paragraph in the Best possible manner choice (C) presents one – sided picture of the paragraph summary is The essence or the gist of the paragraph. Choice (B) and choice (D) are wrong as per the Information mentioned in the paragraph .

GENERAL APTITUDE

07. Raju has 14 currency notes in his pocket consisting of only Rs. 20 notes and Rs.10 notes. total money value of the notes is Rs. 230. The number of Rs.10 notes that Raju has in

(GATE-EC/EE-13)

- (a) 5 (b) 6 (c) 9 (d) 10

ANS: (a)

Let the number of Rs .20 notes be 'a' and Rs .10 notes be 'b' respectively.

$$\text{So } a + b = 14$$

$$20a + 10b = 230$$

$$\text{On solving } a = 9, b = 5$$

Hence number of Rs .10 notes are 5.

Alternate Solution

If we use option (a) Number of ten Rs. notes are 5.

$$\text{Rs } 10 \times 5 = 50 \text{ Rs .}$$

The remaining nine notes have to be 20Rs notes

$$\text{Rs } 20 \times 9 = 180$$

$$\text{And hence } 50 + 180 = 230 \text{ (total money value)}$$

Hence option (a) number of ten Rs. notes are 5.

08. There are eight bags of rice looking alike, seven of which have equal weight and one is slightly heavier. He weighing balance is of unlimited capacity. Using this balance, the minimum number of weightings required to identify the heavier bag is

(GATE-EC/EE-13)

- (a) 2 (b) 3 (c) 4 (d) 8

ANS: (a)

Divide 8 bags into three parts. 3,3 and 2 respectively .Now two cases arises.

Case(i) Firstly if we compare the 3,3 bags on pans of a balance .We can identify on which side is the lighter bag placed .And then we will need only one more weighing for identifying faulty bag. So only two weightings are required.

Case (ii)

Secondly if the 3,3 compared bags are equal then to be sure faulty bag is one among the remaining two bags. So one more weighing is required to find out faulty bag from these two bags. Hence overall two weightings are required. Thus in either cases only two weightings are required.

Alternate Solution

In case of a weighing balance (i.e. beam balance) the following model can be observed

$$1 - 3 \rightarrow 1 \text{ weighing required}$$

$$4 - 9 \rightarrow 2 \text{ weighing required}$$

$$10 - 27 \rightarrow 3 \text{ weighing required}$$

$$28 - 81 \rightarrow 4 \text{ weighing required}$$

And the process will go so on

As there are eight objects .Minimum number of weightings required will be only 2.

GENERAL APTITUDE

09. The data given in the following table summarizes the monthly budges of an average household. The approximate percentage of the monthly budget **NOT** spent on savings is

(GATE-EC/EE-13)

(a)10%

(b)14%

(c)81%

(d)86%

Category	Amount (Rs.)
Food	400
Clothing	1200
Rent	2000
Savings	1500
Other expenses	1800

ANS: (d)

Category	Amount
Food	4000
Clothing	1200
Rent	2000
Savings	1500
Otheres	1800
Total	10500

So total expenses are 10,500 Expenses excluding savings are 9000 Hence required percentage

$$\begin{aligned} &= \frac{9,000}{10,500} \times 100 \\ &= 85.714\% \approx 86\% \end{aligned}$$

10. A and B are friends. They decide to meet between 1 PM and 2 PM on a given day. There is a condition that whoever arrives first will not wait for the other forbore than 15 minutes. The probability that they will meet on the day is

(GATE-EC/EE-13)

(a) $\frac{1}{4}$

(b) $\frac{1}{16}$

(c) $\frac{7}{16}$

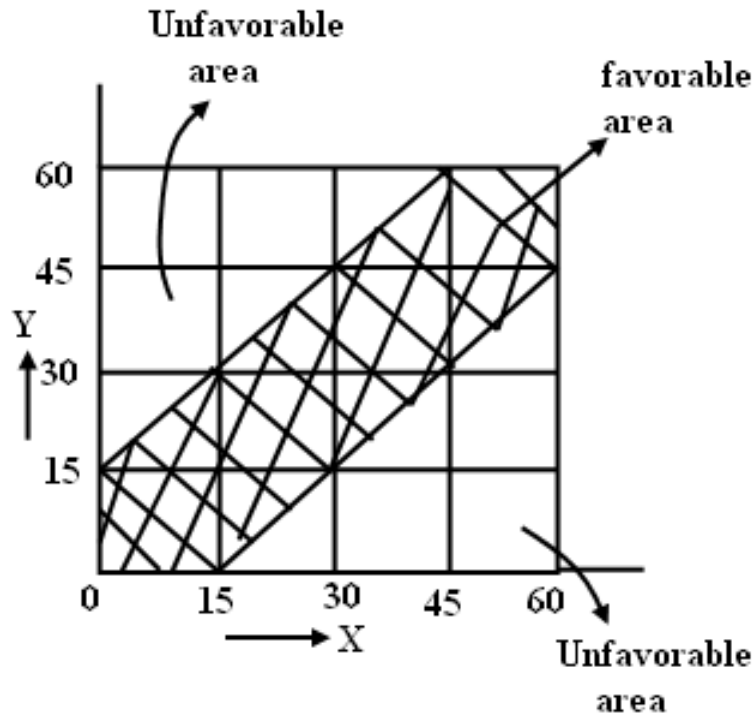
(d) $\frac{9}{16}$

ANS: (c)



GENERAL APTITUDE

Probability that A & B will meet will be given by the graphical representation where shaded region represents favorable area



$$\text{Required probability} = \frac{\text{favourable area}}{\text{total area}} = \frac{7}{16}$$

GENERAL APTITUDE

MECH, CIVIL, CSE- 2012

General Aptitude (GA) Questions

Q. 01 - Q.02 carry one mark each.

01. The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of Rs.50 per unit. The number of units to be produced by the firm such that the profit is maximized is

(GATE-ME/CE/CS-12)

- (a)5 (b)10 (c)15 (d)25

ANS: (a)

Total cost $C = 5q^2$ total sales Revenue

$$S = 50q \text{ Profit } (P) = (S - C) = 50q - 5q^2$$

As the profit have to be maximized

Using options

(a) $q = 5$

$$P = 50 \times 5 - 5 \times 5^2 = 125$$

(b) $q = 10$

$$P = 50 \times 10 - 5 \times 10^2 = 0$$

(c) $q = 15$

$$P = 50 \times 15 - 5 \times 15^2 = -ive$$

(d) $q = 25$

$$P = 50 \times 25 - 5 \times 25^2 = -ive$$

Hence the maximum profit will happen at $q = 5$

Alternate Solution

$$P = 50q - 5q^2$$

$$\text{For max/min } \frac{dp}{dq} = 0$$

$$\Rightarrow 50 - 10q = 0$$

$$q = 10$$

$$\left. \frac{d^2p}{dq^2} \right|_{q=5} = -10 \text{ which is negative}$$

Hence maximum profit will happen at $q = 5$

GENERAL APTITUDE

02. Choose the most appropriate alternative from the options given below to complete the following sentence: (GATE-ME/CE/CS-12)

Despite several----- the mission succeeded in its attempt to resolve the conflict.

- (a) attempts (b) setbacks (c) meetings (d) delegations

ANS: (b)

Setbacks

Despite several setbacks the mission succeeded in its attempt to resolve the conflict. The word 'Despite' indicates that there has to be a contrast in the sentence use of the word 'Setbacks' in the blank indicates that despite problems the mission was successful.

03. Which one of the following options is the closest in meaning to the word given below?

Mitigate

(GATE-ME/CE/CS-12)

- (a) Diminish (b) Divulge (c) Dedicate (d) Denote

ANS: (a)

Diminish

Mitigate means to reduce to lessen etc. So only the word Diminish is close. Rest all choices have no link with the given word. Divulge means to disclose or reveal which has no link the given word.

04. Choose the grammatically **INCORRECT** sentence: (GATE-ME/CE/CS-12)

- (a) They gave us the money back less the service charges of Three Hundred rupees.
(b) This country's expenditure is not less than that of Bangladesh.
(c) The committee initially asked for a funding of Fifty Lakhs rupees, but later settled for a lesser sum.
(d) This country's expenditure on educational reforms is very less.

ANS: (b)

The country's expenditure is not less than that of Bangladesh.

The correct statement should be : The country's expenditure is not less than that of Bangladesh. It should be compared with Bangladesh's expenditure.

05. Choose the most appropriate alternative from the options given below to complete the following sentence: (GATE-ME/CE/CS-12)

Suresh's dog is the one----- was hurt in the stampede.

- (a) That (b) which (c) who (d) whom

ANS: (a)

Suresh's dog is the one that was hurt in the stampede, That is used with restrictive clauses.

GENERAL APTITUDE

Q. 06- Q.10 carry two marks each.

06. Wanted Temporary, Part- time persons for the post of Field Interviewer to conduct personal interviews to collect and collate economic data. Requirements: High School- pass, must be available for Day, Evening and Saturday work, transportation paid, expenses reimbursed. Which one of the following is the best inference from the above advertisement?

(GATE-ME/CE/CS-12)

- (a) Gender- discriminatory
- (b) Xenophobic
- (c) Not designed to make the post attractive
- (d) Not gender- discriminatory

ANS: (d)

Not gender - discriminatory

Choice (A) cannot be considered since there is no gender discrimination mentioned in the argument.

Choice (B) Xenophobic is one who has fear of foreigners', no link with the given argument.

Choice (C) it is wrong to say that the profile has not been designed to make the post attractive, since there are certain features which have been added to make the profile lucrative (which are given towards the end of the advertisement like Transportation paid, expenses reimbursed).

07. A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows height of the arch is

(GATE-ME/CE/CS-12)

- (a) 8 meters
- (b) 10 meters
- (c) 12 meters
- (d) 14 meters

ANS: (b)

$$y = 2x - 0.1x^2$$

For (height) to be maximum $\frac{dy}{dx} = 0$

$$\frac{dy}{dx} = 2x - 2x = 0$$

$$\Rightarrow x = 10$$

$$\left[\frac{d^2y}{dx^2} \right]_{x=10} = -0.2 \text{ (-ive)}, \frac{d^2y}{dx^2} < 0$$

Hence maximum height will be at $x = 10$

$$Y = 2 \times 10 - 0.1 \times (10)^2 = 10$$

Maximum height = 10 meters.

08. An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable. The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

(GATE-ME/CE/CS-12)

- (a) 0.288
- (b) 0.334
- (c) 0.667
- (d) 0.720

GENERAL APTITUDE

ANS: (b)

Let total 100 shock absorbers are supplied

So X supplies = 60 (40% of 100)

Y supplies = 40 (60% of 100)

Reliable supply by X = 96% of 60 = 57.6

Reliable supply by Y = 72% of 40 = 28.8

So the required probability that random chosen reliable shock absorber is made by Y is

Required probability

$$\begin{aligned} &= \frac{28.8}{57.6+28.8} = 0.333 \\ &= 0.334 \end{aligned}$$

09. Which of the following assertions are **CORRECT**? (GATE-ME/CE/CS-12)

P: Adding 7 to each entry in a list adds 7 to the mean of the list

Q: Adding 7 to each entry in a list adds 7 to the standard deviation of the list

R: Doubling each entry in a list doubles the mean of the list

S: Doubling each entry in a list leaves the standard deviation of the list unchanged

(a) P, Q

(b) Q, R

(c) P, R

(d) R, S

ANS: (c)

Adding each and every observation by constant increases the arithmetic mean by same constant. Also multiplying each and every observation by constant multiplies the arithmetic mean by same constant. While the standard deviation will not show similar effect.

Alternate Solution:

$$\frac{x_1+x_2+x_3+\dots+x_n}{N} = \bar{X}$$

$$\frac{(x_1+7)+(x_2+7)+\dots+(x_n+7)}{N} = \bar{x} + 7$$

$$\text{And } \frac{2(x_1+x_2+\dots+x_n)}{N} = 2\bar{x}$$

Hence only P, R are correct.

10. Given the sequence of terms AD CG FK P, the next term is (GATE-ME/CE/CS-12)

(a) OV

(b) OW

(c) PV

(d) PW

ANS: (a)

AD, CG, FK, JP are given terms the first letter in each term A, C, F, J are respectively 1, 3, 6, 10 letter of Alphabetical sequence. A = 1; 1st letter of alphabetical sequence

C = 1+2 = 3; 3rd letter of alphabetical sequence

F = 1 + 2 + 3 = 6; 6th letter of alphabetical sequence

G = 1 + 2 + 3 + 4 = 10; 10th letter of alphabetical sequence

The next term should begin with

1 + 2 + 3 + 4 + 5 = 15th letter of alphabetical sequence. So next term should begin with o.

GENERAL APTITUDE

Also in the series AD, CG, FK, JP, two letters in each term are separated by 2, 3, 4, 5 letters in Between them respectively .Hence next term should be OV as there should be a gap of 6 alphabets Between o and V. Hence answer is OV.

VANI

GENERAL APTITUDE

MECH, CIVIL, CSE- 2011

Q. 01 – Q. 05 carry one mark each.

01. Which of the following options is the closest in the meaning to the word below?

Inexplicable

(GATE-ME/CE/CS-11)

- (a) Incomprehensible (b) Indelible
(c) Inextricable (d) Infallible

ANS: A

02. If $\log(P) = (1/2)\log(Q) = (1/3)\log(R)$, then which of the following options is TRUE?

- (a) $P^2 = Q^3 R^2$ (b) $Q^2 = PR$ (c) $Q^2 = R^3 P$ (d) $R = P^2 Q^2$
(GATE-ME/CE/CS-11)

ANS: (b) $\log(P) = \frac{1}{2} \log(Q) = \frac{1}{3} \log(R)$

$$\Rightarrow \log(P) = \log(Q)^{1/2} \\ = \log(R)^{1/3} = K$$

$$\Rightarrow P = (Q)^{1/2} = (R)^{1/3} = K$$

$$\Rightarrow P = K, Q = K^2, R = K^3$$

Now, only option (B) $Q^2 = PR$ satisfies

$$Q^2 = (K^2)^2 = K^4$$

From – (i)

$$P.R = K * K^3 = K^4$$

From ---(ii)

Here $Q^2 = PR$ holds true

03. Choose the most appropriate word (s) from the options given below to complete the following sentence (GATE-ME/CE/CS-11)

I contemplated _____ Singapore for my vacation but decided against it

- (a) to visit (b) having to visit (c) visiting (d) for a visit

ANS: C

04. Choose the most appropriate word from the options given below to complete the following sentence (GATE-ME/CE/CS-11)

If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or

- (a) hyperbolic (b) restrained (c) argumentative (d) indifferent

ANS: B

05. Choose the word from the options given below that is most nearly opposite in meaning to the given word. (GATE-ME/CE/CS-11)

Amalgamate

- (a) merge (b) Split (c) Collect (d) separate

ANS: B



GENERAL APTITUDE

Q. 06 – Q.10 carry one mark each

06. Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting.

Based on the above passage which topic would not be included in a unit on bereavement ?

(GATE-ME/CE/CS-11)

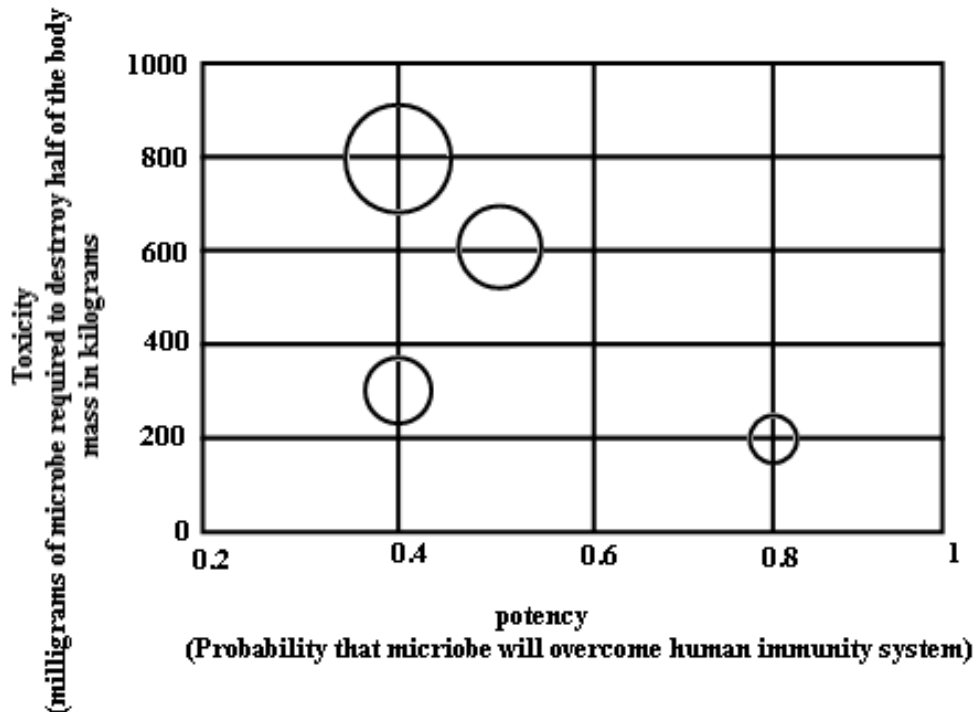
- (a) How to write a letter of condolence
- (b) What emotional stages are passed through in the healing process
- (c) What the leading causes of death are
- (d) How to give support to a grieving friend.

Answer: (c)

Exp: The given passage clearly deals with how to deal with bereavement and grief and so after the tragedy occurs and not about precautions. Therefore, irrespective of the causes of death, a school student rarely gets into details of causes-which is beyond the scope of the context. Rest all are important in dealing with grief.

07. P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below.

(GATE-ME/CE/CS-11)



A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

- (a) P
- (b) Q
- (c) R
- (d) S

GENERAL APTITUDE

ANS: (d)

As we can understand that the danger of a microbe to human being will be directly proportional to potency and growth. At the same time it will be inversely proportional to toxicity defined (more dangerous will a microbe be if lesser of its milligram is required). So level of dangerous. $\propto \frac{P \uparrow \times G \uparrow}{T \downarrow}$ where P, G and T are the potency growth and toxicity as defined in question. So $D_i = \frac{KPG}{T}$ where K is constant of proportionality so level of dangerous of S will be maximum given by $D_S = \frac{0.8 \times \pi (10mm)^2}{200}$ Similar calculation for D_P , D_Q , D_R can be done based on (i) to find out that D_S is maximum and most dangerous among them.

08. The variable cost (V) of manufacturing a product varies according to the equation $V = 4q$, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation $F = 100/q$. How many units should be produced to minimize the total cost (V + F)? (GATE-ME/CE/CS-11)
- (a) 5 (b) 4 (c) 7 (d) 6

ANS: (a)

(T.C) total cost = V + F

$$T.C = 4q + \frac{100}{q}$$

As we have to minimize total cost Using options

(a) $q = 5$, $T.C. = 4 \times 5 + \frac{100}{5} = 40$

(b) $q = 4$, $T.C. = 4 \times 4 + \frac{100}{4} = 41$

(c) $q = 7$, $T.C. = 4 \times 7 + \frac{100}{7} = 42.285$

(d) $q = 6$, $T.C. = 4 \times 6 + \frac{100}{6} = 40.\overline{66}$

Hence T.C. is minimum at $q = 5$, (a) ans *always put options to get answer fast.

Alternate Solution

Or $T.C. = 4q + \frac{100}{q}$

For minimum $\frac{d}{dq} \left[4q + \frac{100}{q} \right] = 0$

$\Rightarrow q^2 = 25$ or $q = 5$

$\frac{d}{dq} \left[4q + \frac{100}{q} \right]_{q=5} > 0$

Hence TC (Total cost) is minimum at $q = 5$ (a) ans

GENERAL APTITUDE

09. A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders. Alternatively, if he uses only trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of the 5th day? (GATE-ME/CE/CS-11)
- (a) 4 (b) 5 (c) 6 (d) 7

ANS: (c)

Let 'y' be the backlog with transporter and 'x' be the number of orders each day. So, as per conditions given in question $4x + y = 28$

$$10x + y = 30$$

Solving (i) and (ii) $x = \frac{1}{3}$ and $y = \frac{80}{3}$

Since we need to find out number of trucks so that no pending order will be there at the end of 5th Day $5x + y = n \times 5$. So we need to find 'n' where 'n' is the number of trucks

required $n = \frac{5x+y}{5} = \frac{5 \times \frac{1}{3} + \frac{80}{3}}{5} = \frac{85}{15} = \frac{17}{3}$. Hence, $5.\overline{66}$ truck will be required. As number of trucks to be natural number hence 6 trucks will be required.

10. A container originally contains 10 litres of pure spirit. From this container 1 litre of spirit is replaced with 1 litre of water. Subsequently, 1 litre of water is replaced with 1 litre of spirit and this process is repeated one more time. How much spirit is now left in the container? (GATE-ME/CE/CS-11)
- (a) 7.58 litres (b) 7.84 litres (c) 7 litres (d) 7.29 litres

ANS: (d)

Shortcut Method

Every time if we take 1 litre of mixture out and replace with water content of pure spirit will keep on reducing by 10%. So final quantity of spirit after 3 such operations are

$$10 \times 0.9 \times 0.9 \times 0.9 = 7.29 \text{ litres}$$

Alternate Solution

$$\frac{\text{Quantity of spirit left after } n^{\text{th}} \text{ operation}}{\text{Initial quantity of spirit}}$$

$$= \left[\frac{a-b}{a} \right]^n = \left[1 - \frac{b}{a} \right]^n \text{ where 'a' is initial quantity of pure spirit and 'b' is quantity taken out and replaced every time. Hence quantity of spirit left after 3rd operation}$$
$$= \text{initial quantity} \times \left[1 - \frac{1}{10} \right]^3 = 10 \times 0.9 \times 0.9 \times 0.9$$
$$= 7.29 \text{ litres}$$

GENERAL APTITUDE

ECE, EEE – 2011

Q. 01 – Q. 05 carry one mark each.

01. The question below consists of a pair of related words followed by four pairs of words.

Select the pair that best expresses the relation in the original pair: (GATE-EC/EE-11)

Gladiator: Arena

(a) Dancer: stage

(b) commuter: train

(c) teacher: classroom

(d) lawyer: courtroom

ANS: A

57. There are two candidates P and Q in an election. During the campaign 40% of the voters promised to vote for P, and rest for Q. However, on the day of election 15% of the voters went back on their promise to vote for P and instead voted for Q. 25% of the voters went back on their promise to vote for Q and instead voted for P. Suppose, P lost by 2 votes, then what was the total number of voters? (GATE-EC/EE-11)

(a) 100

(b) 110

(c) 90

(d) 95

ANS: (a) Let there be overall 'X' candidates initially those who decided to vote for P and Q are $0.4x$ and

$0.6x$ respectively. On the day of election 15% of $0.4x = 0.06x$ went from P's side to Q's side and

25% of $0.6x = 0.15x$ went from Q's side to P's side.

Now after transfer 'P' has $0.4x - 0.06x + 0.15x = 0.49x$

And after transfer 'q' has $0.6x - 0.15x + 0.06x = 0.51x$

Given in the question that P lost by 2 votes $Q - P = 2$ votes

$0.51x - 0.49 = 0.02x = 2$ votes

Hence ($x = 100$) total number of votes are 100

Alternate Solution

Using options:

Lets take option (a) 100 as total number of voters

Initially P and Q has 40 and 60 voters on their sides respectively.

15% went from P's side to Q and 25% went from Q's side to P.

P	Q
(40)	(60)
(-)15% of 40	-25% of (60)
+25% of 60	+15% of (40)
$40 - 6 + 15 = 49$	$60 - 15 + 6 = 51$

Hence Q and P has 51 and 49 respectively difference in number of votes = 2. Hence answer (a).

GENERAL APTITUDE

02. Choose the most appropriate word from the options given below to complete the following sentence: (GATE-EC/EE-11)

It was have view that the country's problems had been by foreign technocrats, so that to invite them to come back would be counter-productive.

- (a) Identified (b) ascertained (c)exacerbated (d) analyzed

ANS: C

03. Choose the word from the options give below hat is most nearly opposite in meaning to the given word: (GATE-EC/EE-11)

Frequency

- (a) Periodicity (b) rarity (b) gradualness (d) persistency

ANS: C

04. Choose the most appropriate word from the options give below to complete the following sentence:

Under ethical guidelines recently adopted by the Indian Medical Association, human genes are to be manipulated only to correct diseases for which----- treatments are unsatisfactory.

- (a) Similar (b)most (c) uncommon (d) available

ANS: C

Q. 06 To Q.10 carry two marks each.

06. The horse has played a little known but verb important role in the field of medicine. Horses were injected with toxins of diseases un till their blood built up immunities. Then a serum was made from their blood. Serums to fight with diphtheria and tetanus were developed this way. (GATE-EC/EE-11)

It can be inferred from the passage, that horses were

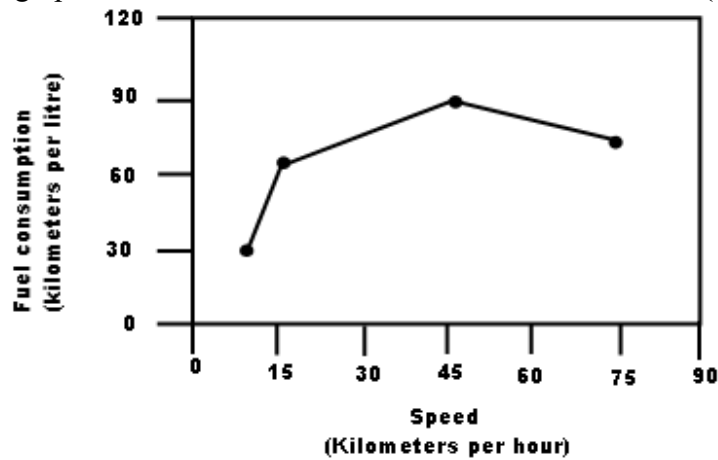
- (a) Given immunity to diseases
(b) Generally quite immune to disease
(c) Given medicines to fight to toxins
(d) Given diphtheria and tetanus serums

Answer:- (B)

Exp: From the passage it cannot be inferred that horses are given immunity as in (A), since aim is to develop medicine and in turn immunize humans. (B) is correct since it is given that horses develop immunity after some time. Refer “until their blood built up immunities”. Even (C) is invalid since medicine is not built till immunity is developed in the horses. (D) is incorrect since specific examples are cited to illustrate=rate and this cannot capture the essence.

GENERAL APTITUDE

07. The fuel consumed by a motorcycle during a journey while traveling at various speeds is indicated in the graph below. (GATE-EC/EE-11)



The distances covered during four laps of the journey are listed in the table below

Lap	Distance (kilometers)	Average speed (kilometers per hour)
P	15	15
Q	75	45
R	40	75
S	10	10

From the given data, we can conclude that the fuel consumed per kilometer was least during the lap

- (a) P (b) Q (c) R (d) S

ANS: (b)

Fuel consumed per km will be least when mileage (kilometers per liter) mentioned on 'y' axis of graph will be maximum irrespective of number of kilometers travelled. From the graph 'y' axis we can observe that mileage (kilometers per liter) is maximum when vehicle is driven at 45 kilometers per hour. Hence the stretch which 'Q' covered at 45 kmph mileage was highest and fuel consumption per kilometers was lowest.

08. Three friends R, S and T shared from a bowl. R took $\frac{1}{3}$ rd of the toffees, but returns four to the bowl. S took $\frac{1}{4}$ th of what was left but returns three toffees to the bowl. T took half of the remainder but returned two back into the bowl. If the bowl had 17 toffees, how many toffees were originally there in the bowl? (GATE-EC/EE-11)
- (a) 38 (b) 31 (c) 48 (d) 41

GENERAL APTITUDE

ANS: (c) Shortcut method

The question can be done orally as R took $1/3^{\text{rd}}$ of toffees initially. Hence total number of toffees have to be multiple of 3. There is only one option © as 48 which is multiple of 3.

Alternate Solution

Using options:

If we use options for option (c) if total number of toffees are 48 R took $1/3^{\text{rd}}$ of 48 and Returned 4. So R has $16 - 4 = 12$ toffees. Out of left over 36 toffees S took $1/4^{\text{th}}$ of 36 and Returned 3. So, S has $9 - 3 = 6$ and remaining 30 will be left in bowl. Now T took $1/2$ of 30 and returned two back. So the bowl has 17 left. *Always solve such questions by options.

09. Given that $f(y) = \frac{|y|}{y}$, and q is any non-zero real number, the value of $|f(q) - f(-q)|$ is
- (a) 0 (b) -1 (c) 1 (d) 2
- (GATE-EC/EE-11)

ANS: (d)

$$f(y) = \frac{|y|}{y}$$

As we know,

$$|y| = y \quad \begin{cases} y \geq 0 \\ y < 0 \end{cases}$$

-y

$$\text{So from (i) } f(y) = \frac{y}{y} = 1 \text{ If } y \geq 0$$

$$f(y) = \frac{-y}{y} = -1 \text{ If } y < 0$$

$$\text{Hence } |f(q) - f(-q)| = |1 - (-1)| = 2$$

Alternate Solution

$$\text{Otherwise if } f(y) = \frac{|y|}{y}$$

$$f(1) = \frac{|1|}{1} = 1$$

$$f(-1) = \frac{|-1|}{-1} = -1$$

For calculating $|f(q) - f(-q)|$

Let, $q = 1$, $|f(q) - f(-q)|$

$$|f(1) - f(-1)| = 1 - (-1) = 2$$

10. The sum of n terms of the series $4+44+444+ \dots$ is
- (a) $(4/81) [10^{n+1} - 9n - 1]$ (b) $(4/81) [10^{n-1} - 9n - 1]$
(c) $(4/81) [10^{n+1} - 9n - 10]$ (d) $(4/81) [10^n - 9n - 10]$
- (GATE-EC/EE-11)

ANS: (c)



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$$4 + 44 + 444$$

If we see first term of series = 4 Hence ,Sum up to 1st term is also = 4

Put ,n =1 (first term) only option (c) satisfies.

$$\frac{4}{81} [10^{n+1} - 9n - 10] = \frac{4}{81} [10^{1+1} - 9 \times 1 - 10]$$
$$\frac{4}{81} \times 81 = 4$$

Hence option (c) is correct answer Always solve these questions by putting values.

Alternate Solution

$$4 + 44 + 444$$

$$\frac{4}{9} [9 + 99 + 999]$$

$$= \frac{4}{9} [(10 - 1) + (10^2 - 1) + (10^3 - 1)]$$

$$= \frac{4}{9} [(10 - 1) + (10^2 - 1) + (10^3 - 1)]$$

$$= \frac{4}{9} [(10 + 10^2 + 10^3 - \dots - 10^n) - (1 + 1 + 1) \dots \dots]$$

$$= \frac{4}{9} \left[\frac{10[10^n - 1]}{9} - n \right]$$

$$= \frac{4}{9} \left[\frac{10^{n+1} - 10 - 9n}{9} \right]$$

$$= \frac{4}{81} [10^{n+1} - 9n - 10]$$

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ECE, EEE, CSE, MECH, CIVIL- 2010

General Aptitude (GA) Questions Q. 01 - Q. 05 carry one mark each.

01. Which of the following options is the closest in meaning to the word below:

Circuitous

(GATE-2010)

- (a) Cyclic (b) indirect (c) confusing (d) crooked

Ans: (b)

Circuitous : Deviating from a straight course $\Rightarrow$ Indirect

- (a) Cyclic : Recurring in cycle
(b) Indirect : Not leading by straight line
(c) Confusing : Lacking clarity
(d) Crooked : For shapes (irregular in shape)

02. The question below consists of a pair of related words followed by four pairs of words.

Select the pair that best expresses the relation in the original pair.

(GATE-2010)

Unemployed : Worker

- (a) Fallow : land (b) unaware : sleeper
(c) wit : jester (d) renovated : house

Ans: (a) Unemployed : Worker $\Rightarrow$ Here one is opposite to other

- (a) Fallow : Land $\Rightarrow$ Fallow means undeveloped land.
(b) Unaware : sleeper $\Rightarrow$ Both are same unaware or asleep
(c) Wit : Jester $\Rightarrow$ Wit means ability to make jokes and jester is a joker
(d) Renovated : House $\Rightarrow$ Renovate means to make better and house can be renovated.

03. Choose the most appropriate word from the options given below to complete the following Sentence:

(GATE-2010)

If we manage to ----- our natural resources, we would leave a better planet for our children.

- (a) Uphold (b) restrain (c) cherish (d) conserve

Ans: (d) (a) Uphold : cause to remain $\Rightarrow$ not appropriate

- (b) Restrain : keep under control $\Rightarrow$ not appropriate
(c) Cherish : Be fond of $\Rightarrow$ not related
(d) Conserve : Keep in safety and protect from harm decay loss or destruction $\Rightarrow$ most appropriate

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04. Choose the most appropriate word from the options given below to complete the following sentence: (GATE-2010)

His rather casual remarks on politics ----- his lack of seriousness about the subject.

- (a) Masked (b) belied (c) betrayed (d) suppressed

Ans: (c) (a) Masked : hide under a false appearance $\Rightarrow$ opposite
(b) Belied : Be in contradiction with $\Rightarrow$ not appropriate
(c) Betrayed : Reveal unintentionally $\Rightarrow$ most appropriate
(d) Suppressed : To put down by force or authority $\Rightarrow$ irrelevant

05. 25 persons are in a room 15 of them play hockey. 17 of the play football and 10 of them play both hokey and football. Then the number of persons playing neither hockey nor football is : (GATE-2010)

- (a) 2 (b) 17 (c) 13 (d) 3

Explanations:-

Ans: (d)

Using the set theory formula $n(A)$: Number of people who play hockey = 15

$n(B)$: Number of people who play football = 17

$n(A \cap B)$: Persons who play both hockey and football = 10

$n(A \cup B)$: Persons who play either hockey or football or both Using the formula

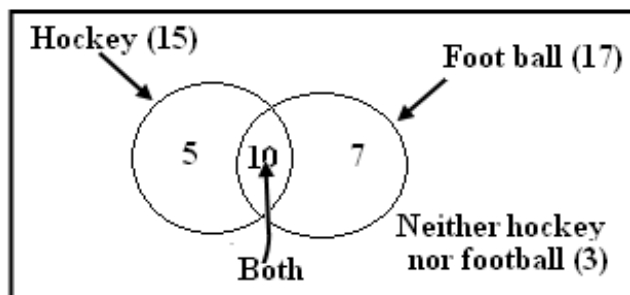
$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B) = 15 + 17 - 10 = 22$$

$$\text{Thus people who play neither hockey nor football} = 25 - 22 = 3$$

Alternative Method

Refer to venn diagram given below



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Q. 06- 10 carry two marks each.

06. Modern warfare has changed from large scale clashes of armies to suppression of civilian populations. Chemical agents that do their work silently appear to be suited to such warfare are useful tools for their cause. **(GATE-2010)**

Which of the following statements best sums up the meaning of the above passage:

- (a) Modern warfare has resulted in civil strife
- (b) Chemical agents are useful in modern warfare.
- (c) Use of chemical agents in warfare would be undesirable.
- (d) People in military establishments like to use chemical agents in war.

Ans: (d)

- (A) Modern warfare has resulted in civil strife .There is no direct consequence of warfare given so it is not appropriate
- (B) Chemical agents are useful in modern warfare : Passage does not say whether chemical agents are useful or not so not appropriate
- (C) Use of chemical agents in warfare would be undesirable .Given that people in military think these are useful underirable is wrong
- (D) People in military establishments like to use chemical agents in war. Correct choice as last statement tells that military people think that chemical agents are useful tools for their cause (work silently in warfare)

07. If $137 + 276 = 435$ how much is $731 + 672$? **(GATE-2010)**
(a) 534 (b) 1403 (c) 1623 (d) 1513

ANS: (c)

$$137 + 276 = 435$$

This is a octahedral addition thus

$$731 + 672(8) = 1623$$

Simply Every number change in to decimal system and also finally change into octal system.

08. 5 skilled workers can build a wall in 20 days; 8 semi- skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled. 6 semi- skilled and 5 unskilled workers, how long will it take to build the wail? **(GATE-2010)**
(a) 20 days (b) 18 days (c) 16 days (d) 15 days

ANS: (d)

Per day work or rate of 5 skilled workers $= \frac{1}{20} \Rightarrow$ Per day work or rate of one skill worker $= \frac{1}{5 \times 20} = \frac{1}{100}$ Similarly per day work or rate of 8 semiskilled workers $= \frac{1}{25} \Rightarrow$ Per day work or rate of 10 unskilled workers $= \frac{1}{30} \Rightarrow$ Per day work or rate of one semi – skill worker $= \frac{1}{10 \times 30}$

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$$= \frac{1}{300} \text{ Thus total per day work of 2 skilled 6 semiskilled and 5 unskilled workers } = \frac{2}{100} + \frac{6}{200} + \frac{5}{300} = \frac{12+18+10}{600} = \frac{40}{600} = \frac{1}{15} \text{ Thus time to complete the work is 15 days}$$

Alternative Method

Let one day work of skilled semi – skilled and unskilled worker be a, b, c units respectively.

$$5a \times 20 = 8b + 25 = 10c \times 30 = \text{total unit of work}$$

$$100a = 200b = 300c$$

$$a = 2b = 3c$$

$$\Rightarrow b = \frac{a}{2} \text{ and } c = \frac{a}{3}$$

Given that 2 skilled 6 semi – skilled and 5 unskilled workers are working .Let they finish the work in 'X' days. $(2a + 6b + 5c) X = 5a \times 20$

= Total units of work

$$\left[2a + 3a \frac{5}{3} \right] x = 5a \times 20$$

$$\frac{20a}{3} \times = 5a \times 20$$

$$x = 15 \text{ days}$$

09. Give digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit number greater than 3000 can be From d? (GATE-2010)
(a) 50 (b) 51 (c) 52 (d) 54

ANS: (b) We have to make 4 digit numbers so the number should be start with 3 or 4 two cases possible.

Case(1) Thousands digit is 3 now other three digits may be any of 2,2,3,3,4,4,4,4

(a) Using 2,2,3 $\Rightarrow 223, 232, 322$

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(b) Using 2,2,4 $\Rightarrow 224, 242, 422$

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(c) Using 2,3,3 $\Rightarrow 233, 323, 332$

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(d) Using 2,3,4 $\Rightarrow 234, 243, 324, 342, 423, 432$

$(3! = 6 \text{ numbers are possible})$

(e) Using 2,4,4 $\Rightarrow 244, 424, 442$

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(f) Using 3,3,4 $\Rightarrow 334, 343, 433$

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$



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(g) Using 3,4,4 $\Rightarrow$ 344,434,443

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(h) Using 4,4,4 $\Rightarrow$ 444

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

Total 4 digit numbers in case 1 = 3 + 3 + 3 + 6 + 3 + 3 + 3 + 1 = 25

Case (2) thousands digit is 4; Now other three digits may be any of 2,2,3,3,3,4,4,4

(a) Using 2,2,3 $\Rightarrow$ 223,232,322

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(b) Using 2,2,4 $\Rightarrow$ 224,242,422s

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(c) Using 2,3,3 $\Rightarrow$ 233,323,332

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(d) Using 2,3,4 $\Rightarrow$ 234,243,324,342

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(e) Using 2,4,4 $\Rightarrow$ 244,424,442

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(f) Using 3,3,3 $\Rightarrow$ 333

$$\left[\frac{3!}{3!} = 1 \text{ number is possible} \right]$$

(g) Using 3,3,4 $\Rightarrow$ 334,343,433

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(h) Using 3,4,4 $\Rightarrow$ 344,434,443

$$\left[\frac{3!}{2!} = 3 \text{ numbers are possible} \right]$$

(i) Using 4,4,4 $\Rightarrow$ 444

$$\left[\frac{3!}{3!} = 1 \text{ number is possible} \right]$$

Total 4 digit numbers in case 2 = 3 + 3 + 3 + 6 + 3 + 3 = 1 + 3 + 1 = 26

Thus total 4 digits numbers using case (1) and case (2) = 25 + 26 = 51

Alternative Method /Shortcut method

As the number is greater than 3000. So thousand's place can be their 3 or 4. Let's consider the following two cases

Case (I) When thousands place is 3 3 a b c

If there is no restriction on number of two's three's and four's. Then each of a,b,c can be filled With 2 or 3 or 4 each in 3 ways So $3 \times 3 \times 3 = 27$ numbers are three. Out of which

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3222,3333 Are invalid as 2 can be used twice & three thrice only so number of such valid numbers Beginning with 3 are $27 - 2 = 25$

Case(II)

When thousand's place is 4 a b c without restriction on number of 2's, 3's and 4's a, b, c (as Explained in case I) can be filled in 27 ways. Out of these 27 numbers 4 2 2 2 is only invalid as Two have to be used twice only so valid numbers are $27 - 1 = 26$ Total numbers from case (I) & case (II) $25 + 26 = 51$

10. Hari (H), Gita (G), Irfan (I) and Saira (S) are siblings (i.e brother and sisters). All were born on st January. The age difference between any two successive sibling (that is born one after ther is less than 3 years. Given the following facts:

- hari's age + Gita's age > Irfan's age + Saira's age.
- The age difference between Gita and Saira is 1 year. However, Gita is not the oldest and Saira is not the youngest.
- There are no twins in what order were they born (oldest first)>?

(a) HSIH

(b) SCHI

(c) IGSH

(d) IHSC

(GATE-2010)

ANS: (b)

Suppose : Hari's age : H, Gita's age : G Saira's age : S, Irfan's age: I

- $H + G > 1 + S$
- Using statement (2) both $G - S = 1$ or $S - G = 1$; can't be oldest and S can't be youngest.
- There are no twins thus using statement (2) either GS or SG possible.

(A) HSIH: not possible as there is I between S and G which is not possible using statement(3)

(B) SGHI : SG order is possible $S > G > H > 1$ and $G + H > S + 1$ (possible) Because if $\{S = G + 1; \text{and } G = H + 1 \text{ and } H = 1 + 2 \text{ then } G + (1 + 2) > (G + 1) + 1\}$

(C) IGSH : according to this $1 > G$ and $S > H$ thus adding these both inequalities we get $1 + S > G + H$ which is opposite of statement (2) thus not possible.

(D) IHSG : according to this $1 > H$ and $S > G$ thus adding both inequalities $1 + S > H + G$ which is opposable of statement (2) thus not possible.

